Topological Fixed-Point Theory (TFPT)

A discrete compiler for the constants of physics

Two inputs — the boundary constant $c_3 = \frac{1}{8\pi}$ and a five-slot carrier — build E_8 and read off the Standard Model: status assessment and reading guide

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In one sentence

Topological Fixed-Point Theory (TFPT) constructs, from exactly **two inputs** — a boundary constant $c_3 = \frac{1}{8\pi}$ (P1) and a five-slot carrier $g_{\text{car}} = 5$ (P2) — a discrete compiler for the Standard-Model skeleton: the gauge group, three families, hypercharges and the flavor matrix, with E_8 ($D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$) as the consistency checksum. The *algebraic core is machine-checkable* and the dimensionless constants follow as fixed points ($\alpha^{-1} = 137.0359992$); *physical readouts* — absolute scales, masses, inflation, gravity and cosmological transfers — run through explicitly *named, status-typed bridges* (v_{geo}, G_{net}, F_{transfer}), not as free outputs. The honest one-line claim is on the status card below.

The TFPT document set — what is treated where

Plain language: TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant $c_3 = \frac{1}{8\pi}$ and the carrier rank $g_{\text{car}} = 5$ — build $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. `introduction` — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (`verification/status_ledger.csv`). *(entry point)*

1. `tfpt_1_architecture_e8` — the two axioms, the derivation map, the EM fixed point α^{-1}, the $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ construction, the QBL theorem chain.

2. `tfpt_2_standard_model` — the SM in one φ_0-ladder formula, flavor from parabolic transport (sheet diamond, dual anchor, Q_+ cohomology), the worked closures.

3. `tfpt_3_e8_audit_bootstrap` — the seven E_8 slices as an audit raster, the cascade bridge, the Möbius bootstrap.

4. `tfpt_4_frontier` — honest status of η_B, m_p/m_e, Koide, dark matter, full QG.

5. `tfpt_5_redteam` — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

Companions: R. `tfpt_research_contracts` — the open interfaces ($v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$) as numbered contracts; **H.** `tfpt_horizon_readouts` — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O.** `origin_theory` — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

You are reading the introduction — the entry point and reading guide.

TFPT status at a glance — read this card the same way in every document

Two inputs, one compiler. From the boundary constant $c_3 = \frac{1}{8\pi}$ (axiom P1) and the five-slot carrier $g_{\text{car}} = 5$ (axiom P2) — jointly the elementary symmetric data of one anchor $a = (1, 1, 2)$ — the discrete compiler $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$ is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ($\alpha^{-1} = 137.0359992$) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

What is closed, and what genuinely remains. The discrete/algebraic layer is closed ([E]); the everything-open residual reduces to *three named interface problems*:

Interface	Question	Status
v_{geo}	the one metrology unit ($= 1/\sqrt{G} = m/\mu$); No-Unit Thm: no compiler scale	primitive [O]
G_{net}	simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle(1, 1)\rangle \cong (E_8)_1$	[E]/[C]
F_{transfer}	one functor, four typed interfaces (Koide, η_B, axion, m_p/m_e)	[C]

Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ($U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$) are retained only for ledger continuity — the live residual is $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$.

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

[E] exact / machine-proven [C] conditional [O] open / axiom [X] kill test

P1 $c_3 = \frac{1}{8\pi}$ **P2** $g_{\text{car}} = 5$ *(two inputs, [O] axioms)*

↓

Anchor $a = (1, 1, 2)$: $e_1, e_2, e_3 = 4, 5, 2$ *([E] exact)*

↓

Discrete compiler: $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$, $h = 30$, φ_0 *([E] exact)*

↓

SM skeleton: three families, hypercharges, flavor matrix R *([E])*

↓

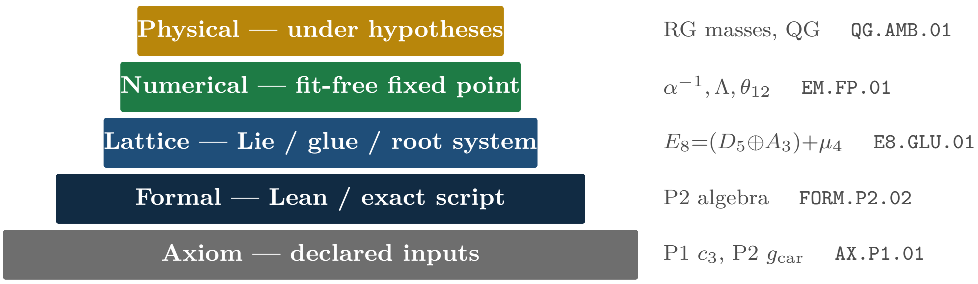
Physical bridges: v_{geo}, G_{net}, F_{transfer} *([C] bridges)*

↓

Frontier readouts: Λ, inflation, η_B, dark matter, QG *([C]/[O])*

Contents

Reading the markers. Claims carry one of *four* reader-facing classes, so one can see at a glance what is proven and what is conditional: [E] *exact / machine-proven* — an exact identity (e.g. $240 = 16 \cdot 5 \cdot 3$), a Lie/lattice theorem (e.g. $D_5 \oplus A_3 + \mu_4 = E_8$), a Lean/script formalisation, or a reproduced numerical fixed point; [C] *conditional* — follows under explicitly named hypotheses (a physical bridge, readout or pending step); [O] *open / axiom* — a declared input or a genuine gap; [X] *kill test* — a falsifiable threshold. The finer per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is kept in `verification/status_ledger.csv`, the single source of truth.



The claim stack (bottom = most certain/fewest, top = conditional). Each claim sits on exactly one level; the bottom two are inputs, the middle two are proved/reproduced, only the top is conditional. The single source of truth for which claim sits where is `verification/status_ledger.csv`.

No-free-pattern rule. To keep the structure from drowning in decorative coincidences, an identity enters the *main text* only if it (1) is a necessary step in a derivation, (2) kills an alternative, (3) is ablation-relevant, (4) links two previously independent modules, or (5) is experimentally tested. Everything else is explicitly demoted to an *audit fingerprint* [E] (clearly boxed as such) — so the load-bearing beams stay visible and the ornament never gets promoted to spine.

Admissible-invariant classes (the harder filter). Sharper still: a *load-bearing* integer may only come from one of seven typed sources — (i) a symmetric function $e_k(a)$ of the anchor $a = (1, 1, 2)$; (ii) a power sum $p_n(a)$ or difference $p_{n+1} - p_n$; (iii) a determinant, spectrum, Smith normal form or trace of (R, Q, K, L); (iv) an anchor quadratic form $u^\top M v$ with $u, v \in \{\mathbf{1}, a\}$; (v) a Plücker coordinate of the anchor plane $\langle \mathbf{1}, a \rangle$; (vi) an E_8 Casimir degree or branching block; (vii) a pre-registered experimental observable. Anything else is an audit fingerprint or a frontier item, never spine. This is what converts TFPT from *pattern hunting* ("look, another number") into a typed compiler: *fewer admissible kinds of explanation*, not more matches.

Reviewer path (read in this order; the rest is support):

1. the architecture: the two axioms $\{c_3, g_{\mathrm{car}}\}$ and the two-engine picture (§1, the figures);
2. the E_8 glue $E_8 = (D_5 \oplus A_3) + \mu_4$ — document 1, Part II, machine-checked (`v1_e8_glue.py`);
3. the α^{-1} fixed point plus ablation (`v3_em_alpha.py`);

4. the **single status ledger** `verification/status_ledger.csv` — the source of truth for every claim's grade, location, dependency and script;
5. the **open gates** as numbered research contracts (`tfpt_research_contracts`: the residual classes R1–R5 and their current reductions) and the honest frontier list (document 4).

Everything carries a claim ID (e.g. `E8.GLU.01`, `EM.FP.01`) that resolves to a row of the ledger. **If the text and the ledger ever disagree, the ledger wins.**

1 What this is about: the jump in construction principle

In plain words

What we found. TFPT reproduces dozens of real numbers of physics (particle masses, the fine-structure constant, dark-energy and inflation values) from a small set of inputs — and the compiler closure says *why* the same handful of small integers $(2, 3, 5, 16, \ldots)$ reappears in every sector: those numbers are not separate lucky hits. A single tiny "machine" — fed by just **two inputs** (a boundary number $\frac{1}{8\pi}$ and a five-slot carrier) — builds the exceptional symmetry E_8, and E_8 is the unimodular "audit hull" that classifies the discrete Standard-Model readout: the constants, the flavor skeleton and the gravity/cosmology regime are then read off by projection and boundary dressing (not "printed" for free — where a scheme or analytic interface is needed, it is named).

Why this is simple. There are not many separate derivations that happen to share numbers — there is *one generator and a handful of corollaries*, and the number of independent structural assumptions is **two**.

How plausible it is. A theory becomes more believable when it *explains more from less.* Here the two inputs are even *over-determined*: the machine reproduces the very two numbers it started from (the dashed bootstrap loop in the architecture diagram, §2), so there is *no free dial left to tune.* The only genuinely free, continuous ingredient that remains is the number π.

Sharper still: one anchor $a = (1, 1, 2)$, and E_8 as a compiler hull ([E])

The two axioms are not even independent: they are the *elementary symmetric polynomials* of the single parabolic anchor $a = (1, 1, 2)$ (the exponents-at-infinity / splitting type $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$):

$$e_1(a) = 4 = |\mu_4|, \qquad e_2(a) = 5 = g_{\text{car}}, \qquad e_3(a) = 2 = |\mathbb{Z}_2|, \qquad c_3 = \frac{1}{2e_1(a)\pi} = \frac{1}{8\pi}.$$

Its *power sums* $p_n(a) = 1^n+1^n+2^n = 2+2^n$ generate the big Lie data directly: $p_1{=}4{=}|\mu_4|$, $p_2{=}6{=}|R^+(A_3)|$, $p_3{=}10{=}\mathcal{A}_\Lambda$, and

$$|R(E_8)| = p_1p_2p_3 = 240, \quad \dim E_8 = p_1p_2p_3 + (p_4 - p_3) = 248, \quad p_4 - p_3 = 8 = \operatorname{rank} E_8,$$

with the binary ladder $p_{n+1} - p_n = 2^n$. So the inputs collapse from $\{c_3, g_{\text{car}}\}$ to $\boxed{a = (1, 1, 2) \text{ plus } \pi}$.
[`verification/v23_anchor_generator.py`]

And a is itself half-forced ([E]/[C]): on $\mathbb{P}^1$ every bundle splits (Birkhoff–Grothendieck), so once $|\mu_4| = 4$, $N_{\text{fam}} = 3$, $\deg E = -4$ are set, the three positive anti-degrees are positive integers summing to 4, and $4 = 2 + 1 + 1$ is the *unique* partition into three positive parts. The genuine remaining axiom is therefore not "why $a = (1, 1, 2)$" but "why the carrier $g_{\text{car}} = 5$ / $(|\mu_4|, N_{\text{fam}}) = (4, 3)$" — that interface (with c_3) is the [O] input layer.

Two readings, one source (no contradiction). P1 and P2 are the *operational compiler inputs*; their anchor provenance is $a = (1, 1, 2)$ plus π. The *single structural bedrock postulate* is `QGEO.SYM.01` (the carrier μ_4 clock is the seam's conformal deck, see `tfpt_research_contracts`): it forces the μ_4 deck, hence $N_{\text{fam}} = 3$, $\deg E = -4$ and the tripartition $4 = 2{+}1{+}1$, of which c_3 and g_{car} are the two elementary readouts. "Two inputs" (the working axioms) and "one postulate" (the geometric bedrock) are the same statement at two layers.

One source, four readings (the cleanest origin). The four-point seam divisor μ_4 on $\mathbb{P}^1$ generates the whole discrete skeleton by four canonical functors:

$$\underbrace{c_3 = \tfrac{1}{8\pi}}_{\text{thermal}} \quad \underbrace{\operatorname{rank} H_1(\mathbb{P}^1 \setminus \mu_4) = 3 = N_{\text{fam}}}_{\text{cycles}\to A_3} \quad \underbrace{h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = 5 = g_{\text{car}}}_{\text{functions}\to D_5}$$

$$\underbrace{(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 \cong (E_8)_1}_{\text{net}}$$

and the function side even generates D_5 itself ([`verification/v197_rr_carrier_clifford_d5.py`]): $\Lambda^{\text{even}}(\mathbb{C}^5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor. So μ_4 is not merely glue: it is the *common source* of A_3, D_5, c_3 and E_8 ([`verification/v189_riemann_roch_carrier.py`]) — the arithmetic is exact [E], the physical reading of the carrier as the seam mode space stays [C].

What E_8 is (and is not). In TFPT E_8 is *not* an unbroken physical gauge group: it is the unimodular *audit / compiler hull* that classifies the admissible discrete charge and residue structures. The Standard Model is a *readout* after projection, not a direct E_8 gauge theory; the Lorentz signature appears only after projection; fermions are not "everything in the adjoint". This is exactly why the known no-go results against literal E_8 world-formulas (Distler–Garibaldi; Coleman–Mandula) do *not* bite: they constrain E_8 as a spacetime gauge symmetry, which TFPT does not claim. The defensible claim is: *TFPT is the unique parabolic compilation of the anchor $a = (1, 1, 2)$ into the E_8 audit hull.*

In one line:

> The anchor generates the syntax; the boundary generates the scale; E_8 checks the consistency.

i.e. the discrete content (carrier $D_5 \oplus A_3 + \mu_4$, the operators R, Q, K, L) flows from $a = (1, 1, 2)$; the continuous content ($c_3 = \frac{1}{8\pi}$, φ_0, α^{-1}, the exponential scales) flows from π; and E_8 is the unimodular *checksum container*, not the world group.

The anchor ratio triad ([`verification/v119_review_validation_2.py`]). The three elementary symmetric values, normalised by the family count, are the theory's three recurring fractions [E]:

$$\frac{e_3}{p_0} = \frac{2}{3} \text{ (Koide branch / sheet ratio)}, \quad \frac{e_1}{p_0} = \frac{4}{3} \text{ (seed gain)}, \quad \frac{e_2}{p_0} = \frac{5}{3} \text{ (carrier branch)}$$

— the canonical flow's critical points are exactly $(2, 5) = (e_3, e_2)$ with inflection $\frac{7}{2} = \frac{e_3+e_2}{2}$: the fractions $\frac{2}{3}, \frac{4}{3}, \frac{5}{3}$ are not scattered numbers but the three normalised anchor coefficients. Further micro-identities [E]: $\Omega_{\text{adm}} = 2p_1p_2 = 48$, $10b_1 = 1 + p_1p_3 = 41$, $\Delta_Y = e_2^2 = p_0^2 + 2 \operatorname{rank} E_8 = 25$, $h(E_8) = e_3p_0e_2 = 30$, $\operatorname{rank} E_8 = p_0 + e_2$.

$a = (1, 1, 2)$ *[E]*

elementary $e_k(a)$
$e_1 = 4 \to |\mu_4|$
$e_2 = 5 \to g_{\text{car}}$
$e_3 = 2 \to |\mathbb{Z}_2|$
$c_3 = \frac{1}{2e_1\pi} = \frac{1}{8\pi}$

power sums $p_n = 2 + 2^n$
$p_0, p_1, p_2, p_3 = 3, 4, 6, 10$
$p_1p_2p_3 = 240 = |R(E_8)|$
$p_4 - p_3 = 8 = \operatorname{rank} E_8$
$\Rightarrow \dim E_8 = 248$

derived motifs
$4, 5, 8, 16, 25,$
$30, 41, 120, 240$
one anchor,
many projections

The companion documents

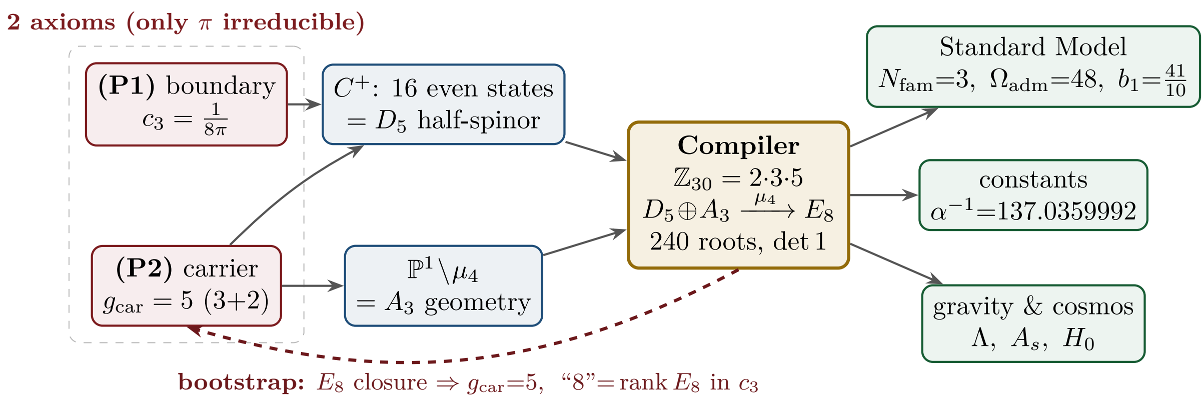
The whole development is consolidated into **eight companion documents** (plus this introduction), organised in layers that build on one another; this introduction is the entry point and reading guide. Each document collects the material of its layer into self-contained parts:

Document	What it contains (former notes now as parts)
`tfpt_1_architecture_e8`	**Core.** The two axioms $\{c_3, g_{\text{car}}\}$, the derivation map and the EM fixed point (Part I); the Coxeter–cyclotomic compiler $\mathbb{Z}_{30}$=2·3·5, the explicit $D_5 \oplus A_3 + \mu_4$ E_8 construction, the machine-checkable E_8 appendix and the group-level glue $(\mathrm{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ (Part II).
`tfpt_2_standard_model`	**Standard Model.** The full SM in one φ_0^{ret}-ladder formula (Part I); the flavor block from parabolic weights and the transport theorem, H2 as a spectral selector (Part II); the worked closures — explicit gap, APS–Fredholm, truncation/n_s, r, Starobinsky M, derived θ_{12}, quark c, the H2 splitting type, Λ-typing (Part III).
`tfpt_3_e8_audit_bootstrap`	**E_8 audit & bootstrap.** The seven E_8 slices as an audit raster ($E_6\times A_2$ reads the flavor matrix) (Part I); the old cascade D=60−2n as the same even-integer spine (Part II); the Möbius bootstrap — g_{car}=5 and the "8" in c_3 as overdetermined fixed points, only π irreducible (Part III).
`tfpt_4_frontier`	**Frontier.** The honest status of η_B, m_p/m_e, Koide, dark matter and quantum gravity — the genuine handle, the precision, and what is *not* forced onto the ladder.
`tfpt_5_redteam`	**Red Team.** The adversarial audit of the five load-bearing reductions (Targets A–E): the declared attacks, what survives each, what each one reduces to, and the explicit kill tests.
`tfpt_research_contracts`	**Research contracts.** The open gates as numbered contracts with lemma chains and closing theorems: (U_{wall}) (now reduced to the selector triangle: $(d,n)\Rightarrow R$ columnwise, residue = *two* line pairings, the third integrality-forced) and $G_{\text{metric}}/G_{\text{net}}$ (the index-4 seam-net identification — one theorem, three doors, now realised at the finite Lie level).
`tfpt_horizon_readouts` **(App. H)**	**Appendix H — horizon unit system (reframe, not new physics).** One seam constant $c_3 = \frac{1}{8\pi}$ as the universal horizon thermal code: Hawking/de Sitter/Unruh, BH thermodynamics, Page, scrambling, Nariai bound, $v_{\text{GW}}=c$ — standard physics in seam units, two compiler fingerprints (1920=$\lvert W(D_5)\rvert$, $\lvert\mu_4\rvert$=4); plus the Nariai/anchor surface, the dual anchor and the resummed seam clock (v101–v138).
`origin_theory`	**Origin Theory.** Why no free *dimensionless compiler dial* remains beyond the anchor and π (the dimensionful v_{geo} and transfer physics stay typed): the $(g_{\text{car}}, N_{\text{fam}})$ skeleton, the triply-forced 8, the order-30 Coxeter cycle, one boundary transport for flavor and horizon, the gapped unique attractor; **[E]** core plus one honestly-typed **[C]** cyclic interpretation.

The five formerly "research-level" problems are now *closed, reduced or typed* inside the documents where they belong — the group-level closure $(\mathrm{Spin}(10)\times SU(4))/\Delta\mathbb{Z}_4$ in document 1 (closed), and the H2 splitting type, the Starobinsky scalaron mass, the θ_{12} texture and the quark c in document 2 (the first two closed; the quark *ratios* now closed combinatorially (v49/v71), only the *absolute* scale a U_{point} anchor; θ_{12} the seam-misalignment residual). Eight companion files (plus this introduction), each result where it logically lives.

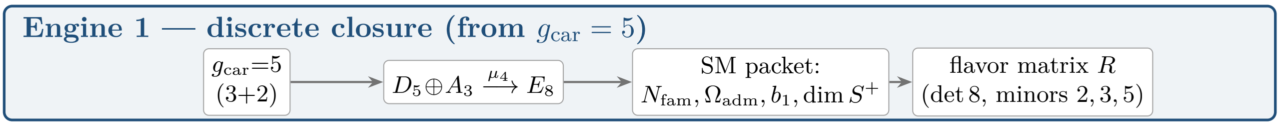
2 The architecture at a glance

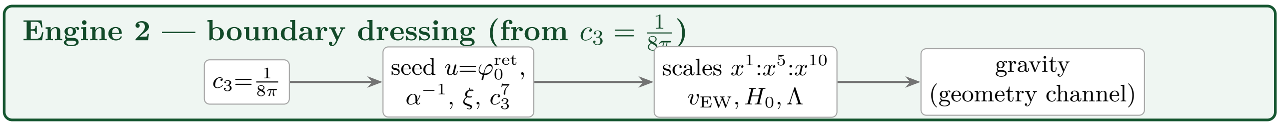
The diagram below shows the whole pipeline: *two* inputs on the left, the discrete compiler in the middle, the physics on the right. Everything in between is a consequence.



Key statement (the loop closes): previously D_5, A_3 and E_8 sat *as given* Lie structures in the background. Now they are *intermediate products* of the two inputs — *and the red back-channel is the new part*: the E_8 closure feeds back and *fixes the inputs*. $g_{\text{car}} = 5$ is the unique rank-fill/Coxeter/Pascal solution $(2^{g-1} = \binom{g}{0}+\binom{g}{1}+\binom{g}{2})$, and the integer 8 in $c_3 = \frac{1}{8\pi}$ is rank $E_8 = h(D_5) = \det R$. **Inputs and output form an overdetermined fixed point** — the structure does not "prove itself"; it has fewer degrees of freedom than independent consistency conditions. The only thing *not* produced by the loop is the continuous π (Möbius/Gauss–Bonnet), the lone irreducible primitive. So it is not a one-way arrow chain but a closed, overdetermined self-consistency loop.

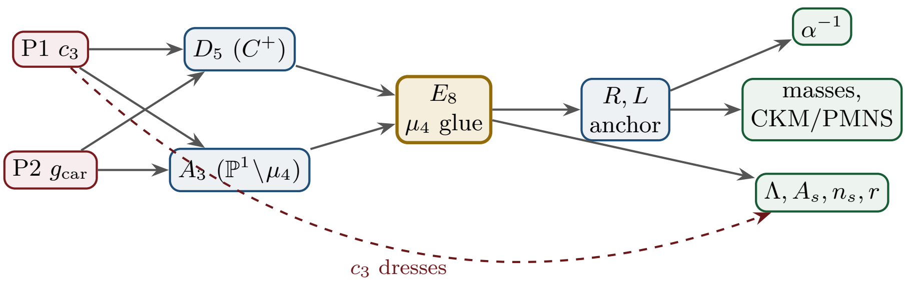
The master story is two engines. Read from the two axioms, the whole theory factorises into exactly *two* engines — a *discrete* closure and a *boundary* dressing. Gravity is not a third block; it is the geometry channel of Engine 2.





3 The dependency DAG and the proof ledger

The whole theory is a directed acyclic graph: two axioms at the source, the compiler in the middle, the observables as sinks. The DAG makes the attack surface explicit — each arrow is a named theorem or a declared input, and each node fails in exactly one way.



Node	Definition	Inputs	St.	Output	Failure mode
P1	RP boundary kernel, c_3=1/(8π)	— (axiom)	[O]	c_3	no reflection-positive seam
P2	5-slot carrier $g_{\rm car}$=5	— (axiom)	[O]	carrier 3+2	wrong family/charge lattice
D_5	C^+ even-Hamming = half-spinor 16	P2	[E]	$\dim S^+$=16, Y	group/matter mismatch
A_3	$\mathbb{P}^1 \setminus \mu_4$ four-puncture geom.	P1	[E]	$N_{\rm fam}$=3, $\mathbb{Z}_6$	wrong multiplicity
E_8	μ_4 glue: disc=$\mathbb{Z}_4$, q-sum= 2	D_5, A_3	[E]	240, 248	not even-unimodular
R, L	residue + winding matrix, anchor (1, 1, 2)	A_3	[E]	det 8, minors (2, 3, 5)	wrong D_6 branch
α^{-1}	root of $F_{U(1)}(\alpha)$=0	c_3, M=41	[E]	137.0359992	no/second root (excl.)
masses	$\varphi_0^{\rm ret}$-ladder $\lambda_Y^L \Lambda$	R, L, c_3	[E]/[C]	9 masses, mixings	hierarchy mismatch
θ_{12}	TBM + seam ε=c_3	R, L	[E]/[C]	0.307	seam-misalign. lemma fails
Λ, A_s	seam det. / R^2 scalaron	c_3	[C]	Λ, A_s, n_s, r	ambient RP fails

Marker key: [E] exact / machine-proven · [C] conditional (holds under named hypotheses) · [O] open / axiom · [X] falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

Reading the table top-to-bottom is reading the theory: two red axioms in, everything else a consequence with exactly one failure mode each.

The ledger itself is versioned: each row of `verification/status_ledger.csv` carries a claim ID, its dependencies, its verification script, and `active/canonical_status/supersedes` fields, so claims that were later superseded (e.g. the early "flavor gate open" rows, now *ratios closed* via `FLAV.RIGID`) are marked inactive — the canonical status is the only one read as current.

The compact status formula: three honest layers

$$\underbrace{\text{compiler}}_{\text{closed}} \;\Big|\; \underbrace{\text{admissible IR physics}}_{\text{conditionally closed (RP, gap)}} \;\Big|\; \underbrace{\text{strict physical TOE}}_{\text{open}}$$

TFPT 5.0 closes the discrete compiler, the algebraic Standard-Model readout, the EM fixed point, the admissible gapped IR sector and the $R + R^2$ spectral-action shadow. It does *not* yet certify a strict physical TOE. The remaining items have now collapsed sharply: the R-bridge amplitude normalisation U_{point} *reduces to* v_{geo} (Gate 1 closed, v75 — the same anchor as $1/G$), and the ambient metric measure is *holographically reduced* to a finite seam-*boundary* measure (Gate 2 tier B, v76) — which is in turn the rigorously-constructed $(E_8)_1$ lattice net (v77), so G_{metric} is imported into existing RCFT rigor rather than built anew. So the only genuine residuals are *one* dimensionful scale v_{geo} — the *dimensional-analysis floor* that no theory escapes (v78), shared by flavor and gravity — the boundary net identification, and the named QFT/cosmology transfer interfaces (F_{frontier}).

Two points that are *not* in the residual (closed, but easy to mis-list as open). *(i) Parameter-freeness is a theorem under the named hypotheses:* the boundary transport is gapped ($\Delta = 6\log\frac{3}{2} > 0$), so by Perron–Frobenius its iteration has a *unique* attracting fixed point — the constants are *selected*, not tuned ([**E**] for the attractor, [`verification/v56_unique_attractor.py`]; the *physical identification* of the transport operator stays [**C**]), reinforcing the static bootstrap web ($g_{\text{car}} = 5$ forced three ways *given* the E_8 closure rank — a consistency loop, see red team Target B — [`verification/v6_bootstrap.py`], [`verification/v14_carrier_uniqueness.py`]). *(ii) Newton's constant is not an* independent *input:* fixing the physical Λ branch ($(8\pi)^2\delta_{\text{top}} = \frac{3}{4\pi^2}$) pins $\bar{M}_{\text{Pl}}$ *relative to* Λ, so G_N is read out *after* the branch choice + *one* dimensionful induced-gravity anchor ([**E**]/[**C**], [`verification/v60_lambda_metrology_branch.py`]), not predicted from pure numbers. What stays irreducible is then just π, the discrete E_8-closure fixed point (self-consistent, *not* a free axiom; the bootstrap of `tfpt_3`), and *one* dimensionful scale v_{geo} — and that single scale is shared by *both* the flavor sector ($U_{\text{point}} \to v_{\text{geo}}$, v75) and gravity ($1/G$, v68): the two $[A]$ anchors are *one* (dimensional analysis; no metre from pure mathematics).

The hard reduction: two research gates plus typed interfaces

After the compiler closure, the entire residual is

$$\text{Rest} \quad = \quad (U_{\text{wall}}) \quad \oplus \quad (G_{\text{metric}}) \quad \oplus \quad (F_{\text{frontier}})$$

— one parabolic wall-selection, one quantum-gravity measure, and a set of deliberately typed QFT/cosmology interfaces. *Sharper aliases* (current state of the reductions; the historical gate names are kept for ledger continuity): $(U_{\text{wall}}) \to v_{\text{geo}}$ — the flavor ratios are *closed* (Plücker/Readout Rigidity, v49/v71/v75), what remains is *one dimensionful unit*, not a flavor gap; $(G_{\text{metric}}) \to G_{\text{net}}$ — reduced to a single boundary-net theorem, *"the seam–Calderón net is the index-4 μ_4 simple-current extension of the carrier net"* (holomorphy then follows, v89/`GATE.METRIC.06`); since the QBL chain (v108–v113) the *carrier choice rides on the same theorem*: the certified scalar kernel exists iff it pairs the sheets (v110), the certified channel counts the code identically (v112), pair transport is minimally complete (v111), and the one quasi-free kernel — rank $= c$: 5 carrier block, 8 seam hull — is the defining datum of the free net the gate already posits (v113), so closing G_{net} also closes the carrier selection [**E**]; $(F_{\text{frontier}}) \to F_{\text{transfer}}$ — named QFT/cosmology *transfer* interfaces (Koide source→pole, η_B Boltzmann, axion relic, m_p/m_e), never compiler powers. *Latest reductions (v134–v148):* the flavor selector residue R4′ is now "derive *one map*" — $d = \frac{3}{2}a - 2\cdot\mathbf{1}$ is pure anchor data, (d, n) forces R *columnwise* (v136/v139), the third pairing is *integrality-forced* mod 11 (v142), and the pairing *values* are atom identities given the exponent-duality lift, including the new closure $|\mu_4| + g_{\mathrm{car}} = N_{\mathrm{fam}}^2$ (v145); the sheet/parity question `GATE.QGEO` has *no discrete freedom left* — the $\mathbb{Z}_3$ choice is *derived* (v141) and the *full* Möbius D_4 of the seam curve matches the integer model parity by parity (v146); R2's Q-system identification holds at the finite Lie level (v143), with the remaining net statement *scoped*: the odd glue sectors are Ramond modules, invisible to the untwisted NS module, and the glue choice is the R-projection (v148); and the seam quantum clock R1 is *typed closed* — edge-sector budget matches Volkov–Wipf exactly (v138), the det-ratio cancellation is derived in-family (v144), the ring sum *is* the Born-squared Gaussian zero-mode integral, and the quantum bend is det′-clean (v147). Beyond the five classes: *both* dual normals are now anchor data ($n = (p_2, p_0, e_2)(a)$ in the cusp frame, v149), R3's mechanism is exhibited at the gapped-model level (v150), the Calderón transfer answered (the DtN kernel is conically *clean*, v151), and its normalisation shown to be the dimensionful anchor, not a separate gap (v152). Two closing theorems then re-type the survivors: the **No-Unit Theorem** (v153) makes v_{geo} an *irreducible metrology primitive* — a dimensionless compiler cannot make an absolute scale, and $U_{\mathrm{point}} \sim v_{\mathrm{geo}}$, $1/G \sim v_{\mathrm{geo}}^2$, $m/\mu = e^{3/4}$ are one unit in three dimensions — and the **Simple-Current Extension Theorem** (v154) states the central G_{net} closing as $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$ (index 4, c=8, μ=1 $\Rightarrow$ holomorphic $\Rightarrow$ E_8), exact, with only the seam-side identification left. *That seam-side identification is then driven to bedrock (v175–v181):* net existence and full-cone reflection positivity are discharged to [E] (v175), the realisation is assembled as one central theorem (v176) and split into the marks + kernel obligations whose finite cores close (v177/v178); the two then unify into one conformal-realisation premise (v179), which reduces — via uniformisation, Kerékjártó and the order-4 Möbius classification — first to a seam-*isometry* premise (v180) and finally, by equivariant uniformisation, to the strictly weaker conformal-*symmetry* / deck premise `QGEO.SYM.01` (v181): the carrier μ_4 clock is the conformal deck of the seam. Below that the residual is *definitional* (the seam's conformal structure *is* the μ_4-deck structure of the carrier), not a missing theorem — the single, irreducible structural residual of the whole theory. It is since sharpened to its *non-circular* form (v194–v198): the bedrock is cracked to the *state-invariance* $\omega \circ \rho = \omega$ (the DtN principal symbol commutes *exactly* with the clock; Tomita–Takesaki gives $[\rho, \Lambda_\Sigma] = 0$ with no conformal-covariance assumption, removing the Bisognano–Wichmann circularity) — the maximally-operational form of the same postulate. The status card:

Item	Status	Reading
parameter-freeness (self-consistency)	[E]	gapped transport ⇒ *unique* Perron–Frobenius attractor (v56); bootstrap web g_{car}=5 forced (v6/v14)
Newton constant G_N	[E]/[C]	not *independent*: branch pins $\bar{M}_{\text{Pl}}$ rel. to Λ (v60); readout after branch + one dim. anchor
ratio $c_u/c_d = \frac{55}{117}$	[E]	$g_{\text{car}}\|\text{Pl}_{1,a}(K)\|_1/(N^2_{\text{fam}}\Delta_Q)$; rigid on the derived stratum (v49/v71), no solve
(U) flavor gate	[O]$\to v_{\text{geo}}$	**Gate 1 closed**: $U_{\text{point}}\to v_{\text{geo}}$ (ratios + Grand Mass Volume, v75); same anchor as $1/G$
$R + R^2$ gravity	[E]/[C]	covariant $f(R)$ field equation closed in regime; heat-kernel grounded (G2)
physical QG sector	[E]/[C]	gap-decoupled admissible measure (Δ_{eff}=1.648>0)
ambient QG measure	[O](reduced)	**Gate 2 tier B**: holographically reduced to a finite seam-*boundary* measure (v76); IR tier closed (decoupling); strict-TOE cert. = the simple-current extension $(D_5)_1{\otimes}(A_3)_1{\rtimes}\langle(1,1)\rangle \cong (E_8)_1$ (v154); the free-bulk premise (A) is itself a fixed-point theorem with the cone eliminated (cone gap = one-particle gap $(2/3)^6$) and factors into the already-open A_2 net-existence + `GATE.QGEO` — zero new gates (v160–v165)
v_{geo} metrology unit	[O](primitive)	No-Unit Theorem (v153): a dimensionless compiler makes no absolute scale; $U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v^2_{\text{geo}}$, m/μ a ratio — one unit, irreducible
Koide	[C]	$Q_{\text{src}} + \varphi_0/24$ as source→pole transfer conjecture
axion DM	[C]/[O]	$f_a = M_{\text{scal}}/128$ as a scenario
η_B	[C]	readout closed; leptogenesis Boltzmann solve missing
m_p/m_e	[O]	cross-sector QCD/EW ratio
$N_\star$	[C]	reheating input; scalaron-reheating point $N_\star$=51.4 (v86); frozen band $[50, 60]$
$a = (1, 1, 2)$	[E]/[C]	forced by $4 = 2 + 1 + 1$, conditional on the carrier
carrier selection g=5 (QBL)	[E]/[C]	theorem chain v108–v113: sheet-odd kernel, self-counting channel, transport degree 2, one quasi-free kernel (rank $= c$); residue = the G_{net} premise itself — gate closes ⇒ carrier [E]
selector establishment R4′	[E]/[C]	$(d, n) \Rightarrow R$ columnwise (v136); both normals *anchor data* — $n = (p_2, p_0, e_2)(a)$ in the cusp frame (v145/v149); residue = *one* assignment
sheet/parity `GATE.QGEO`	[E]/[C]	H^1 characters = $\text{Spec}(Q_+) = A_3$ exponents (v137); $\mathbb{Z}_3$ choice *derived* (v141); full Möbius D_4 matched parity by parity (v146); residue = the module identification only
seam quantum clock R1	[E]/[C]	typed closed: resummed clock = entropy power law (v124–v133); VW edge match (v138); det-ratio in-family (v144); ring sum = Born^2 Gaussian zero-mode integral, bend det′-clean (v147); residue = measure + reference
E_8 Lean	[E]	hardening, not a blocker (script-certified)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`*.*

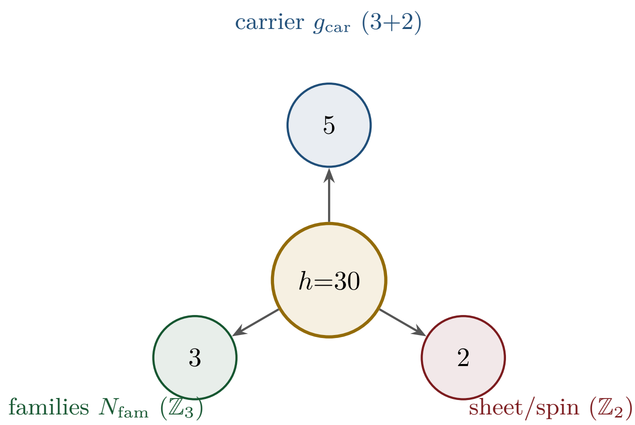
Only *two* research gates remain — (U_{wall}) and G_{metric}; the frontier items are typed interfaces, not structural holes. Of these, (U_{wall}) is the only genuine *physics* gate left: G_{metric} is now heat-kernel grounded (G2: the spectral action gives $R + R^2$ structurally) and *gap-decoupled* (G5: Δ_{eff}=1.648>0 isolates the closed admissible sector from the un-built ambient), so its residual — the ambient projective limit (G6) — is a *strict-TOE completion target*: its absence does not affect the bounded IR claim, but it does block certification as a strict physical TOE. Both are written up as numbered *research contracts* (lemma chains + closing theorem + machine-certifiability) in the companion **`tfpt_research_contracts`**, with priority on the selector-triangle pairings (the v_{geo} scale anchor) and the G_{net} inclusion theorem. [`verification/v36_spectral_action_g2.py`]

Reduction in one line

The number of *structural* assumptions is **two axioms**. Everything else ($N_{\text{fam}}, \Omega_{\text{adm}}, b_1, \alpha^{-1}$, the flavor matrix, the E_8 glue, the scale grammar) is a *consequence. And even those two tighten:* the bootstrap note shows $g_{\text{car}} = 5$ and the integer 8 in c_3 are *overdetermined* E_8*-closure fixed points* (rank-fill, Coxeter-match, integer-glue/norm all give 5; $8 = h(D_5) = \operatorname{rank} E_8 = \det R = \varphi(30)$), leaving *one* genuine continuous primitive: π from the Möbius boundary. So the *discrete* core has no freely tunable fundamental numbers — the bootstrap is *not creation from nothing* (two inputs remain), but an overdetermination of those inputs. (This concerns the discrete core only; dimensionful masses, the QG measure and frontier items remain typed-open.) [`verification/v6_bootstrap.py`] [`verification/v14_carrier_uniqueness.py`] [`verification/v23_anchor_generator.py`]

4 The compiler: why $30 = 2 \cdot 3 \cdot 5$

The Coxeter number of E_8 is $h = 30 = 2 \cdot 3 \cdot 5$. Exactly these three prime factors are the three discrete atoms of the theory — that is the point of the compiler:

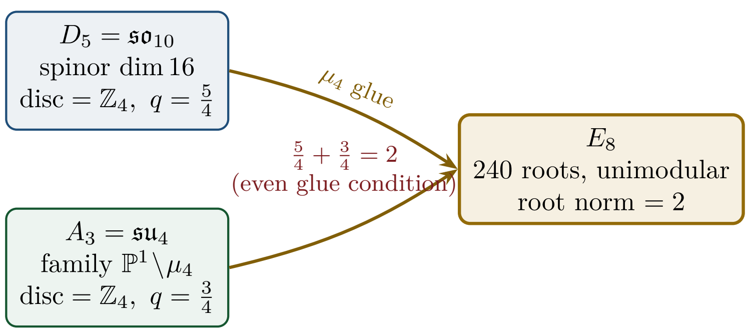


From this the compiler reads, among others:

- $\dim S^+ = 1 + 10 + 5 = 16$ (one generation) as the even exterior algebra of the 5-slot carrier — the same Pascal row $\binom{5}{0}, \binom{5}{1}, \binom{5}{2} = 1, 5, 10$ appears in family, action ladder and the E_8 root count. **[E]**
- $|R(E_8)| = 16 \cdot 5 \cdot 3 = 240$ and $\dim E_8 = 240 + (5 + 3) = 248$. **[E]**
- the $\mathbb{Z}_{30}$ Coxeter element $\leftrightarrow$ the cyclotomic polynomial Φ_{30}; the corner rotation $z \mapsto iz$ on $\mathbb{P}^1 \backslash \mu_4$ is the A_3 Coxeter element. **[E]**

5 The μ_4 glue: how E_8 is really built

The decisive, now *proved* step: D_5 and A_3 have the *same* discriminant group $\mathbb{Z}_4$, and their discriminant-form norms are exactly two TFPT constants which add up to the E_8 root norm.



Glue theorem (proved)

$\mathrm{disc}(D_5) = \mathrm{disc}(A_3) = \mathbb{Z}_4$, glue index $|\mu_4| = 4$, and $q(D_5) + q(A_3) = \frac{5}{4} + \frac{3}{4} = 2 =$ the E_8 root norm. Hence $E_8 = (D_5 \oplus A_3) + \mu_4$-glue is a *closed lattice-theoretic* construction — not a blind "positing of 248". **[E]** [`verification/v1_e8_glue.py`]

6 How derivation now works: the derivation map

What can one concretely *compute* with $\{c_3, g_{\mathrm{car}}\}$? The following map summarises what is now a consequence rather than an assumption.

Sector	How it now arises	Status
gauge group / families	carrier 3+2, $\dim S^+$=16, N_{fam}=3, Ω_{adm}=48	[E]
hypercharge	roots of the charge polynomial $bsY^2 + (s-b)Y - 1$	[E]
$U(1)$ index b_1	$b_1 = \dfrac{g_{\mathrm{car}}\, 2^{g_{\mathrm{car}}-2} + 1}{10} = \dfrac{41}{10}$	[E]
fine structure α^{-1}	unique root of $F_{U(1)}(\alpha) = 0$	[E]
flavor matrix	transport theorem (H1 proved, H2/H3 force the matrix)	[E]/[C]
electroweak scale	$v_{\mathrm{EW}} \sim e^{-\alpha^{-1}/5}$ (carrier divides α^{-1} by 5)	[E]
cosmological constant	$\Lambda \sim e^{-2\alpha^{-1}}$ (density quotient; see typing note)	[E]/[C]
Hubble scale	$H_0 \sim \sqrt{\Lambda}$	[C]
inflation amplitude	$A_s = \dfrac{N_\star^2}{24\pi^2} c_3^7$ in the R^2 branch	[C]
E_8	$D_5 \oplus A_3 + \mu_4$-glue	[E]

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

Notable: the scale grammar is *one* exponential engine. The same number $\alpha^{-1} \approx 137$ generates the electroweak scale (divided by 5 = carrier), the cosmological constant (times 2) and — via the square root — the Hubble scale. One constant becomes many orders of magnitude.

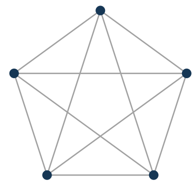
Typing note on Λ (a deliberate danger zone). "Λ" denotes *three* distinct objects that must not be conflated: (i) the seam-determinant exponent $e^{-2\alpha^{-1}}$ (an exact [E] readout), (ii) the dimensionless density quotient $\rho_\Lambda/\bar{M}_{\mathrm{Pl}}^4$ (which carries a prefactor *branch*, hence [C]), and (iii) the observed curvature, to which (ii) is *anchored*, not predicted. The exponential is exact; the absolute dark-energy density remains a typed cosmology interface (full disambiguation in `tfpt_2_standard_model`).

Cosmology is the Pascal action grammar of the carrier

With $x = v_{\mathrm{geo}}/(5\beta_{\mathrm{rad}}^2 \bar{M}_{\mathrm{Pl}})$ the cosmological readouts are a *power tower* on the five-slot carrier graph K_5:

$$\frac{H_0}{\bar{M}_{\mathrm{Pl}}} = x^5 R_H, \qquad \frac{\rho_\Lambda}{\bar{M}_{\mathrm{Pl}}^4} = x^{10} R_\Lambda, \qquad (1, 5, 10) = \binom{5}{0}, \binom{5}{1}, \binom{5}{2}$$

EW is the unit x^1, Hubble is the product over the $\binom{5}{1} = 5$ vertices, Λ is the product over the $\binom{5}{2} = 10$ edges — the *same* Pascal triple that reads one generation, the action ladder and the E_8 root count $16 \cdot 5 \cdot 3 = 240$. "Hubble = slot product, Λ = pair product." [E]



$\binom{5}{1} = 5$ vertices — action x^1 (Hubble slot)
$\binom{5}{2} = 10$ edges — Λ density pair x^{10}
$\binom{5}{3} = 10$ triangles — dual sector
$\binom{5}{4} = 5$ complements — inverse sector
$\binom{5}{5} = 1$ determinant — closure

K_5

K_5 on the five carrier slots; the action powers x^1, x^5, x^{10} are its $\binom{5}{k}$ faces (mnemonic; the physical scale map stays [C]).

The entire fermion spectrum in one formula

Masses, Yukawa, CKM, PMNS and neutrinos all follow from *one* master formula with *one* seed:

$$\hat{m}_{f,j} = \frac{v_{\text{geo}}}{\sqrt{2}}\,\lambda_Y^{L_{f,j}}\,\Lambda_{f,j}, \qquad \lambda_Y = \sqrt{\varphi_0^{\text{ret}}(1-\varphi_0^{\text{ret}})}, \qquad \varphi_0^{\text{ret}} = \frac{1}{6\pi} + \frac{3}{256\pi^4}.$$

The word-lengths $L_{f,j}$ are the *fixed residue matrix of the compiler*, whose characteristic polynomial $t^3 - 9t^2 + 10t - 8$ carries only compiler numbers ($\mathrm{tr} = 9 = N_{\text{fam}}^2$, $\det = 8 = h(D_5)$, principal 2-minors 2,3,5 with product $30 = h(E_8)$). Through the φ_0^{ret}-ladder $\hat{m} = \frac{v_{\text{geo}}\pi}{\sqrt{2}} c\,(\varphi_0^{\text{ret}})^k$ all nine word-lengths are fixed and the *charged-lepton* amplitudes are closed; the *quark* mass *ratios* are closed too (integer Plücker readouts on the derived selector stratum, v49/v71), with only the *absolute* amplitude scale reducing to the (U_{wall}) anchor. **SM status:** the discrete packet, the dimensionless flavor skeleton, the charged-lepton source masses and the quark *ratios* are closed; the *absolute* quark scale and the full PMNS completion are reduced/conditional; $m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s$ are RG scheme-layer (m_H rests on the closed UV quartic $\frac{1}{16\pi^2}$); and the solar angle is *realized by a TFPT texture* and reduced to the seam-misalignment lemma ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$; see below). *(Full derivation: document 2, `tfpt_2_standard_model`.)* [`verification/v18_quark_yukawa.py`] [`verification/v20_lepton_c_derivation.py`] [`verification/v46_grand_mass_volume.py`]

7 Physical, geometric, topological: why the architecture is plausible

The closure works on three levels at once: *physically*, α^{-1} is the fixed point of *one* equation and one generator drives many scales; *geometrically*, the C^+ spinor and the puncture geometry $\mathbb{P}^1 \setminus \mu_4$ produce D_5 and A_3, glued to E_8; *topologically*, the common discriminant $\mathbb{Z}_4$ (the μ_4 winding) forces the glue and Φ_{30}/Coxeter drives the sectors — with exactly 2 axioms and the rest a consequence. **Why the compiled architecture is plausible.** It is (i) *parsimonious* (few independent assumptions $\Rightarrow$ low overfitting risk), (ii) *rigid* (the glue condition $5/4 + 3/4 = 2$ and the discriminant $\mathbb{Z}_4$ leave little freedom), and (iii) *checkable* (machine-checkable E_8 appendix, explicit fixed-point equation for α^{-1}). More explained from less — exactly the criterion that makes a theory more probable.

8 What role E_8 now plays exactly

E_8 is no longer the starting point but the *output format*:

- **Bookkeeping:** E_8 is the unimodular container in which carrier (D_5) and family (A_3) fit together consistently. Its numbers $240, 248$ are carrier traces, not inputs.
- **Consistency test:** unimodularity ($\det 1$) is a hard condition; it closes only because the discriminant-form norms $5/4$ and $3/4$ (both already present in TFPT) add up to 2. E_8 thus tests the internal coherence of the two atoms.
- **Not an end in itself:** E_8 does not explain "everything"; it is the smallest even unimodular hull of the datum $D_5 \oplus A_3$. The physics sits in the *two inputs*, not in E_8.

New — the secondary atlas. Beyond that, E_8 is an *audit container*: each maximal subalgebra projects a TFPT module. Document 3 (`tfpt_3_e8_audit_bootstrap`, Part I) shows the seven slices of 248 explicitly and labelled. The strongest new hit is that $E_6 \times A_2$ reads the flavor matrix:

$$248 = \|R\|_F^2 + \det R + 2\,(\mathbf{1}^\top R\,a)\,N_{\text{fam}} = \underbrace{78}_{\dim E_6} + \underbrace{8}_{\dim A_2} + 2 \cdot 27 \cdot 3,$$

with $\|R\|_F^2 = 78 = \dim E_6$, $\det R = 8 = \dim A_2 = h(D_5)$. The discipline rule is: *every load-bearing number must appear in at least one E_8 projection* — this makes the structure falsifiable rather than numerological. (Audit level; all identities machine-verified.)

9 What this means for the theory-of-everything question

The state of the five questions — closed, reduced or typed

1. **Flavor residual — ratios closed; only the absolute scale is the wall-selection.** Mehta–Seshadri supplies the equivalence-class framework (stable parabolic $\leftrightarrow$ unitary representation); the algebraic selector R, the Q spectrum and the splitting type are fixed (now *derived*: $\det R = 8$ lattice, $\mathrm{Spec}(Q_+)$ from D_4, `v69/v70`; since `v136/v139` the dual pair (d, n) forces R *columnwise*, d is pure anchor data, and $\mathrm{Spec}(Q_+) = (1, 2, 3)$ is the A_3-exponent grading on the seam curve's cohomology, `v137`), and the charged-lepton amplitudes are closed. The quark mass *ratios* ($c_u/c_d = \frac{55}{117}$ etc.) are then *constant* on this discrete selector stratum (Readout Rigidity, `v49/v71`) — integer Plücker readouts, no transcendental solve. Only the *absolute* amplitude scale reduces to the selected *polystable* point on the parabolic stability wall (U_{point}, an anchor; Research Contract 1). [**E**]/[**O**]
2. **Gravity — exact in the R^2 regime.** Spectral action $\to R{+}R^2$, BRST/Hodge kills the gauge directions, FRW reduction $\to$ Starobinsky; the only remainder is the *controlled* low-curvature truncation ($R^{n\geq 3} \to R^2$, valid for curvature $\ll M^2$). [**E**] in the regime
3. **Strong CP — decoupled.** $\theta_{\mathrm{eff}}{=}0$ follows from three structural facts (γ_5-Hermiticity, polar structure, sheet involution) + reflection positivity — *without* a mass gap. [**E**] (given RP)
4. **Full dynamics — the admissible transfer model is closed.** The mass gap is the *explicit* discrete gap $6 \log \frac{3}{2}$ of the C_6 transfer operator (eigenvalues $(k/3)^6$), which supports the OS reconstruction path on the admissible sector; the continuum and metric-coupled theory still require the stated RP, cluster and limit assumptions. [**E**] (transfer model) / [**C**] (continuum)
5. **Axioms P1/P2 — P2 machine-verified.** The Lean proof builds cleanly (`1886 jobs`, 0 `sorry`, only kernel axioms); the P2 algebra is a theorem, P1 a typed interface. They remain declared inputs. [**E**]/[**O**]

In other words: the theory has shrunk from "many open audit points" to "*standard steps* (cluster expansion, Fredholm, truncation bound) + two declared axioms". The five questions are now *closed, reduced or typed* rather than open-ended — a qualitative jump in discipline, not a claim of completion. *(Details and proof chains: document 2, `tfpt_2_standard_model`, Part III.)*

The closure ledger: what is done, and what genuinely remains

A complete theory must close the discrete structure, the dimensionless constants, the mass spectrum, gravity, and the quantum measure. The current state, graded honestly:

TOE piece	**Status**	**If not complete: what closes it**
gauge group, N_{fam}, hypercharge, b_1	**[E]**	— complete
$E_8 = (D_5 \oplus A_3) + \mu_4$, group closure	**[E]**	— complete
bootstrap $(g,\mu)=(5,4)$, "8" in c_3	**[E]**	— complete (reverse glue $\mu^2-5\mu+4=0$)
fine structure α^{-1}	**[E]**	— **existence+uniqueness proved**; value is a numerical fixed point (ledger `EM.FP.01`), Ward origin **[C]**
fermion masses (φ_0^{ret}-ladder)	**[E]**/**[O]**	leptons exact; quark *ratios* closed (Plücker, v49/v71); absolute scale = anchor
mixings $\theta_{13}, \theta_{23}, \theta_{12}$	**[E]**/**[C]**	θ_{12} cond. derived; θ_{23} maximal
flavour matrix / H2	**[E]**	dictionary *exact* (hexagon = twisted family spectrum, addresses pinned, margins forced, v118–v122); R forced columnwise by (d,n) (v136/v139); pairing values = atom identities (v142/v145); residue = *one* lift map
seam quantum clock (R1)	**[E]**/**[C]**	typed closed: reduced edge-sector clock (v124–v133); external VW edge = 2× reduced budget, full 4d det = firewall (v138); det-ratio cancellation derived in-family (v144)
inflation A_s, n_s, r, scalaron M	**[C]**	scalaron $M = c_3^{7/2}\bar{M}_{\text{Pl}}$ exact; n_s, r are bands over $N_\star \in [50, 60]$ (reheating point 51.4, v86)
genuinely open — the actual TOE remainder		
dim. EW/QCD masses	**[C]**	standard RG threshold matching (no new physics)
QFT measure (admissible sector)	**[E]**	**closed: RP from cusp spectrum, OS reconstruction**
ambient QFT + full gravity (metric sector)	**[O]**	*not constructed here* (frontier doc); conditionally reduces to one named hypothesis (ambient RP) on the relative spectral-action measure
analytic boundary origin of π ($c_3 = \frac{1}{8\pi}$)	**[C]**	**hardened:** $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K) = 1/(8\pi)$ **(Gauss–Bonnet)**; "8" = rank E_8; π stays the one primitive
η_B, m_p/m_e, Koide, dark matter	**[O]**	frontier handles only (see document 4)

Marker key: ***[E]*** *exact / machine-proven ·* ***[C]*** *conditional (holds under named hypotheses) ·* ***[O]*** *open / axiom ·* ***[X]*** *falsifiable kill test. Per-claim fine type in* `verification/status_ledger.csv`.

The honest bottom line: the discrete core, the dimensionless constants, the *mass-hierarchy exponents* and the mixings are *closed*; the nine masses follow from one φ_0^{ret}-ladder (lepton rationals exact, quark $O(1)$ digits still a conditional finite evaluation). The *admissible-sector* QFT measure is closed (a finite gapped transfer system, RP from the cusp spectrum, OS-reconstructed); the *ambient/metric-sector* measure — i.e. *full quantum gravity* — is *not* constructed here and stays **[O]** (frontier doc), reducing conditionally to a single named hypothesis (ambient reflection positivity). The seam seed is *hardened*: $c_3 = 1/(\lvert\mathbb{Z}_2\rvert \oint_{S^2} K\, dA) = 1/(8\pi)$ is the one-sided Gauss–Bonnet normalisation of the seam sphere, with the discrete "8" equal to rank E_8 and only the Gauss–Bonnet π left as the single irreducible primitive. What remains for a full TOE is then the ambient-RP hypothesis (full QG), plus the standard-RG dimensionful masses and the honest frontier items. No load-bearing discrete parameter remains free, and π is the lone continuous primitive.

10 Does this make the theory more probable?

Plausibility balance. *Pro:* (a) two inputs instead of many; (b) α^{-1} to 2.9×10^{-10} (CODATA-2022, 1.9σ) as a *fixed point*, not a fit; (c) E_8 constructed, not postulated; (d) one compiler generates several sectors ⇒ fewer degrees of freedom, more predictive power.

Contra/open: what remains are only *standard steps* (cluster expansion in the thermodynamic limit, the Fredholm property, the truncation bound) and the two declared axioms P1/P2; on the SM side the dimensionful EW/QCD masses $(m_W, m_Z, m_H, \sin^2\theta_W, \alpha_s)$ sit on the RG scheme layer. The solar angle, previously the one open SM number, is now *conditionally derived* ($\frac{1}{3} - \frac{\varphi_0^{\text{ret}}}{2}$, modulo the seam-misalignment lemma). A final verdict "it maps reality" requires this short list plus the near-term experimental tests (below).

Net: the parsimony (Kolmogorov-like compression: many numbers from little) and the high precision of individual closures raise the a-posteriori plausibility substantially — without hiding the open points.

11 Predictions for running and upcoming experiments

Because *one* generator now sits behind the numbers, the theory makes a set of concrete, falsifiable statements that running or near-term experiments are actively testing. Each row gives the TFPT value, the *new* derivation behind it, the experiment, and an honest status.

Observable	**TFPT value & derivation**	**Experiment (running / upcoming)**	**St.**
solar angle $\sin^2\theta_{12}$	prediction of record $\frac{1}{3}-\frac{\varphi_0^{\rm ret}}{2}=0.306747$; TBM + seam misalignment $\varepsilon=\frac{3}{4}\varphi_0^{\rm ret}=c_3+36c_3^4$ (cond.)	**JUNO**: first 59.1 d run 0.3092±0.0087 (compatible), precision phase pending	[E]/[C]
reactor angle $\sin^2\theta_{13}$	$\varphi_0^{\rm ret}e^{-5/6}=0.0231$; seed × carrier-trace $e^{-\gamma}$, $\gamma=\frac{5}{6}$	Daya Bay / RENO / JUNO (PDG 0.0220)	[E]
atmospheric $\sin^2\theta_{23}$	$\approx\frac{1}{2}$ ($\mu\tau$-symmetric limit; octant not selected)	NOvA / T2K / **DUNE** (octant ambiguity; NuFIT 6.0 pre-JUNO baseline)	[C]
fine structure α^{-1}	137.0359992 as the unique root of $F_{U(1)}(\alpha)=0$ (no fit, no second branch)	CODATA-2022 (data through 2022) 137.035999177(21): dev. 2.9×10^{-10} (1.9σ of its uncertainty)	[E]
tensor ratio r	$r=\frac{12}{N_\star^2}\in[0.0033,0.0048]$ (frozen band, $N_\star\in[50,60]$); scalaron-reheating point 0.0045 ($N_\star$=51.4, v86)	now r<0.036 (BICEP/Keck BK18); **CMB-S4** ($\sigma_r\le5\times10^{-4}$, obs. ∼2033)	[C]
tilt n_s	$n_s=1-\frac{2}{N_\star}\in[0.960,0.967]$ (same band); scalaron-reheating point 0.9611 ($N_\star$=51.4, v86)	Planck (0.9649±0.0042); CMB-S4 sharpens	[C]
amplitude A_s	$\frac{N_\star^2}{24\pi^2}c_3^7$; 2.0×10^{-9} at $N_\star$=55 (the measured A_s prefers fast reheating, v86)	Planck (2.1×10^{-9})	[C]
neutron EDM	$\theta_{\rm eff}=0\Rightarrow d_n$ essentially null (structure + RP)	**PSI nEDM** / SNS (running)	[E]
CP phase $\delta_{\rm CKM}$	$\delta=\frac{\pi}{3}+3\lambda^2=66.96°$ (frozen of record, v88); survives at +0.98σ vs $\gamma_{\rm PDG}$	**LHCb** / Belle II γ combination; decision at $\sigma_\gamma\le0.96°$	[E]
cosmic birefringence β	$\beta=\frac{\varphi_0^{\rm ret}}{4\pi}=0.2424°$ (seam readout)	ACT DR6 hint 0.215°±0.074° (systematics not yet understood — not a validation target); **Simons Obs.** / LiteBIRD	[C]
$0\nu\beta\beta$ / ordering	Majorana branch; normal-ordering preference	**LEGEND / nEXO**, DUNE, KATRIN	[C]
flavor invariants	$\det R=h(D_5)=8$, principal minors $(2,3,5)$, $\chi_R=t^3-9t^2+10t-8$ (exact)	any future CKM/PMNS global fit must satisfy these	[E]
H_0 vs. Λ	$v_{\rm EW}\sim e^{-\alpha^{-1}/5}$, $\Lambda\sim e^{-2\alpha^{-1}}$, $H_0\sim\sqrt{\Lambda}$: one engine	SH0ES / **DESI** / Planck (Hubble tension)	[C]

*Marker key: **[E]** exact / machine-proven · **[C]** conditional (holds under named hypotheses) · **[O]** open / axiom · **[X]** falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.*

The two decisive near-term tests

(1) JUNO and θ_{12}. JUNO is *taking data since August 2025* and targets sub-percent $\sin^2\theta_{12}$. TFPT commits to the single prediction of record $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.306747$ (misalignment $\varepsilon = \frac{3}{4}\varphi_0^{\rm ret} = c_3 + 36c_3^4$). The first 59.1-day JUNO run reports $\sin^2\theta_{12} = 0.3092 \pm 0.0087$ — *compatible*, but the precision phase is still pending; this is a first compatibility check, not yet the decisive test. A converged central value away from ~ 0.307 at high significance falsifies the seam mechanism.

(2) r. The R^2 scalaron with M fixed by c_3^7 predicts a small $r = 12/N_\star^2$: the frozen band $r \in [0.0033, 0.0048]$ over $N_\star \in [50, 60]$, with the scalaron-reheating point $r = 0.0045$ ($N_\star$=51.4, v86) — already below the current bound $r < 0.036$ (BICEP/Keck BK18) and within reach of CMB-S4 ($\sigma_r \leq 5 \times 10^{-4}$, first observations ∼2033–34). A detection of $r \gtrsim 0.01$ would be incompatible with the R^2 branch on which $M_{\rm Pl}$ and A_s rest.

Structural (non-experimental) checks. Independent of any apparatus, the machine-checkable E_8 appendix (all 240 roots, Gram matrices, discriminant groups) lets anyone verify that the two TFPT atoms really generate E_8; and the discipline rule "*every load-bearing number appears in at least one E_8 projection*" (atlas note) turns the number stock into a falsifiable raster rather than numerology. The numerology charge is additionally *quantified* by an explicit null model: under the declared formula grammar the alignment density is anomalously high — a random theory of equal formula complexity essentially does not reproduce the scorecard. This is a *grammar-internal null-model bound*, not a Bayesian probability that TFPT is true; the explicit figure and its controls are stated in the verification appendix [E] [`verification/v100_numerology_null_mc.py`].

Freeze file: committed kill criteria

To prevent post-hoc rescue, the predictions above are frozen with explicit kill criteria in the machine-readable registry `verification/freeze_file.csv`. The decisive entries:

- **solar** $\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} \approx 0.3067$ — a JUNO central value clearly away from 0.307 kills the seam-misalignment mechanism;
- **tensor** $r \approx 0.0040$ — any robust $r \gtrsim 0.01$ kills the R^2 branch carrying $M_{\rm Pl}, A_s$;
- **ordering** normal, small $m_{\beta\beta}$ — inverted ordering or large $m_{\beta\beta}$ kills the Majorana branch;
- **strong CP** $\theta_{\rm eff} = 0$ — a solid nEDM signal above SM background falsifies the structural cancellation;
- $w = -1$ — a robust $w \neq -1$ kills the single-engine dark-energy readout.

The proton/electron ratio is explicitly *not* claimed as a compiler power (only fails if mis-asserted as one).

Blind-prediction registry (frozen 2026-06-09, machine-enforced). Beyond the kill criteria, the *values* themselves are now frozen: `verification/predictions_frozen.json` registers every dimensionless prediction of record at 25 significant digits *before* JUNO / CMB-S4 / LiteBIRD / DESI decide, and [`verification/v84_frozen_registry.py`] re-derives each decimal from the two axioms on every suite run (formula↔value lock; covered by `manifest.sha256`; ledger `REG.FREEZE.01`). In particular the registry admits exactly *one* θ_{12} prediction of record (the seed $\frac{1}{3} - \frac{\varphi_0^{\rm ret}}{2} = 0.306747$) — the seam and non-linear variants are typed as derived variants of the same texture and can never be silently promoted afterwards. r and n_s enter only as *bands* over the declared external $N_\star \in [50, 60]$, never as point predictions.

What likely follows next.

- the quark mass *ratios* are now closed combinatorially (integer Plücker readouts on the derived selector stratum, v49/v69/v71); the only residual is the *absolute* amplitude scale, an anchor ($U_{\rm point}$) of the same nature as the one dimensionful scale — no new physics;
- further constants as carrier traces (analogous to $b_1 = 41/10$), since the compiler systematically produces traces over S^+;
- a sharper geometry↔gravity bridge once the Hodge/R^2 branch is lifted from [C] to [E].

Conclusion

TFPT's sectors form an impressive collection of closures sharing common numbers. The compiler closure explains *why* they are the same numbers: a small discrete generator $\mathbb{Z}_{30} = 2 \cdot 3 \cdot 5$ builds, from two inputs $\{c_3, g_{\text{car}}\}$ via $D_5 \oplus A_3 + \mu_4$, the E_8 container and outputs the Standard Model, the constants and the scale grammar from there. The theory thereby becomes *more parsimonious, more rigid and more checkable.* Precisely nameable gaps remain — one geometric flavor realisation, the conditional gravity/QFT closures, and two declared axioms — but the path to closure is now short and concrete.

Appendix: computational verification (reproducibility)

Every claim in this document marked [E] (exact identity), [E] (Lie/lattice theorem) or [E] (numerical fixed point) is re-derived from the two axioms $\{c_3, g_{\text{car}}\}$ alone by a small, self-contained Python suite in the repository folder `verification/`. The inline references `[verification/...]` throughout the documents point to the exact script; the table below is the master map (its source is the shared fragment `tex-artefacts/verification.tex`).

Script	What it machine-checks
`v1_e8_glue.py`	glue theorem $E_8{=}(D_5{\oplus}A_3){+}\mu_4$: disc $=\mathbb{Z}_4$, $q(D_5){+}q(A_3){=}\frac{5}{4}{+}\frac{3}{4}{=}2$, $240{=}16{\cdot}5{\cdot}3$, $248{=}240{+}8$
`v2_carrier_pascal.py`	$g_{\text{car}}{=}5$ unique Pascal solution; $16{=}1{+}5{+}10$, $N_{\text{fam}}{=}3$, rank $E_8{=}8$, $\Omega_{\text{adm}}{=}48$, $b_1{=}\frac{41}{10}$
`v3_em_alpha.py`	EM closure: unique root of $F_{U(1)}(\alpha){=}0$, $\alpha^{-1}{=}137.0359992168$, existence/uniqueness, inverse test, ablation
`v4_flavor_matrix.py`	residue matrix R: det $=8$, minors $\{2,3,5\}$, $\chi_R{=}t^3{-}9t^2{+}10t{-}8$, SNF diag$(1,1,8)$, $\lVert R\rVert_F^2{=}78$, $\sum L{=}40$
`v5_e8_cascade.py`	cascade $D_n{=}60{-}2n$: endpoints, exponent rungs $\to 240$, IR tail 46080, variance 6552
`v6_bootstrap.py`	reverse glue $\mu^2{-}5\mu{+}4{=}0$; $g_{\text{car}}{=}5$ three ways; $8{=}\,$rank $E_8{=}h(D_5){=}\varphi(30)$
`v7_gravity_cosmo.py`	scalaron exponent 7, $M_{\text{scal}}^2/\bar{M}_{\text{Pl}}^2{=}c_3^7$, A_s, n_s, r, Ω_b, $\sin^2\theta_{12}$
`v8_horizon.py`	horizon code: $\frac{1}{2\pi}{=}4c_3$, $1920{=}\lvert W(D_5)\rvert$, S_{dS}, Page, $\beta_{\text{rad}}{=}0.2424°$
`v9_neutrino_texture.py`	solar angle: explicit $\mu\tau$ Majorana texture $\to \sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\text{ret}}}{2}$, $\theta_{23}{=}45°$, $\theta_{13}{=}0$
`v10_projection_involution.py`	Q,Σ algebra: $K{=}R{+}Q\Sigma$, $L{=}R{+}Q(I{+}\Sigma)$; χ_Q, χ_K; det ladder $3{\cdot}4{\cdot}8{\cdot}20{=}1920{=}\lvert W(D_5)\rvert$; $a^\top Ka{=}41$
`v11_unique_KQ.py`	K, Q are the *unique* nonneg-int matrices with their compiler budgets, spectrum and monotone rows (enumeration)
`v12_mass_generation_polynomials.py`	sector/generation polynomials of K and the anchor-block det ladder $(9, 10, 16, 40)$
`v13_open_gates.py`	gate closures: $M{=}41{=}10b_1{=}\sum L{+}N_\Phi$; $Q_+ = A_3$ exponents, $Q_-^2{=}N_{\text{fam}}$
`v14_carrier_uniqueness.py`	$g_{\text{car}}{=}5$ unique ($N_{\text{fam}} \in \mathbb{Z}^+$, rank $E_8{=}8$); split $(3,2)$ from $b{+}s{=}5, b^2{+}s^2{=}13$; $\operatorname{Tr} Y{=}\operatorname{Tr} Y^3{=}0$
`v15_bootstrap_classification.py`	$D_5{\oplus}A_3$ unique familyful cyclic-glue of E_8 (16-spinor + 3 families), glue $\mathbb{Z}_4$
`v16_solar_dual_anchor.py`	$a^\top R^{-1}{=}a^\top L^{-1}{=}(-\frac{1}{2},-\frac{1}{2},1)$; $\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0^{\text{ret}}}{2}$; full PMNS

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Script	**What it machine-checks** ***(continued)***
`v17_hexagonal_resolvent.py`	finite resolvent $D_y^{-1}=(\sum y^{5-m}\delta^m U_6^m)/(y^6-\delta^6)$ (quark c backbone)
`v18_quark_yukawa.py`	quark source ratios $\frac{m_u}{m_d}=0.470085$, $\frac{m_c}{m_s}=13.61$, $\frac{m_t}{m_b}=40.80$ exact; lepton $c=(\frac{16}{7},\frac{4}{3},\frac{7}{6})$; full hierarchy
`v19_monodromy_moduli.py`	exact pole $z_\star=\frac{794-7\sqrt{9961}}{2187}$ (3^7); D_4 $SU(3)$ monodromy constructible but D_4-fixed locus positive-dim $\Rightarrow$ exact quark c reduce to (U)
`v20_lepton_c_derivation.py`	lepton c's derived: $\delta=\frac{1}{2}$ resolvent $c=\lvert\mu_4\rvert^w/(\frac{5}{4}-\cos\frac{r\pi}{3})$ $\Rightarrow$ $\frac{16}{7},\frac{4}{3}$; product $\frac{2^{g_{\mathrm{car}}}}{N_{\mathrm{fam}}^2}=\frac{32}{9}$ $\Rightarrow$ $\frac{7}{6}$
`v21_solar_product_quark.py`	solar coeff $=q(A_3)=\frac{3}{4}$ (glue-norm, $q(D_5)+q(A_3)=2$), $\sin^2\theta_{12}=\frac{1}{3}-\frac{\varphi_0^{\mathrm{ret}}}{2}$; product $\frac{2^{g_{\mathrm{car}}}}{N_{\mathrm{fam}}^2}$; quark law fails ($\frac{1}{7}$ vs 0.724)
`v22_open_gates_audit.py`	residual-gates audit contract: exact reduction of each open gate (A2,B3,B4,B5,B6,C7) machine-pinned (forced part vs. named residual)
`v23_anchor_generator.py`	anchor $a=(1,1,2)$: $e_k=(4,5,2)$, $c_3=\frac{1}{2e_1\pi}$; $p_n=2+2^n\Rightarrow 240, 248$, binary ladder; inputs $\to\{a,\pi\}$
`v24_quark_ratio_closure.py`	quark ratios $\frac{55}{117},\frac{34}{47},\frac{3}{26}$ from TFPT blocks; mass ratios match $<0.03\%$; rational c gauge, Λ_q in $O(1)$
`v25_frontier_conjectures.py`	conjectures: Koide $Q_{\mathrm{src}}+\frac{\varphi_0^{\mathrm{ret}}}{24}\approx\frac{2}{3}$; axion $f_a=M_{\mathrm{scal}}/128\to m_a\approx 23.8\,\mu\mathrm{eV}$
`v26_flavor_frontier_unification.py`	the "11" is not uniquely forced ($\Rightarrow$ [C]); flavor frontier unifies to one (U) gate; cusp config on the parabolic stability wall
`v27_wall_representative.py`	explicit balanced wall rep W_{wall} (cols = perm$(0,1,2)$, rows $w=(2,1,1)$, pardeg 0); selector det $R=8$, Spec$(Q_+)=\{1,2,3\}$
`v28_gravity_fR.py`	$R+R^2$ closed: $f(R)$, $M=c_3^{7/2}\bar{M}_{\mathrm{Pl}}$ scalaron, $N_\star=57\to n_s,r,A_s$; full metric measure open; $248c_3^2<6\log\frac{3}{2}$
`v29_research_contract_certs.py`	research-contract certificates: $C_U^{(1)}$ wall enumeration (1296→144→5 orbits; symmetry alone does not pick W_{wall}) and $G5$ gap-dominance $2\Vert V\Vert<\Delta$
`v30_d4_character_variety.py`	$C_U^{(2)}$: full D_4-fixed $SU(3)$ character variety is positive-dimensional $\Rightarrow$ selector needs the $R(\rho)$ dictionary (H2; exact since v118–v122, residue = the R4′ selector establishment)
`v31_R_dictionary.py`	$R(\rho)$ characterized: case A alive (irreducible $\Rightarrow$ mixing); R integer but ρ-invariants continuous $\Rightarrow$ $R(\rho)$ is the transcendental non-abelian-Hodge map (residual = Hitchin solve)
`v32_rh_splitting.py`	RH route: $\sum_k U^kA_0U^{-k}=4\,\mathrm{diag}(A_0)$ exact $\Rightarrow$ splitting $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2\Leftrightarrow\mathrm{diag}(A_0)=(\frac{1}{2},\frac{1}{4},\frac{1}{4})$ (algebraic); $\prod M_k=I$ + R-bridge stay open (c_u/c_d not forced)
`v33_explicit_flat_bundle.py`	explicit valid (U_{wall}) flat bundle (RH solve): cusp + $\mathcal{O}(-2)\oplus\mathcal{O}(-1)^2$ + $\Vert M_\infty-I\Vert\sim10^{-9}$ ($\prod M_k=I$) + irreducible (case A)
`v34_h2_bridge_attempt.py`	H2 bridge: explicit per-puncture M_k (cusp, $\prod M_k=I$); honest negative — natural extraction $\neq$ lepton amplitudes, the $\Gamma^{\min}$ dictionary is the remaining input (c_u/c_d not obtained)
`v35_quark_qcd_boundary.py`	the open gates are the discrete↔continuous boundary: quark cross-ratio cleanness tracks generation/QCD sensitivity (compiler hands off to continuous physics)
`v36_spectral_action_g2.py`	G2: spectral action $\to R+R^2$ ($a_2\sim-R/3$, $a_4\big\vert_{R^2}\sim R^2/72$); $M^2/\bar{M}_{\mathrm{Pl}}^2=6(4\pi)^2/f_0$, c_3^7 fixes f_0; gap-decoupling $\Delta_{\mathrm{eff}}=1.648>0\Rightarrow$ admissible IR closed under RP, ambient (G6) = strict-TOE completion target
`v37_plucker_anchor.py`	anchor-plane Plücker apparatus: $\Vert\mathrm{Pl}(K)\Vert_1=11$, $\Vert\mathrm{Pl}_R(K)\Vert_1=26$, $\sum\mathrm{Pl}_R(L)=60$; pencil $\det(K+xQ)=3x^3+7x^2+6x+4$; $K_e-K_u=a$, $L_e-L_u=\mathrm{Exp}(A_3)$; lepton ring $c_ec_\tau=\lvert\mathbb{Z}_2\rvert c_\mu$; Koide $u\to\frac{53}{54}u$ ([C])

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Script	**What it machine-checks *(continued)***
`v38_uwall_killswitch.py`	($U_{\rm wall}$) U2 kill-switch hardened: D_4-cusp rep irreducible at all 400 sampled points (commutant=1), mixing ≥ 0.35, reducible locus measure-zero $\Rightarrow$ case A generic, ($U_{\rm wall}$) survives; amplitude dictionary still research-level
`v39_uwall_selectors.py`	($U_{\rm wall}$) selector side closed on the explicit point: splitting type = anchor $a=(1,1,2)$, pardeg $=0$; $\det R=8=n\cdot a$, $\mathrm{Spec}(Q_+)=\{1,2,3\}=3\alpha+1$ read off the bundle; only the amplitudes c_u/c_d ($\Gamma^{\min}$ Hitchin) remain
`v40_harmonic_metric.py`	($U_{\rm wall}$) harmonic metric = FINITE linear algebra (polystable$\Rightarrow$unitary$\Rightarrow\Phi=0$): unique invariant form $H>0$, unitarised $\widetilde{M}_k$ cusp-class with $\lvert\widetilde{M}_k\rvert\in\{0,\frac{1}{2},\frac{1}{\sqrt{2}}\}$; the feared Hitchin PDE collapses to unitarisation + combinatorial R
`v41_leg_assignment.py`	($U_{\rm wall}$) final leg test: amplitudes = $\delta=\frac{1}{2}$ resolvent (closed); $\Lambda_\mu>1$ rules out holonomy moduli; harmonic holonomy $\lvert\mathrm{diag}\rvert=(0,\frac{1}{2},\frac{1}{2})$ supplies $\delta=\frac{1}{2}$ (suggestive); honest negative on literal c_u/c_d reproduction
`v42_exterior_leg.py`	($U_{\rm wall}$) Exterior Leg Lemma: u,d share $(r,w)=(1,1)$ so any scalar leg gives $c_u/c_d\equiv1$; the u/d split is the Λ^2F area $\mathrm{Pl}_{\mathbf{1},a}(K)$, $\lVert\cdot\rVert_1=11$ generated $\Rightarrow \frac{5\cdot11}{9\cdot13}=\frac{55}{117}$ [E]; phys. identification [C]
`v43_exterior_bridge.py`	($U_{\rm wall}$) Λ^2F bridge: $\Lambda^2(\widetilde{M})=\widetilde{\widetilde{M}}$ (exterior leg $=\bar{\mathbf{3}}$) confirmed; continuous action $\neq$ integer $\mathrm{Pl}(K)$ $\Rightarrow$ bridge is the discrete H2 invariant ($\det R=8$, $\mathrm{Spec}(Q_+)$ confirmed), not continuous
`v44_carrier_exterior.py`	($U_{\rm wall}$) Lie-level grounding: $16=\Lambda^{\rm even}(5)=1+10+5$; $u^c\in\Lambda^2(5)$, $d^c\in\Lambda^4(5)$ $\Rightarrow$ exterior degrees $(2,4)$, sum$=6=\lvert R^+(A_3)\rvert$, diff$=2=\lvert\mathbb{Z}_2\rvert$; the exterior leg is the carrier's own Λ^2 grading (no special math)
`v45_family_exterior.py`	($U_{\rm wall}$) the "11" is Pascal: $\lvert\mathrm{Pl}(K)\rvert=(1,4,6)=\{\Lambda^0,\Lambda^1,\Lambda^2\}(\mu_4)$; $\lVert\mathrm{Pl}(K)\rVert_1=11=\sum_{k\leq2}\binom{4}{k}=16-g_{\rm car}$ (cumulative ext of μ_4 up to $\deg u^c=2$); $15=\dim\mathfrak{su}(4)=10+5=N_{\rm fam}g_{\rm car}$
`v46_grand_mass_volume.py`	absolute mass-scaling is [E], not a product rule: sector dets $(\varphi_0^{\rm ret})^6,(\varphi_0^{\rm ret})^9,(\varphi_0^{\rm ret})^{10}$ (exponents $\lvert R^+A_3\rvert, N_{\rm fam}^2,\mathcal{A}_\Lambda$) $\Rightarrow$ $\det M_{\rm SM}\sim(\varphi_0^{\rm ret})^{25}=(\varphi_0^{\rm ret})^{g_{\rm car}^2}$; Q-rows $(4,5,6)$ sum $15=\dim A_3$; $25+15=40=\sum L$
`v47_selection_theorem.py`	Boundary Carrier Selection (Thm A): $\mu_4\to A_3, N_{\rm fam}=3$; 16-spinor$\to D_5$; $q(D_5)+q(A_3)=2$, glue $4=\lvert\mu_4\rvert$; only $D_5\oplus A_3$ meets all four conditions ($D_8,E_7+A_1,E_6+A_2,A_4+A_4,A_8$ excluded) [E]/[C]
`v48_em_ward.py`	EM boundary Ward (Thm C): $F_{U(1)}=$ Maxwell α^3 $-$ Calderón $2c_3^3\alpha^2$ $-$ transport $8b_1c_3^6\log\frac{1}{\varphi_{\rm seam}}$; $8b_1=\frac{4}{5}M=\frac{164}{5}$, $-\frac{5}{4}=q(D_5)$ [E]; Ward origin [C]
`v49_readout_rigidity.py`	Readout Rigidity (Thm U2): $c_u/c_d=\frac{55}{117}$ is constant on the discrete selector stratum $\mathcal{S}_{8,Q}$ $\Rightarrow$ no point-uniqueness needed; $(U_{\rm wall})=U_{\rm unitary}+U_{\rm H2}+U_{\Lambda^2}+U_{\rm point}$, only $U_{\rm point}$ open
`v50_q_geometry.py`	Q geometry (Thm Q): $Q_+=$ diag$(3,2,1)$-block $\chi=(t-1)(t-2)(t-3)$, Q_- $\chi=t(t^2-3)$; Q unique under rows$(4,5,6)$/cols$(9,5,1)$/spectra; geom. origin [C]
`v51_boundary_half_step.py`	glue norms $=(g_{\rm car},N_{\rm fam})/\lvert\mu_4\rvert$: $q(D_5)=\frac{5}{4}$, $q(A_3)=\frac{3}{4}$; four ops forced $\to$ {sum$=2$ root norm, diff$=\frac{1}{2}=\lvert\mathbb{Z}_2\rvert/\lvert\mu_4\rvert=\delta$, prod$=\frac{15}{16}$, ratio$=\frac{5}{3}$}. So $\delta=\frac{1}{2}$ is the carrier$-$family count over the glue index, not chosen [E]
`v52_pencil_endpoints.py`	$K+xQ$ pencil endpoints: $P(-1)=2=\lvert\mathbb{Z}_2\rvert$, $P(0)=4=\lvert\mu_4\rvert$, $P(1)=20=\det L$, $P(2)=68=2p_5(a)$; $\det(K-Q)=2$, $\mathrm{tr}=N_{\rm fam}$ (audit ext. of v37) [E]
`v53_compiler_core.py`	Big picture: the whole integer skeleton from $(g_{\rm car},N_{\rm fam})=(5,3)$ — rank $E_8=8$, $\lvert\mathbb{Z}_2\rvert=2$, Pythagorean $\Delta_Y=g_{\rm car}^2=25=N_{\rm fam}^2+\lvert\mathbb{Z}_2\rvert\cdot$rank $E_8=9+16=\dim S^+$ (diff. of squares); anchor char-poly $(t-1)^2(t-2)$ unique [E]
`v54_seam_horizon_keystones.py`	The "8" in $c_3=\frac{1}{8\pi}$ overdetermined and gravitationally aligned ($2\lvert\mu_4\rvert=$ rank $E_8=h(D_5)=\varphi(30)=8\pi$ Hawking/Einstein); one boundary transport operator (spec $\{1,(2/3)^6,(1/3)^6\}$, $\lambda_2=(2/3)^6$) governs *both* SM flavor gap and horizon Page recovery [E]

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Script	What it machine-checks *(continued)*
`v55_coxeter_cycle.py`	Computed E_8 Coxeter element: order exactly $30{=}\lvert\mathbb{Z}_2\rvert N_{\rm fam}g_{\rm car}$, eigenphases = exponents = totatives of 30, so rank $E_8{=}8{=}\varphi(30){=}$ live phases of the order-30 cycle; one α^{-1} sets EM, S_{dS} and Λ ($S_{dS}\rho_\Lambda{=}32\pi^4$) **[E]**
`v56_unique_attractor.py`	Gapped boundary transport (gap $6\log\frac{3}{2}{>}0$) $\Rightarrow$ *unique* attracting fixed point, geometric rate $(2/3)^6$ (two starts $\to$ same direction); Coxeter in $\lvert\mu_4\rvert{=}4$ planes; entropy reset = rank-1 projector **[E]**
`v57_horizon_crosslinks.py`	Horizon cross-links: $c_3{=}\frac{1}{8\pi}$ is the Einstein/Jacobson coefficient ($1/4{=}1/\lvert\mu_4\rvert$ entropy density); Hod QNM $\omega_R/T_H{=}\ln 3{=}\ln N_{\rm fam}$; Hawking power denom $1920{=}\lvert W(D_5)\rvert$; one α^{-1} for EM/S_{dS}/Λ/Hubble **[E]** (+ **[C]** identification)
`v58_seam_horizon_chain.py`	Seam-horizon chain (precise: seam = abstract normaliser, locally realised as a horizon): one-sided S^2 Gauss–Bonnet $c_3{=}1/(2{\cdot}4\pi){=}\frac{1}{8\pi}$; KMS $4c_3\kappa$; $S_{BH}{=}M^2/2c_3$; RT in seam units. Seam–Horizon Theorem (area-law from the seam kernel) **[O]** open
`v59_area_law_evidence.py`	Area-law necessary condition met: a reflection-positive Gaussian (Calderón-type) kernel yields an area law (gapped block-EE saturates; gapless $\sim\frac{c}{3}\log L$), by the *same* gap that fixes the attractor; $8{=}\lvert\mathbb{Z}_2\rvert\lvert\mu_4\rvert$. The c_3 coefficient stays **[O]** open
`v60_lambda_metrology_branch.py`	Λ branch selection: $(8\pi)^2\delta_{\rm top}{=}\frac{3}{4\pi^2}$, mis-scale $2c_3/\delta_{\rm top}{=}\frac{64\pi^3}{3}{=}661.47 \Rightarrow G_N$ pinned as output; 123 orders = 119.028+3.920; ACT DR6 birefringence $0.215°{\pm}0.074°$ vs $0.2424°$ (0.37σ) **[E]**
`v61_cft_bridge.py`	Boundary-CFT mirror: level-1 WZW central charges $c(E_8){=}8$, $c(D_5){=}5$, $c(A_3){=}3$ = (rank $E_8, g_{\rm car}, N_{\rm fam}$); conformal embedding $c_{\rm coset}{=}0$; $SU(3){\leftrightarrow}SU(4)$ reconciliation $N_{\rm fam}{=}\lvert\mu_4\rvert{-}1$, $4{\to}3{+}1$ (one geometry $\mathbb{P}^1\backslash\mu_4$, two sides) **[E]**
`v62_data_scorecard.py`	TFPT vs 2024/25 data: α^{-1}, $\sin^2\theta_{12}$, $\beta_{\rm rad}$ match; $\sin^2\theta_{13}$ (2.0σ) and Starobinsky n_s vs DESI ($\sim 2\sigma$) mild tensions; θ_{23} open; $r{\approx}0.004$ future falsifier **[E]**
`v63_seam_engineering_index.py`	Exact Seam-Engineering Index $\Xi{=}2\lVert V\rVert/\Delta{=}\frac{31}{24\pi^2\log(3/2)}{\approx}0.323$ ($2\lVert V\rVert{=}\frac{31}{4\pi^2}$, $248{=}\dim E_8$, $31{=}1{+}h^\vee$), $\Delta_{\rm eff}{\approx}1.648$; four-class possibility taxonomy **[E]**/**[O]**
`v64_causal_boundary_nogo.py`	Causal Boundary Engineering *conditional no-go*: gapped transfer has no periodic orbit (discrete no-CTC); RP+OS+gap $\Rightarrow$ ANEC $\Rightarrow$ Topological Censorship $\Rightarrow$ no traversable shortcut; only ER=EPR bridge survives **[E]** (core)/**[C]** (chain)
`v65_falsification_layer.py`	Prediction layer with explicit failure thresholds: 3 core ($v_{\rm GW}{=}c$, no 4th gen, seam=8), 4 observational (β, r, n_s, θ_{12}), no-go + unitarity; $N_\star$ demoted to **[C]** input; none violated **[E]**
`v66_e8_casimir_degrees.py`	Organizing simplification: the compiler atoms *are* the E_8 Casimir degrees $\{2,8,12,14,18,20,24,30\}$ ($2{=}\lvert\mathbb{Z}_2\rvert$, $8{=}$rank, $14{=}\dim G_2$, $20{=}\det L$, $24{=}\lvert W(A_3)\rvert$, $30{=}h(E_8)$); $\sum{=}128{=}2^7$ (dS prefactor), $\sum\exp{=}120{=}\lvert R^+(E_8)\rvert$ **[E]**
`v67_area_law_coefficient.py`	Central-theorem *structural closure*: Fursaev–Solodukhin $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow S{=}2\pi c_3A{=}\frac{1}{4}A$ ($2\pi c_3{=}\frac{1}{4}$); $c_3{=}\frac{1}{8\pi}$ is the *unique* value with replica-coeff = BH $\frac{1}{4}$. Reduces to the one coeff $k{=}c_3/2$ (then *forced*, v73) **[E]**
`v68_seeley_dewitt_residual.py`	Residual *resolved as an anchor*: $a_2{=}{-}\frac{d}{192\pi^2}R \Rightarrow$ the *absolute* $\frac{1}{16\pi G}$ is UV-sensitive ($\propto d\Lambda^2 f_2$), so the *scale* $1/G$ is the one dimensionful anchor (the dimensionless $k{=}c_3/2$ is forced, v73); cutoff-indep. R^2/scalaron derived **[E]**
`v69_d4_q_geometry.py`	D_4-equivariant Q-geometry (gate **[C]**/**[E]**): on $H_1(\mathbb{P}^1\backslash\mu_4){=}B_1{\oplus}E$, $Q_+{=}3w{+}1{=}\mathrm{diag}(1,2,3)$ (cusp weights on $\mu_4{=}\mathbb{Z}_4$ eigenspaces), $Q_-{=}E$-coupling $\sqrt{N_{\rm fam}} \Rightarrow t(t^2{-}3)$; Σ-split $=D_4{=}\mathbb{Z}_4{\rtimes}\mathbb{Z}_2$ **[E]**
`v70_q_integer_lift.py`	Q integer-lift sharpened: $\mathbb{Z}_4$ puncture action R unimodular (eigenvalues $\{-1,i,-i\}$); $\det Q{=}3{=}N_{\rm fam}$, SNF $\mathrm{diag}(1,1,3) \Rightarrow$ residual = one lattice invariant $\det Q{=}N_{\rm fam}$, not closable by D_4 alone **[E]**
`v71_simple_r_bridge.py`	*Simple* R-bridge (gate #3): selector stratum $\mathcal{S}_{8,Q}$ now fully derived ($\det R{=}8{=}$rank E_8 lattice + $\mathrm{Spec}(Q_+){=}\{1,2,3\}$ from v69) $\Rightarrow$ quark *ratios* are pure integer Plücker readouts ($\frac{55}{117}$, no monodromy, v49 rigidity); residual $U_{\rm point}{=}$absolute norm. = an anchor **[E]**/**[O]**

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Script	What it machine-checks *(continued)*
`v72_q_det_from_cusp.py`	$\det Q{=}N_{\text{fam}}$ derived from the cusp class: $\det Q{=}\lvert\operatorname{coker} Q\rvert{=}$ cusp-class order = denominator of the weights $\{0, \frac{1}{3}, \frac{2}{3}\}$ (same data as Spec Q_+); $\operatorname{coker} Q{=}\mathbb{Z}/N_{\text{fam}}{=}$ triality deck group ⇒ integer-lift residual closed **[E]**
`v73_k_c3_half.py`	Central coefficient $k{=}c_3/2$ *forced*: $(1/2)$ variational $\times$ $1/(\lvert\mathbb{Z}_2\rvert 2\pi\chi(S^2))$ Gauss–Bonnet topology ($\chi(S^2){=}2$), both cutoff-indep ⇒ Fursaev–Solodukhin $S{=}A/4$; only the absolute 4D Newton scale stays the anchor (v68) **[E]**
`v74_compiler_micro_lemmas.py`	Spine quotient ladder $(\frac{2}{3}, \frac{3}{4}, \frac{4}{5}, \frac{5}{4}, \frac{4}{3}$ = adjacent quotients of $(2, 3, 4, 5))$; pencil differences $(2, 16, 48){=}(\lvert\mathbb{Z}_2\rvert, \dim S^+, \Omega_{\text{adm}})$ sheet→1gen→3gens; anchor QF $a^\top K a{-}\mathbf{1}^\top K\mathbf{1}{=}41{-}25{=}16{=}\dim S^+$ (EM budget = mass vol + one gen); solar dual anchor $L{=}R{+}61e_1^\top$, $a^\top R^{-1}\mathbf{1}{=}0$ (Sherman–Morrison rank-one invariance) **[E]**
`v75_upoint_to_vgeo.py`	**Gate 1 complete:** $U_{\text{point}}{\to}v_{\text{geo}}$. Ratios closed (v49/v71) + lepton c exact (v20) + sector products = φ_0^{ret}-powers (v46); (ratios, product)⇔(individuals) bijection ⇒ absolute amplitudes fixed up to *one* scale v_{geo} = the same anchor as $1/G$ (v68); two $[A]$ → one **[E]**/**[O]**
`v76_gmetric_reduction.py`	**Gate 2:** Tier A (IR) closed under RP+gap (Decoupling Thm: $\Delta_{\text{eff}}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{=}1.648{>}0$, $\lVert V\rVert{\le}248c_3^2{=}\frac{31}{8\pi^2}$); Tier B (strict TOE) holographically reduced from bulk to a finite seam-*boundary* measure (conditional: RP+tightness ⇒ closes) **[E]**/**[C]**
`v77_e8_conformal_net.py`	**G6 route:** the seam boundary measure = the $(E_8)_1$ *lattice net* (rigorous, holomorphic $c{=}8$); level-1 $c(E_8){=}\frac{248}{31}{=}8$, $c(D_5){=}5$, $c(A_3){=}3$, embedding 5+3=8 (coset $c{=}0$); imports $G6$ into RCFT/conformal-net rigor **[E]**/**[C]**
`v78_vgeo_floor.py`	v_{geo} **floor:** no pure number is dimensionful ⇒ v_{geo} irreducible by logic; all scales are ratios × one $\bar{M}_{\text{Pl}}$; $S_{dS}\rho_\Lambda{=}32\pi^4$ pins it from one measurement; theory at the minimum (one scale + π) **[E]**/**[O]**
`v79_review_identities.py`	Four validated review identities + one firewall: hypercharge Lucas–Binet $D_n{=}(3^n{-}({-}2)^n)/5{=}1, 1, 7, 13, 55, 133$ (roots = carrier charges); Inverse Anchor Thm ($\mathbf{1}^\top M^{-1}\mathbf{1}{=}1/\text{atom}$, $a^\top M^{-1}a{=}1$ for R, K, L); MacWilliams $(1, 10, 5)$; $6Y^2{-}Y{-}1{=}(2Y{-}1)(3Y{+}1)$. Rejects m_p/m_e, η_B near-formulas (firewall) **[E]**
`v80_operator_pencil_geometry.py`	Operator-pencil geometry: Anchor Singularity Thm $\det B_{K+xQ}{=}(N_{\text{fam}}x{+}\lvert\mathbb{Z}_2\rvert)(N_{\text{fam}}x{+}g_{\text{car}}){=}(3x{+}2)(3x{+}5)$ ($\frac{2}{3}$ Koide/gap = anchor-plane singularity); buildup chain $(3, 10, 16, 20)$; type checker $\det B_{Q,K,R,L}{=}(9, 10, 16, 40)$; audit: differential spectrum, $F_4{\times}G_2$ shadow $248{=}52{+}14{+}26{\cdot}7$ **[E]**/**[C]**
`v81_singular_pencil_matrices.py`	Double cover $y^2{=}\det B_{K+xQ}{=}(3x{+}2)(3x{+}5)$: Koide $x{=}{-}\frac{2}{3}$ and carrier $x{=}{-}\frac{5}{3}$ are the two branch points (deck $2{=}\lvert\mathbb{Z}_2\rvert$; disc${=}81{=}N_{\text{fam}}^4$; separation=1 transport period). Clearing matrices $C_{2/3}{=}3K{-}2Q$ ($D_5{\oplus}A_3$ glue, $B{=}(5, 7)^\top(9, 11)$, $\sum{=}240$), $C_{5/3}{=}3K{-}5Q$ (neutral, $\chi{=}(\lambda{+}2)(\lambda^2{+}\lambda{+}6)$); scalaron 7= mixed anchor disc$/N_{\text{fam}}$ **[E]**
`v82_koide_attractor_splitting.py`	Koide attractor *forced*: the branch-divisor-preserving Möbius map fixing $q{=}2, 5$ is unique, multiplier $(2/3)^6{=}\lambda_2$ of the established transfer operator ⇒ $F'(2){<}1$ attractor, $F'(5){>}1$ repeller; three postulates → one **[C]**. Splitting trichotomy disc 81/49/40: the clean split is non-generic (anchor-first) **[E]**/**[C]**
`v83_e8net_holomorphic_uniqueness.py`	Holomorphy pins $(E_8)_1$: #primaries = det Cartan (D_8:4 vs E_8:1), and a holomorphic $c{=}8$ chiral CFT is the lattice theory of the *unique* even unimodular rank-8 lattice (Minkowski–Siegel mass $1/\lvert W(E_8)\rvert$); carrier Pascal truncation $K{=}(g{-}1)/2$ = half-spinor split **[E]**
`v84_frozen_registry.py`	Blind-prediction registry frozen 2026-06-09 (25 digits, machine-enforced lock formula↔value): exactly *one* θ_{12} prediction of record; r, n_s as $N_\star$ *bands*; conditional entries carry their hypothesis by name **[E]**
`v85_master_cover.py`	Master-cover theorem: $\det B$ is GL(2)-covariant on span$\{K, Q\}$ ⇒ exactly *one* double cover up to Möbius (disc = $N_{\text{fam}}^4\det(G)^2$); the disc-81 family is its orbit (${-}\frac{8}{3}{=}{-}\text{rank}E_8/N_{\text{fam}}$ = carrier − one period); μ_4 is *not* a 4:1 cover (ladder of double covers); branch trace fixes only the scalaron *scale* **[E]**

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Script	What it machine-checks *(continued)*
`v86_nstar_reheating.py`	$N_\star$ band → point [C]: $\Gamma{=}4M^3/48\pi\bar{M}^2{=}128\,\mathrm{GeV}$, $T_{\rm reh}{=}9.6{\times}10^9\,\mathrm{GeV}$, $N_\star(0.05/\mathrm{Mpc}){=}51.4 \Rightarrow n_s{=}0.9611$, $r{=}0.0045$ (inside the frozen band); honest tensions: n_s $+0.9\sigma$, A_s dichotomy (slow point $-11.4\sigma \Rightarrow$ near-instant reheating or exponent fails) [C]
`v87_bulk_uniqueness_reduction.py`	Red-team Target A residual (ii) conditionally closed: holomorphic ⇒ Rep = Vect ⇒ 2D bulk *unique* (LR/KLM/BKLR); machine contrast $SO(16)_1$ admits *six* modular invariants (incl. both E_8 pairings) ⇒ Target A = one residual [E]/[C]/[O]
`v88_cp_phase_audit.py`	Target-D CP residual quantified: frozen $\delta{=}\pi/3{+}3\lambda^2$ survives at $+0.98\sigma$ vs $\gamma_{\rm PDG}$; the $0.07°$-near alternative $\pi/3{+}2\lambda^2$ recorded as look-elsewhere trap, *not* adopted; decision at $\sigma_\gamma{\le}0.96°$ [E]
`v89_carrier_index_lemma.py`	Carrier Index Lemma: KLM $\mu_A{=}[B{:}A]^2\mu_B \Rightarrow$ Jones index $[(E_8)_1{:}(D_5)_1{\times}(A_3)_1]{=}4{=}\|\mu_4\|$ (glue order *is* the index); μ-additivity $16/4^2{=}1 \Rightarrow$ holomorphy *follows*; Gate A (H)⇔(H′) index-4 extension [E]
`v90_conical_defect_chain.py`	Fursaev–Solodukhin factor 4π *derived* (smoothed-cone Gauss–Bonnet, exactly smoothing-independent); replica $S{=}4\pi kA$, $k{=}c_3/2 \Rightarrow S{=}A/4 \Leftrightarrow c_3{=}\frac{1}{8\pi}$ (sympy-solved); residual isolated to “seam determinant ⇒ EH form” [E]/[O]
`v91_spine_tetrahedron.py`	Spine tetrahedron $\{2,3,4,5\}$ = anchor + zeroth moment; edges/faces/volume = the integer grammar, volume $120{=}\|R^+(E_8)\|$; $240{=}\|\mu_4\|\|E(K_4)\|\|E(K_5)\|$ with K_6 negative control; dual cuts typed tautological; sub-grammar incompleteness honest [E]
`v92_glue_uniqueness.py`	Glue uniqueness: of all 15 subgroups of the discriminant form exactly two Lagrangian glues (the two *chiralities*, swapped by either single spinor conjugation, fixed by the simultaneous one) and exactly one halfway extension = $SO(16)_1$; tower carrier$\to SO(16)_1 \to (E_8)_1$, nothing else [E]
`v93_koide_relaxation_toy.py`	Koide relaxation toy (P2 narrowed): basin lemma (every physical configuration in the attractor basin); exact rate $(2/3)^6$; source at $\rho{=}{-}\varphi_0^{\rm ret}/24$ to 0.8% ([C]); honest negatives: flow time $t{=}2.84 \notin \mathbb{Z} \Rightarrow$ the missing object is a *continuous* generator [E]/[C]
`v94_sheet_diamond.py`	Sheet diamond: all four flavor operators on *one* surface $M(s,t){=}R{+}Q\,\mathrm{diag}(s,t,t)$ (pencil = the cut $(1{+}x, x{-}1)$; diagonal-cut disc 153 negative control); winding line $\det B{=}2\det$ for all s; cofactor seam normal $(5,-9,6)$; Plücker canonicalisation: 11 *and* 26 both live in K; spine-quotient firewall *rejected* [E]
`v95_centered_diamond.py`	Centered cross: $Q{=}U{+}V$, $R/L{=}C{\mp}U$, $K/F{=}C{\mp}V$ (five objects → three); $\mathrm{Spec}\,V{=}\{0,1,2\}{=}N_{\rm fam}{\cdot}\mathrm{cusp}$; $\det C{=}14$, $\sum C{=}31{=}2^{g_{\rm car}}{-}1$; $\mathrm{Pl}_R(C){=}7(2,3,1) \parallel \mathrm{Pl}_R(L){=}10(2,3,1)$; G_2 reading audit-typed [E]
`v96_branch_kernel_selection.py`	Branch-kernel selection (P1 unblocked): rank-1 anchor blocks at the branch points have *integer* kernels (carrier kernel = $\mathbf{1}$); collapse lemma ⇒ $(-1,1,0)$: up = −down, *lepton pairing* = 0 (leptons on the ramification); sector zero ladder (up $-\frac{3}{2}$ doubly, down 0 & $-\frac{9}{5}$); anchor-placement controls [E]
`v97_sheet_conjugation_bridge.py`	Sheet-conjugation bridge (P1 → one [C]): the anchor forces the cusp conjugation uniquely ⇒ integer T_A, $\det{=}{-}1$; $a{=}e_2{+}e_3$ (conjugation-symmetric); one deck action on $\mathbb{R}^3/\,\mathrm{span}\{\mathbf{1},a\}$; $\langle T_A,\Sigma\rangle{=}D_4$; sheet-index lemma (intertwiner dets even, min $2{=}\|\mathbb{Z}_2\|$); remaining [C]: Q_+ grading = A_3 discriminant grading [E]/[C]
`v98_discriminant_dictionary.py`	Dictionary *derived* (P1 closed modulo GATE.QGEO): $G{=}T_A\Sigma$ acts integrally on the cusp basis ($Ge_1{=}{-}e_1$, $Ge_3{=}e_2$, $Ge_2{=}{-}e_3$ = $B_1{\oplus}E$); cusp-0 line = self-conjugate $\mathbb{Z}_4$-class-2 line; $T_AGT_A^{-1}{=}G^{-1}$ $(k{\mapsto}{-}k)$; $q_{A_3}(1){=}q_{A_3}(3){=}\frac{3}{8}$, $q_{A_3}(2){=}\frac{1}{2}$; reflection classes = glue-swap vs glue-fix parities ⇒ P1 carries no separate [C]; residual = the one existing Q-geometry gate [E]

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Script	What it machine-checks *(continued)*
`v99_koide_flow_time.py`	Koide flow time: canonical generator $\dot{q}=(\Delta/N_{\text{fam}})(q-2)(q-5)=(\Delta/N_{\text{fam}})\det B(q)$, time-1 map $=F$ [E]; non-integrality of t=2.84 is *data-fragile* (Q crosses $\frac{2}{3}$ at $+0.43\sigma(m_\tau)$, t passes every integer ≥ 3 within $+0.5\sigma$) [E]; conditional steps: n=2 excluded (-2.9σ), n=3=N_{fam} $\Rightarrow$ m_τ=1776.9427 MeV ($+0.14\sigma$), decision at $\sigma(m_\tau)\sim$0.01 MeV [C]; ln(46080) scale reading destroyed by controls (firewall)
`v100_numerology_null_mc.py`	Numerology null test (look-elsewhere corrected): exact census of the full declared formula grammar over the 13 scored frozen-registry observables (conservative data windows) gives joint formula-fishing probability $\prod p_i = 10^{-25.8}$, robust to budget/window variation; $2\times2\cdot10^5$ Monte-Carlo pseudo-theories reach at most 5/13 hits vs. TFPT 13/13; power controls (seed perturbation, data shuffle) collapse the scorecard; all 94 500 $F_{U(1)}$ variants solved, exactly *one* hits CODATA $\Rightarrow$ total $\leq 10^{-30.7}$, *conditional on the declared grammar* [E]
`v101_horizon_anchor.py`	The maximal black hole is the anchor (SdS in seam units): Nariai cubic $t^3-3t+2=(t-1)^2(t+2)$, roots $(1,1,-2)$ = traceless anchor; $S_{\text{Nariai}}/S_{dS}=\frac{2}{3}=\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ (per horizon $\frac{1}{3}$); interpolation $(x^2+1)/\Phi_3(x)$; three-sheet total $\lvert\mathbb{Z}_2\rvert S_{dS}$ conserved; $Q_{\text{geom}} \in [\frac{3}{8},\frac{1}{2}]$ = the $SU(4)_1$ weights; mass-line cover $(1-3m)(1+3m)$, deck = horizon swap; τ_{BH}=128 $g_{\text{car}} M^3/c_3$; six atom landings, zero free parameters [E]/[C] (bulk reading)
`v102_seam_orientation.py`	One orientation: the anchor is the *stationary repeller* in both sectors — flavor relaxation = gradient flow of a cubic potential with critical points = the branch points, curvatures $V''(2)=+\Delta$, $V''(5)=-\Delta$ (= the gap), inflection at scalaron/2, Lyapunov rate Δ constant; SdS entropy has Nariai as its *unique* stationary point, curvature $\frac{2}{9}=\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}^2$; evaporation = entropy ascent away from the anchor [E]; "one variational principle of the seam" stays a reading [C]
`v103_trisection_normal_form.py`	Trisection normal form (the canonical coordinate): SdS cubic $\Leftrightarrow$ $\cos 3\theta=-3m$ (angle trisection; $\mathbb{Z}_3$ deck = triality); $m=\cos\psi/N_{\text{fam}}$ and $S_{\text{tot}}/S_{dS}=\frac{4}{3}-\frac{2}{3}\cos\frac{2\psi}{3}$ — *one* cosine of glue atoms; canonical curvature $(\frac{2}{3})^3=(\lvert\mathbb{Z}_2\rvert/N_{\text{fam}})^{N_{\text{fam}}}$ (Koide constant to the family power); invariant slope $d\sigma/dm\rvert_N=-\frac{8}{9}=-\operatorname{rank} E_8/N_{\text{fam}}^2$; bridge: flavor rate $(2/3)^{2N_{\text{fam}}}$ vs gravity curvature $(2/3)^{N_{\text{fam}}}$, missing = one *clock* [E]/[C]
`v104_nariai_clock.py`	The *classical* Nariai clock is the anchor (third appearance): static pin $\phi=\rho$ solves the dS_2 static-patch equation with $m^2=-2\Lambda=-\lvert\mathbb{Z}_2\rvert\Lambda$ exactly (Ginsparg–Perry tower: one negative mode); in Hubble units $\chi_{\text{clock}}=(\lambda-1)(\lambda+2)$ = the anchor quadratic, eigenvalues $\{1,-2\}$; Nariai cubic = $(t-1)\cdot\chi_{\text{clock}}$; entropy-deviation rate $2H=\lvert\mathbb{Z}_2\rvert H$; honest (2/3)-test *negative* for the classical clock — the quantum clock is the one remaining [C] of the seam variational principle [E]/[C]
`v105_residual_inventory.py`	The residual inventory: *one constant* ($2/3=\lvert\mathbb{Z}_2\rvert/N_{\text{fam}}$ exact in seven places across both sectors), *one anchor* (three dynamical appearances: configuration, Nariai roots, clock spectrum), *relocation theorem* (cosmological quantum clock suppressed by the Λ hierarchy, 121.5 orders $\approx 2\alpha^{-1}/\ln 10$ $\Rightarrow$ the clock must be the seam transfer operator — one [C] identification), and the *complete machine-pinned gap list*: exactly five structural objects + one scale + π; a sixth gap would fail the script [E]/[C]
`v106_review_validation.py`	External-review validation: seed normal form $\varphi_0^{\text{ret}}=\frac{\lvert\mu_4\rvert}{N_{\text{fam}}}c_3+\Omega_{\text{adm}}c_3^{\lvert\mu_4\rvert}$ exact [E]; hypercharge moment $\operatorname{Tr} X^2$=120=5! computed from the charge table; factorial spine $5!=\lvert R^+(E_8)\rvert=\sum$exponents, 240=2·120, 1920=2^4·5!; degree-2 inventory (Pascal K=2 unique at g=5, glue norms sum 2, $c_3=\frac{1}{2}\times\frac{1}{4\pi}$, $\binom{5}{2}$=10); named [C] hypothesis *Quadratic Boundary Locality* ($\Rightarrow$ K=2 forced; would upgrade the Pascal selection to [E]); audit 240=16×15 non-unique [E]/[C]

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Script	What it machine-checks *(continued)*
`v107_quantum_clock_target.py`	Quantum-clock target made quantitative: the classical Ginsparg–Perry clock decays on $\{0,-1,-2\} = -\operatorname{Spec}(V) = -N_{\mathrm{fam}}\times$cusp (the transfer operator's own grading) [E]; exact target identity $(1/3)^6 = \big((2/3)^6\big)^{\log_{3/2} 3}$ — required bend $\log_{9/4} 3 = 1.355$ per weight step; coupling landscape $\kappa = c/(24\pi)\cdot\Lambda/\bar{M}_{\mathrm{Pl}}^2$: 10^{-122} cosmological vs $1/(3\pi)$ at seam curvature (the only viable, borderline-nonperturbative regime) [E]; identification stays [C]
`v108_pascal_ladder.py`	Pascal Ladder Theorem: $2^{g-1} = \sum_{k\le K}\binom{g}{k} \Leftrightarrow g = 2K+1$ (odd-g midpoint identity + even-g straddle) $\Rightarrow$ carrier selection $g=5$ *exactly equivalent* to degree bound $K=2$ — the Target-B residual = the one QBL hypothesis; neighbour worlds: $K=1$ one-family, $K=3$ nine-family, $K=4$ inconsistent; only $K=2$ also satisfies $g+N_{\mathrm{fam}}=8$ (v14 overdetermination) [E]/[C]
`v109_sheet_pairing.py`	Sheet-Pairing Lemma (QBL anchored): character-exact multisets from $(\pm\frac{1}{2})^5$ — no scalar within a sheet (zero-weight mult of $S^+\otimes S^+ = 0$; odd g flips parity); $S^+\otimes S^- = \Lambda^0\oplus\Lambda^2\oplus\Lambda^4$ exactly (256=1+45+210, zero-modes $(1,5,10)$ = the code); sheet-diagonal = odd forms $(2\Lambda^1+2\Lambda^3+\Lambda^5)$; tower tops at Λ^{2K}, $K=2 \Rightarrow$ the seam scalar kernel is necessarily a *sheet pairing* [E]/[C]
`v110_calderon_sheet.py`	Calderón-sheet selection theorem: a Calderón involution ε on $S^+\oplus S^-$ certifies a scalar two-point datum *iff* it is sheet-odd; sheet-odd $\Rightarrow$ exactly $2=\lvert\mathbb{Z}_2\rvert$ invariant kernels = the two Lagrangian glues; ladder genericity ($g=3,5,7$): sheet-oddness forces $K=(g-1)/2$, *not* $g=5$ (anti-overclaim); QBL residue split into *interface* ("ε sheet-odd", natural from the one-sided collar) + *analytic core* (slot-degrees ≤ 2) [E]/[C]
`v111_quadratic_transport.py`	Quadratic-Transport Theorem: in the integer Jordan–Wigner model linears are sheet-odd, the 45 quadratics ($=\mathfrak{so}(10)$, the Λ^2 term of 1+45+210) sheet-even, generate the code from the vacuum *and* span the full $\operatorname{End}(S^+)=256$ at word length 2 $\Rightarrow$ *transport degree exactly 2* (degree ≤ 1 generates nothing) — the transport half of the QBL core is a theorem; $g=3$ control: degree selected, not rank [E]/[C]
`v112_selfcounting_channel.py`	Self-Counting Channel: $w \mapsto (w,-w)$ is a bijection code states $\leftrightarrow$ neutral certified kernels, so #kernels $= 2^{g-1}$ *exactly*; graded by pair-degree the same set has sizes $\binom{g}{m} \Rightarrow$ the Pascal closure $2^{g-1} = \sum_{m\le K}\binom{g}{m}$ is *two countings of one set* (an identity, not a requirement); Hodge fold $\binom{g}{2j} = \binom{g}{\min(2j,g-2j)}$; QBL residue = one input ("a single scalar 2-point kernel"), selection carried by rank-8+integrality [E]/[C]
`v113_quasifree_kernel.py`	One kernel is the whole net: carrier $(D_5)_1\times(A_3)_1 = SO(10)_1\times SO(6)_1 = 16$ free Majoranas, $c=5+3=8$ at every tower level (only the glue grows, index $2\times2=4=\lvert\mu_4\rvert$); Wick/Pfaffian exact (all 210+210 higher correlators = Pfaffians of the one kernel); kernel = Calderón involution, rank $P=5=g_{\mathrm{car}}$, seam level $8=\operatorname{rank}E_8$ (c = rank of the kernel); vacuum unique $\Rightarrow$ QBL input = R2 premise: gate closes $\Rightarrow$ carrier choice closes [E]/[C]
`v114_torsion_delta.py`	Torsion normal form + $\delta=\frac{1}{2}$ theorem: flatness of the (U_{wall}) $\mathbb{Z}_4$-family $\Leftrightarrow$ $(MU)^4=\mathbf{1}$ (μ_4 = order of the twisted cusp generator); on the involutive branch ($T^2=\mathbf{1}$) the cusp trace forces diag $M=(0,\frac{i}{2},-\frac{i}{2})$ with spec $\{1,\omega,\omega^2\}$ automatic $\Rightarrow$ $\delta=\frac{1}{2}$ exact and constant (v20's hand value = torsion identity); census: $\{1,i,-i\}$ branch varies, v33 point in its $d_1=0$ slice; residue = bundle-side branch selection [E]/[C]
`v115_anchor_residue.py`	The anchor pins the residue: μ_4-average = diagonal projection $\Rightarrow$ anchor splitting $\Leftrightarrow$ diag $A_0=a/\lvert\mu_4\rvert$; off-weights uniquely $(8,0,5)/144$ $(=(\operatorname{rank}E_8,0,g_{\mathrm{car}})/(\lvert\mu_4\rvert N_{\mathrm{fam}})^2$, total $13=\Delta_Q)$; *exact* residue $A_0^\star$ with $\chi=\lambda(\lambda-\frac{1}{3})(\lambda-\frac{2}{3})$ is the RH solution ($\lVert M_\infty-\mathbf{1}\rVert\sim10^{-10}$); anchor forces $d_1=0 \Rightarrow$ harmonic diagonal anchor+torsion-forced [E]

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Script	What it machine-checks *(continued)*
`v116_resonance_uniqueness.py`	Resonance theorem (v115 [N]→[E], linear — no Gröbner): twisted μ_4 averages collapse to $B_1{=}4a_{12}E_{12}{+}4\overline{a_{13}}E_{31}$; anchor exponents $\mathrm{diag}(2,1,1)$ give resonance gap 1 with obstruction $4\overline{a_{13}} \Rightarrow M_\infty{=}\mathbf{1} \Leftrightarrow a_{13}{=}0$; + $(8,0,5)/144$ + Diagonaleichung ⇒ flat anchor locus = *one* gauge orbit = exact $A_0^\star$ — $U_{\rm wall}$ bundle datum algebraically unique; sharpness control $\lVert M_\infty{-}\mathbf{1}\rVert \sim 8.5\lvert a_{13}\rvert$ [E]
`v117_monodromy_weyl_a3.py`	The monodromy is $W(A_3)$: the unitarised holonomy of $A_0^\star$ is *exact* (entries in $\frac{1}{2}\mathbb{Z}[i]$), $\mathrm{diag}\,\widetilde{M}_0{=}(0,-\frac{i}{2},\frac{i}{2}) \Rightarrow d_1{=}0$, $\delta{=}\frac{1}{2}$ are matrix entries; $\langle U,\widetilde{M}_0\rangle$ enumerated exactly: order 24, statistics $(1,9,8,6)$, characters $(3,-1,0,1)$ = $S_4{=}W(A_3)$ (standard⊗sign) — the flavor wall realises the A_3 glue factor as its Weyl group; ODE match $3.5{\times}10^{-10}$ [E]
`v118_hexagon_family_dictionary.py`	First exact H2 piece: $\mathrm{spec}(\widetilde{M}_0)\cup\mathrm{spec}(-\widetilde{M}_0) = \mu_6$ (hexagon = family spectrum + sheet twist, $\mathbb{Z}_6{=}\mathbb{Z}_2{\times}\mathbb{Z}_3$); $\det(\mathbf{1}{-}t\widetilde{M}_0){=}1{-}t^3 \Rightarrow \det_\mp{=}(\frac{7}{8},\frac{9}{8})$; lepton coefficients = resolvent determinants ($c_e{=}\lvert\mu_4\rvert/(2\det_-)$, $c_\mu{=}N_{\rm fam}/(2\det_+)$, $c_\tau{=}\lvert\mu_4\rvert\det_-/N_{\rm fam}$, product = $\lvert\mu_4\rvert/\det_+{=}\frac{32}{9}$); $\langle U,\widetilde{M}_0,-\mathbf{1}\rangle$ has order $48{=}\Omega_{\rm adm}$; the address table stays open [E]/[C]
`v119_review_validation_2.py`	Second review validation: anchor ratio triad $(e_3,e_1,e_2)/p_0{=}(\frac{2}{3},\frac{4}{3},\frac{5}{3})$ = (Koide branch, seed gain, carrier branch; flow critical points $(2,5){=}(e_3,e_2)$); the 121 audit lemma $(\mathbf{1}{+}a)^T R(\mathbf{1}{+}a){=}121{=}11^2{=}\lVert\,\mathrm{Pl}(K)\rVert_1^2$ (orderings $\{105,121,135\}$, only canonical = 121; audit, not load-bearing); micro-identities $\Omega_{\rm adm}{=}2p_1p_2$, $10b_1{=}1{+}p_1p_3$, $\Delta_Y{=}e_2^2{=}p_0^2{+}2\,\mathrm{rank}$; flavor surface = v94/v95 (presentational) [E]
`v120_address_table.py`	Address table (frozen v18/v20 integers): lepton words $(8,5,3){=}(\mathrm{rank}\,E_8,g_{\rm car},N_{\rm fam})$; addresses = divmod by the hexagon ($p_2{=}6$): $e{\to}(2,1)$, $\mu{\to}(5,0)$, $\tau{\to}(3,0)$, $\sum r{=}p_3$; sheet parity: τ sits at the real twisted eigenvalue -1 (= the product-rule site); quark sums: up = p_3, down = $p_1{+}p_3$, all quarks = $24{=}\lvert W(A_3)\rvert$, all nine = $40{=}p_1p_3{=}a^TRa$; top at the vacuum site $(0,0)$; the assignment derivation stays open [E]/[C]
`v121_address_pinning.py`	Address pinning: $L{=}R{+}2U$ (v95) ⇒ the word table = the residue operator R + the generation-1 winding, R column 1 = the anchor a; margins = atoms (rows (e_1,rank,p_3), columns $(e_1,\Delta_Q,g_{\rm car})$); *census theorem*: among all hexagon matrices with atom margins (17 of them) exactly *one* has $\det{=}8$ — and it is R (also unique with SNF $(1,1,8)$; all 17 determinants distinct) ⇒ the address table carries *zero* residual information; the H2 residue = the six atom margins [E]/[C]
`v122_margin_theorem.py`	Margin theorem: the *established* selectors — the D_4 annihilator $n{=}(5,-9,6)$ with $n^TR{=}(8,0,0)$ (v94), $\det R{=}8$, $\det K{=}4$ (the v71 quartet) — pin R *uniquely* among hexagon matrices ($64\to12\to1$) ⇒ the atom margins, the anchor column and all sum rules are *consequences* (their input status is revoked); bonus: $\det L{=}20$ holds automatically on the locus (the quartet is redundant); H2 introduces *no* new residual class [E]/[C]
`v123_inventory_update.py`	Residual inventory updated (post v110–v122): thirteen new ledger rows machine-pinned; R2 carries *three* loads (metric + carrier + QBL: one theorem, three doors); the table re-pins to *five* classes — R1 clock, R2 index-4 net, R3 EH, R4′ selector establishment (replacing the *discharged* H2 dictionary), R5 Q realisation — + $v_{\rm geo}$, π; a sixth class fails the script [E]
`v124_resummed_clock.py`	Resummed clock (the R1 bend in closed form): transfer eigenvalues = exactly $(1{-}n/N_{\rm fam})^{p_2} \Rightarrow \mathrm{rate}(n){=}{-}p_2\ln(1{-}n/N_{\rm fam})$ reproduces $\{0,\Delta,6\ln 3\}$, the bend $\log_{3/2}3$ in one line; the linearisation carries slope $\lvert\mathbb{Z}_2\rvert{=}2$ relative to the classical GP clock $\{0,-1,-2\}$; the pole at $n{=}N_{\rm fam}$ forces the three-level spectrum; the R1 target is now closed-form [E]/[C]
`v125_glue_qsystem.py`	Glue Q-system: $\mathbb{C}[\mathbb{Z}_4]$ satisfies all Longo Q-system axioms exactly (associativity, unit, Frobenius, specialness $mm^*{=}\lvert\mathbb{Z}_4\rvert\mathbf{1}$) ⇒ Jones index $4{=}\lvert\mu_4\rvert$ (the v89 value); locality = isotropy ($q(k(1,1)){=}k^2\in\mathbb{Z}$, v92); halfway sub-Q-system $\mathbb{C}[\mathbb{Z}_2]$ = the $SO(16)_1$ step, $4{=}2{\times}2$; R2 sharpens from "find" to "identify" [E]/[C]

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Script	What it machine-checks *(continued)*
`v126_clock_wall_bridge.py`	Clock and wall share one spectrum: the resummation weights $n/N_{\mathrm{fam}}=\{0,\frac{1}{3},\frac{2}{3}\}$ = exactly spec $A_0^\star$ (the parabolic MS weights); $\lambda_n=(1-\alpha_n)^{p_2}$; checkpoint 1 passed *classically* ($p_2/N_{\mathrm{fam}}=2=\lvert\mathbb{Z}_2\rvert$ = the v104 entropy rate $2H$); $\det(\mathbf{1}-A_0^\star)=\frac{2}{9}$ = the entropy curvature (v102); the quantum content of R1 is isolated in the geometric tail [E]/[C]
`v127_ring_resummation.py`	The clock is a log determinant: rate $=-p_2\,\mathrm{Tr}\ln(\mathbf{1}-\alpha P)$ — six identical ring (RPA) towers, *one per hexagon site*, coupling = the parabolic weight; the k=1 ring = the classical term, the k=2 ring = the first quantum correction; $\kappa_{\mathrm{seam}}=\frac{1}{3\pi}=\alpha_1/\pi$ (audit); the wall = the RPA instability at $\alpha\to 1$; v107's "O(1) resummation" is thereby *typed* (ring/log-det), its derivation open [E]/[C]
`v128_graded_hull.py`	Graded hull + zero-mode multiplicity: the 240 E_8 roots explicit in $D_5\oplus A_3$+glue coordinates, coset decomposition 240=52+64+60+64; the grading is exactly additive on *all* 6720 root-pair sums ⇒ E_8 is a $\mathbb{Z}_4$*-graded* Lie algebra over the carrier (grading = glue = Q-system); ring multiplicity $6=2(2l+1)\rvert_{l=1}$ = the sheet-doubled GP zero modes [E]/[C]
`v129_entropy_power_law.py`	The clock is an entropy power law: $\lambda_n=(S_n/S_{dS})^{p_2}$ with the three Nariai entropy levels $S/S_{dS}\in\{1,\frac{2}{3},\frac{1}{3}\}$ (v101); rate $=p_2\ln(S_{dS}/S)$; triple identity $\alpha=n/N_{\mathrm{fam}}=$ MS weight = entropy-deficit share ⇒ the RPA coupling is nowhere free; orientation consistent (attractor = max entropy); R1 thermodynamic: $\Gamma\propto(S/S_{dS})^{p_2}$ (a Gibbons–Hawking power law, $h=N_{\mathrm{fam}}$ audit) [E]/[C]
`v130_born_square.py`	Born square: amplitude $\sim(S/S_{dS})^{n_{\mathrm{zero}}/2}=(S/S_{dS})^3$ (one $S^{1/2}$ per zero mode), probability $=\lvert A\rvert^2=(S/S_{dS})^6$ ⇒ $h=n_{\mathrm{zero}}/2=3=N_{\mathrm{fam}}$, $p_2=2h$ (the Born rule) — the v129 weight audit is *derived*; the 3 = the dimension of the l=1 space = the number of proper conformal Killing vectors of S^2; the ×2 = the horizon pair (the v101 deck); the R1 residue = the standard zero-mode measure [E]/[C]
`v131_measure_is_area.py`	The measure is the area: $\lVert Y_{1m}\rVert^2=r^2=A/(4\pi)$ *exactly* (all three l=1 modes integrated) ⇒ the per-mode Jacobian = the mode norm $\sim S^{1/2}$ ⇒ amplitude S^3, Born S^6 — every exponent factor with a derivation origin; the 4π = the same Gauss–Bonnet primitive as in c_3, cancelling in all ratios; the R1 residue = *one* det-ratio cancellation $\det'_n/\det'_{dS}=1+O(\text{deficit})$ [E]/[C] *(cancellation since derived within the SdS family, v144)*
`v132_detratio_anomaly.py`	Det-ratio anomaly = the Koide constant: $\zeta(0)\rvert_{\det'}$ of the GP sphere operator $-\Delta-2$ (the 3 zero modes removed) $=a_1-3=\frac{7}{3}-3=-\frac{2}{3}=-\lvert\mathbb{Z}_2\rvert/N_{\mathrm{fam}}$ exactly ($a_1=\frac{1}{N_{\mathrm{fam}}}+\lvert\mathbb{Z}_2\rvert$; numerical continuation $\sim 10^{-7}$) — the eighth 2/3 appearance, the first as a *spectral* anomaly; consequence $\det'(cL)=c^{\zeta(0)}\det'(L)$; the R1 residue = *one* $\zeta(0)$ budget statement (the 4d sum) [E]/[C]
`v133_zeta_budget.py`	The ζ budget, both routes exact (Euclidean Nariai = $S^2\times S^2$): the reduced route gives $-\frac{2}{3}$ per sector, total $-\frac{4}{3}=-e_1/p_0$ (= minus the seed gain — two of the three triad values); the naive 4d route gives $a_2=\frac{161}{45}$, $\zeta(0)=-\frac{109}{45}$ (*no* atom; continuation $\sim 10^{-8}$); zero modes 3+3 = the horizon pair ⇒ the budget structurally favours the *reduced* seam reading; remainder = graviton/ghost shares (Volkov–Wipf class) [E]/[C]
`v134_dual_anchor.py`	Dual anchor lemma: $d{:=}a^\top R^{-1}=a^\top L^{-1}=(-\frac{1}{2},-\frac{1}{2},1)$, $d{\cdot}\mathbf{1}=0$, $d{\cdot}a=1$, $(1,1,-2)=-2d$ — the inverse flavor response *is* the Nariai root (v101); the invariance is exact via Sherman–Morrison (⇔ orthogonality to $R^{-1}\mathbf{1}=\frac{1}{4}(1,1,-1)$; controls: $\mathbf{1},e_1,n$ do *not* lie on the plane); (d,n) = the dual normal pair of the flavor boundary; a third purely *algebraic* leg of the flavor↔horizon bridge [E]/[C]
`v135_det_surface.py`	Determinant surface: $\det M(s,t)=3s(t+1)(t+2)+t^2+5t+8$ — two s-independent iso-volume walls $t=-2\Rightarrow 2=\lvert\mathbb{Z}_2\rvert$, $t=-1\Rightarrow 4=\lvert\mu_4\rvert$ ($\det K=4$ is a *layer*); winding line $6s+8\Rightarrow(8,14,20)$, $\det F=32=2^{g_{\mathrm{car}}}$; $\det B(s,0)=2\det M(s,0)$ exactly, B-wall only at $t=-2$; row-sum axes $C=(7,11,13)$, $U=(3,3,3)$, $V=(1,2,3)=\mathrm{Spec}(Q_+)$; micro-identities $e_1^2+e_2^2=41=10b_1$, $p_1^2+p_2^2=52=$ the carrier coset root count (v128) [E]/[C]

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Script	**What it machine-checks** *(continued)*
`v136_dual_normal_selector.py`	Dual-normal selector: (d,n) pins R *columnwise* — $d{\cdot}c_j{=}a_j$, $n{\cdot}c_j{=}8\delta_{1j}$, kernel = primitive $(6,8,7)$, census: each column unique (1 of 216) ⇒ R4′ shrinks from 6^9 to three column problems; the $A_0^\star$ zero mode carries $\sigma{=}(2,-9,5){=}(\lvert\mathbb{Z}_2\rvert,-N_{\rm fam}^2,g_{\rm car})$ ($\lVert v\rVert^2{=}\frac{16}{9}$, $16{=}\dim S^+$; adj $=\frac{2}{9}P_v$ = the SdS curvature); $n{\cdot}\sigma{=}121{=}11^2$; $W(A_3)$ orbit control: the naive lift $\sigma{\to}n$ fails — a genuine mechanism, not a group element [E]/[C]
`v137_qplus_cohomology.py`	The Q_+ grading is cohomology: $H^1(\mathbb{P}\setminus\mu_4){=}\chi_1{\oplus}\chi_2{\oplus}\chi_3$ ($\omega_k{=}z^{k-1}dz/(z^4{-}1)$, character i^k, residues $\zeta^k/4$); degrees $(1,2,3){=}\mathrm{Spec}(Q_+){=}$ the A_3 exponents ($h{=}4{=}\lvert\mu_4\rvert$); cusp weights $\{0,\frac{1}{3},\frac{2}{3}\}{=}\mathrm{spec}(A_0^\star)$ ⇒ MS weights = cusp classes = characters = Q_+: *one* spectrum, four readouts; the QGEO residue = the naturality of the canonical map [E]/[C]
`v138_vw_firewall.py`	Volkov–Wipf firewall (R1 typed closed): external 4d value $\alpha_{\rm PI}{=}{-}\frac{98}{45}{=}\frac{22}{45}{-}\frac{8}{3}$ (arXiv:2506.02142 Tab. 4.1 = VW hep-th/0003081); **edge match**: $\alpha_{\rm edge}{=}{-}\frac{8}{3}{=}2{\times}({-}\frac{4}{3})$ — two identical edge copies (the horizon pair), per copy exactly the reduced seam budget (v133, the seed gain); bulk $\frac{22}{45}$ = the graviton gas (not the clock); difference $-\frac{38}{45}$ no atom ⇒ typing: the seam clock = an edge object, the full 4d determinant = the gravity firewall [E]/[C]
`v139_selector_triangle.py`	Selector triangle: $d{=}\frac{3}{2}a{-}2\cdot\mathbf{1}$ exactly (pure anchor data, no R) ⇒ the first selector is *derived*; the frame $(\mathbf{1},a,\sigma)$ has $\det{=}11{=}\lVert\,\mathrm{Pl}(K)\rVert_1$, and n is the *unique* covector with atom pairings $(2,8,121){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8,11^2)$; on the constrained line the pairing is $11(11{+}t)$, a square only at the true n; review lift exact (audit): $n{=}\mathrm{reverse}(\sigma){+}\lvert\mu_4\rvert e_3$; R4′ residue = three pairings [E]/[C] *(since v142: two — the third pairing is integrality-forced)*
`v140_canonical_map.py`	The canonical map exists and is rigid: the plain deck U (characters $(0,1,3)$) cannot match H^1; the *sheet-twisted* $-U$ ($(2,3,1)$) and the cusp exponential i^{Q_+} ($(1,2,3)$) both carry the H^1 character set $\{1,2,3\}$ (the v118 sign twist); multiplicity-freeness ⇒ Schur-rigid equivariant isomorphism (unique up to scalars); i^{Q_+} pulls the Euler degree back to Q_+, $-U$ to $Q_+\circ\rho$ — the QGEO residue collapses to *one* $\mathbb{Z}_3$ choice [E]/[C] *(the choice since derived, v141)*
`v141_deck_selection.py`	Deck Selection Theorem (R5): the established integer deck $G{=}T_A\Sigma$ (v97/v98) pairs the $Q_+{=}1$ line with the self-conjugate character 2 ⇒ of the three $\mathbb{Z}_3$ rotations only the sheet-twisted $(2,3,1)$ = chars$(-U)$ survives; i^{Q_+} excluded (same spectrum, wrong grading pairing); residual freedom = the established sheet $\mathbb{Z}_2$ (v92); Euler degree = $Q_+\circ\rho$ [E]/[C]
`v142_frame_integrality.py`	Frame Integrality (R4′): pairing triples of *integer* covectors against $(\mathbf{1},a,\sigma)$ form an index-11 sublattice ($x{-}3y{+}z\equiv 0 \bmod 11$, $11{=}\lVert\mathrm{Pl}(K)\rVert_1$); with $(x,y){=}(\lvert\mathbb{Z}_2\rvert,\mathrm{rank}\,E_8)$ the third pairing is forced $\equiv 0 \bmod 11$, primitivity forces even t; Cramer reductions for the two line pairings; R4′: three pairings → two [E]/[O] *(since v145: the values are atom identities — residue = one lift map)*
`v143_graded_frobenius.py`	Graded Frobenius (R2 at the finite Lie level): coset duality $-C_k{=}C_{-k}$ exact on all 240 roots ⇒ Killing pairing nondegenerate per $g_k\times g_{-k}$; glue average = carrier projector (index $\lvert\mathbb{Z}_4\rvert{=}4$); each sector one $W(D_5){\times}W(A_3)$ orbit = simple current; fusion = $\mathbb{C}[\mathbb{Z}_4]$ (v125); residual = the one conformal-net statement [E]/[C]
`v144_detratio_family.py`	Det-ratio cancellation within the SdS family (R1, the v131 step): e_2-rigidity gives $r_br_c{=}1{-}\Delta^2/3$ exactly ⇒ det′-ratio = $(1{-}\Delta^2/3)^{4/3}$ (via $\zeta(0){=}{-}\frac{2}{3}$, v132) — no first-order term in the split; deficit coefficient $\frac{32}{27}{=}\lvert\mu_4\rvert(\frac{2}{3})^3$ (audit); finite-weight absorption stays open with the $6\to\frac{14}{3}$ obstruction stated sharply [E]/[C] *(the bend since shown det′-clean, v147)*
`v145_pairing_atoms.py`	R4′ values are *atom identities*: with the lift $n{=}w_0(\sigma){+}\lvert\mu_4\rvert e_3$ ($w_0{=}$ the A_3 exponent duality), $n\cdot\mathbf{1}{=}\lvert\mu_4\rvert{-}\lvert\mathbb{Z}_2\rvert{=}\lvert\mathbb{Z}_2\rvert$, $n\cdot a{=}\lvert\mu_4\rvert a_3{=}8$ via the new closure $\lvert\mu_4\rvert{+}g_{\rm car}{=}N_{\rm fam}^2$ ($4{+}5{=}9$, $\sigma\perp w_0(a)$), $n\cdot\sigma{=}\lVert\sigma\rVert^2{-}N_{\rm fam}^2{+}\lvert\mu_4\rvert g_{\rm car}{=}121$; residue = derive the *one* lift map [E]/[C]

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Script	**What it machine-checks *(continued)***
`v146_moebius_d4.py`	Möbius D_4 realisation (R5 at its floor): the full dihedral $\langle z\mapsto iz,\ z\mapsto 1/z\rangle$ on $H^1(\mathbb{P}^1\backslash\mu_4)$ matches the integer model $\langle G, T_A, \Sigma\rangle$ exactly — ι^* swaps $\chi_1\leftrightarrow\chi_3$, fixes χ_2 with $+1$, det $=-1$ ($=T_A$ in the cusp basis); $\delta\iota=\Sigma$ class; parities match (v98) — the `GATE.QGEO` premise reduces to the module identification [E]/[C]
`v147_clock_gaussian.py`	The clock is a Gaussian zero-mode integral (R1): variance ratio $(1-\alpha)$ (norm = area, v131) $\Rightarrow$ Born^2 Γ-ratio $(1-\alpha)^{p_2}$, rate $=-p_2\ln(1-\alpha)=$ the v127 ring sum term by term; the bend $\log_{3/2}3$ lives between the two Nariai levels of *one* geometry $\Rightarrow$ *det′-clean*; the $6\to\frac{14}{3}$ obstruction is localised to the α=0 reference [E]/[C]
`v148_fock_census.py`	NS/R sector census (R2 scoped, corrected): the untwisted (NS) module of the 16 Majoranas carries only the *even* classes $\{0,2\}\times\{0,2\}$ — the odd glue sectors $k(1,1)$ are *Ramond* modules; the R zero-mode module (2^8=256) splits 128+128 into the odd sectors of the *two* Lagrangian glues (v92): choosing the glue *is* choosing the R-projection (128= one $SO(16)$ chirality), $248=120_{\mathrm{NS}}+128_{\mathrm{R}}$; R2 is irreducibly a twisted-sector net statement [E]/[C]
`v149_cusp_normal.py`	Both dual normals are *anchor data* (R4′): in the cusp frame n pairs to $(6,3,5)=(p_2,p_0,e_2)(a)$ — one anchor invariant per Q_+ line ($d=\frac{3}{2}a-2\cdot\mathbf{1}$ in the generation frame, v139); equivalent to the $(2,8,121)$ frame data and the w_0-lift (v145); audit: $\sigma\to(5,1,2)$, sum $14=\dim G_2$; residue = *one* assignment [E]/[C]
`v150_replica_eh_model.py`	The EH form from the replica variation, at the gapped-model level (first R3 movement): conical deficit exact and t-independent (image sum $(N^2-1)/(12N)=C(2\pi/N)$); the gap makes the ζ-variation *finite and cutoff-independent*, $\Delta\log\det'=2C(\gamma)\ln m$; linearisation $=(\ln m/12\pi)\int\sqrt{g}R$ — the EH form; target equation k=$c_3/2\Leftrightarrow\ln m=\frac{3}{4}=q(A_3)$ (audit); Calderón transfer + normalisation stay open — `SEAM.THEOREM.01` narrows, does *not* close [E]/[O] *(Calderón transfer since answered — the DtN kernel is conically clean, v151)*
`v151_bfk_split.py`	BFK split (the Calderón transfer of v150 answered): the Kac corner constant is boundary-condition *independent* (C_N=C_D, derived from reflection doubling) $\Rightarrow$ the conical deficit of the Calderón/DtN jump determinant *vanishes identically*, C_{DtN}=$C_{\mathrm{cone}}(\gamma)-2C_D(\gamma/2)$=0; via Burghelea–Friedlander–Kappeler the seam-reduced action inherits the v150 EH form verbatim — the EH content sits in the *local* half determinants, the nonlocal kernel is conically clean; residue = the $q(A_3)$ normalisation + the kernel premise [E]/[O]
`v152_norm_is_anchor.py`	R3's normalisation *is* the dimensionful anchor: the EH coefficient is $k=\ln(m/\mu)/(12\pi)$ — $\ln m$ alone is scale-ambiguous, only the ratio m/μ is physical; k=$c_3/2\Leftrightarrow m/\mu$=$e^{3/4}$ is the induced-gravity anchor $1/G$ (v68), *not* a separate gap; the irreducibles stay {one scale, π}; audit: the log-ratio $=\frac{3}{4}=q(A_3)$ is the target value, NOT a derivation [E]/[O]
`v153_no_unit_theorem.py`	No-Unit Theorem (v78/v152 formalised): dimensionless data are scale-invariant under $L\mapsto\lambda L$, a mass / $1/G$ is not, so a dimensionless compiler *cannot* select an absolute scale; the three residuals collapse — $U_{\mathrm{point}}\sim v_{\mathrm{geo}}$, $1/G\sim v_{\mathrm{geo}}^2$, $m/\mu=e^{3/4}$ (a ratio); v_{geo} is an irreducible metrology primitive, not an open gap; irreducibles stay {one unit, π} [E]/[O]
`v154_simple_current_theorem.py`	Simple-Current Extension Theorem (the central G_{net} theorem, assembled): A=$(D_5)_1\otimes(A_3)_1$ extended by the isotropic glue L=$\langle(1,1)\rangle$ has index $\lvert L\rvert$=4, c=5+3=8, $\mu(B)$=$\mu(A)/\lvert L\rvert^2$=16/16=1 $\Rightarrow$ holomorphic $\Rightarrow B$=$(E_8)_1$ (unique even-unimodular rank-8; $SO(16)_1$ excluded); finite realisation in hand (v125/v128/v143/v148); residue = the seam-side identification [E]/[C]
`v155_quasifree_boundary.py`	Quasi-free in $\Rightarrow$ quasi-free out (Python-only): the boundary marginal of a Gaussian bulk is the compression $P\Gamma P$ (a contraction, hence a valid covariance), so the G_{net} analytic premise (v110(b)/v113) *follows* from 'the seam bulk is a reflection-positive free field' — the *same* premise as v59 (area law) and v150/v151 (EH); the seam-side identification reduces to ONE shared physical premise [E]/[O]

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Script	What it machine-checks *(continued)*
`v156_seam_net_construction.py`	Constructive route (Route 3) to its exact endpoint: the DtN/Calderón operator of the 2d Laplacian is $\omega_k{=}\lvert k\rvert$ (free chiral dispersion); 16 free Majoranas give $c{=}8{=}5{+}3$; the E_8 currents are $248{=}120_{\mathrm{NS}}{+}128_{\mathrm{R}}$; and the $(E_8)_1$ character $E_4/\eta^8 = q^{-1/3}(1{+}248q{+}4124q^2{+}\dots) = j^{1/3}$ (verified to order 4) makes $B \cong (E_8)_1$ a *checked* identity, not an assertion; residual = the free-bulk premise (v155) + net existence [E]/[C]
`v157_rigid_fixed_point.py`	Freeness as a *rigid fixed point* (simplicity-first reframing of premise A): (i) the DtN/Calderón symbol is universally $\lvert k\rvert$ (Lee–Uhlmann, bulk-detail-independent — the free dispersion is not a premise); (ii) a holomorphic $c{=}8$ theory has *no* marginal $(1,1)$ field (all $\bar{h}{=}0$) $\Rightarrow$ rigid, isolated, no room for an interaction; (iii) the same $\lvert\mathbb{Z}_2\rvert$ that halves 8π in c_3 IS the Ramond projection giving $\mu{=}1$; (iv) holomorphic $c{=}8$ unique = $(E_8)_1$ = free. Premise (A) shrinks from 'prove Gaussianity' to 'the gapped flow reaches the holomorphic fixed point' [E]/[C]
`v158_fixed_point_stable.py`	The free chiral $c{=}8$ fixed point is *stable* (exact dimension count, the finite core of (A)): Majorana $h{=}\frac{1}{2}$, bilinear current $h{=}1$, quartic $h{=}2$; the relevant window $(0,1)$ for bosonic operators is *empty*, the $h{=}1$ currents are chiral (not $(1,1)$-marginal, v157), the lowest interaction (quartic) is $h{=}2{>}1$ irrelevant $\Rightarrow$ isolated stable fixed point; (A) reduces to a basin-of-attraction statement [E]/[C]
`v159_pyrate_gauge_crosscheck.py`	PyR@TE cross-check of the F_{transfer} gauge inputs: the carrier/SM content gives $(b_1,b_2,b_3) = (\frac{41}{10},-\frac{19}{6},-7)$ by the textbook one-loop formula; $b_1{=}\frac{41}{10}$ holds *three* independent ways — carrier algebra $10b_1{=}g_{\mathrm{car}}2^{g_{\mathrm{car}}-2}{+}1{=}41$, the $U(1)$ hypercharge index $\sum \frac{3}{5}Y^2$, and the external RGE generator PyR@TE 3 (`arXiv:2007.12700`) which writes $\beta_{g_1}{=}(\frac{41}{10})g_1^3$ verbatim; the $M{=}41$ split $10b_1{=}40_{(\mathrm{ferm})}{+}1_{(\mathrm{Higgs})}$ reproduces v13 Gate 1; evidence in `verification/pyrate`. The transfer functor's gauge inputs are not free knobs [E]/[C]
`v160_seam_gaussianity_from_pf.py`	Seam Gaussianity as a *conditional* fixed-point theorem (the proposed closure of premise A, honestly typed): (1) quasi-free $\Rightarrow$ all higher connected cumulants $\kappa_{2n}{=}0$ (signed set-partition/Pfaffian inversion, exact on a pure Majorana covariance); (2) Wick functoriality $\Rightarrow$ a kernel-preserving transport fixes the whole quasi-free state (v113); (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a SECOND fixed ray, breaking uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0 — the value $(2/3)^6$ only sets the rate. **The load** [O]: v56's gapped PF uniqueness lives on the 3-dim scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the seam correlation cone has $\binom{16}{4}{=}1820$, $\binom{16}{6}{=}8008$ cumulant directions $\Rightarrow$ (A) reduces to the single internal statement 'the admissible TFPT transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum' [E]/[O]
`v161_one_particle_reduction.py`	The cone reduces to one particle (discharging v160's scope premise via Bogoliubov second quantisation): a quasi-free transport is $T{=}\Gamma(t)$, the second quantisation of a one-particle contraction t; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$, so (1) $\mathrm{spec}(\Gamma(t)\vert_{\deg m})$ = products of m one-particle eigenvalues [E]; (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly [E]; (3) the eigenvalue-1 sector = wedges of the fixed subspace (disconnected Wick powers), no independent fixed higher cumulant, uniqueness closed by v113 (one polarization $\Leftrightarrow$ one state) [E]; (4) quasi-freeness *forced* not assumed — the $120{=}\dim\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant [C]. **Residual** [O]: v160's three full-cone premises collapse to ONE 16-dim one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open [E]/[O]

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Script	What it machine-checks *(continued)*
`v162_seam_transport_identification.py`	The final identification, forced as far as it goes — the v161 one-particle statement carries *no* new open content: (1) the gap $(2/3)^6$ is *forced* — cusp weights $=\mathrm{spec}(A_0^\star)=\{0,\frac{1}{3},\frac{2}{3}\}$ (parabolic residue, v115), transfer eigenvalue $(1-\alpha)^{p_2}$, $p_2{=}6{=}\lvert R_+(A_3)\rvert$, all carrier data **[E]**; (2) μ_4-equivariance forces the generator diagonal in the cusp basis — $\sum_k U^k X U^{-k}=\lvert\mu_4\rvert\,\mathrm{diag}(X)$, commutant of U = the diagonal algebra — so the gapped μ_4-equivariant generator has spectrum = the cusp weights, gap $(2/3)^6$ **[E]**; (3) the cusp grading *is* the A_3 grading on the 16 Majoranas (v137) **[E]**; (4) the fixed space = the rank-8 Calderón polarization (v113/v156) **[E]**. **Closure** [C]/**[O]**: with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the *already-open* A2 (net existence) and R1 (seam clock, `HOR.CLOCK`) — so (A) factors exactly into A2 + R1, adds *no* new open item, and is closed conditionally on them (a unification, not an unconditional proof) **[E]**/**[O]**
`v163_rg_stability_flavor.py`	F_{transfer} scale-tagging: the lepton-mixing seeds ($\sin^2\theta_{12}{=}\frac{1}{3}{-}\frac{\varphi_0}{2}{=}0.3067$, $\sin^2\theta_{13}{=}\varphi_0 e^{-5/6}{=}0.0231$) are *RG-stable* — PMNS running is y_τ^2-suppressed ($\sim 1.8{\times}10^{-5}\ll$ precision), so seam value = measured value; the quark mass-ratio readouts ($m_t/m_b{\sim}41$) run through $y_t^2\sim O(10$–$15\%)$ and carry a scale tag. The only mixing-rotating Yukawa term $\frac{3}{2}YY^\dagger Y$ carries $y^2_{\max}$ (PyR@TE SM RGEs) **[E]**/[C]
`v164_qcd_scale.py`	the QCD confinement scale is parameter-free from the carrier $b_3{=}{-}7$: running $\alpha_s(M_Z)$ down (two-loop, n_f thresholds) gives $\alpha_s(m_b){\sim}0.22$, $\alpha_s(m_c){\sim}0.38$ and confinement ($\alpha_s{\sim}1$) at $\sim$0.5 GeV — so the QCD-scale *numerator* of m_p/m_e is independently sourced (dimensional transmutation), while the ratio 1836 stays the honest 'Open / not forced' frontier non-claim ($m_p{=}O(1)\Lambda$ is lattice) **[E]**/**[O]**
`v165_premise_a_closure.py`	Premise (A), the honest maximal closure — the two residual gates pinned to their floor, so (A) carries *zero* independent open content: (1) A2 (net existence) closes by *assembly* of established mathematics [C]— 16 free Majoranas give a free-fermion net (Buchholz–Mack–Todorov / Böckenhauer–Evans), the holomorphic $c{=}8$ lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B){=}16/16{=}1$, $248{=}120{+}128$, character $E_4/\eta^8{=}j^{1/3}$ (v156); the only TFPT input (seam Calderón kernel = free-fermion two-point function) is done via DtN $=\lvert k\rvert$ (v156/v157); (2) R1 (seam clock) reduces to `GATE.QGEO` [C]— μ_4-equivariance forces diagonal-in-cusp, the flat μ_4-equivariant generator is *unique* $=A_0^\star$ (v115/v116) so the gap $(2/3)^6$ is forced, residual = 'seam boundary = the μ_4-punctured sphere $\mathbb{P}^1\backslash\mu_4$' = `GATE.QGEO` (cohomology established, v137); (3) the irreducible core stays $\{\pi, v_{\mathrm{geo}}\}$, a *theorem* by the No-Unit Theorem (v153). **Final accounting**: (A) = A2 assembly [C] + R1=`GATE.QGEO` + the $\{\pi, v_{\mathrm{geo}}\}$ core $\Rightarrow$ ZERO independent open gates; a unification + re-typing, not an unconditional proof **[E]**/**[O]**
`v166_higgs_free_seam.py`	the Higgs quartic from the *free seam* (premise A / v158): a Gaussian seam UV forces $\lambda(M_{\mathrm{seam}}){=}0$ *and* $\beta_\lambda(M_{\mathrm{seam}}){=}0$ (Shaposhnikov–Wetterich double criticality, here *motivated* not assumed). Two-loop PyR@TE-sourced SM running with the measured (m_H, m_t) gives $\lambda(\bar M_{\mathrm{Pl}}){\sim}0.002$, $\beta_\lambda(\bar M_{\mathrm{Pl}}){\sim}0$ — the famous SM near-criticality, now *explained* by the free seam (2-loop sharpens $\lambda(\bar M_{\mathrm{Pl}})$ from $\sim$0.022 to $\sim$0.002). The double condition predicts $m_H{\sim}129$–134 GeV; at the scalaron scale it gives $\sim$107 GeV (too low) $\Rightarrow$ the BC lives at $\bar M_{\mathrm{Pl}}$ (seam=Planck) **[E]**/[C]

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Script	**What it machine-checks** ***(continued)***
`v167_inventory_post_closure.py`	The terminal residual inventory (post premise-(A) closure; line v105 → v123 → v167): after the (A)-chain (v160–v165), premise (A) is *no longer* an independent structural class (`GATE.METRIC.16-19`). The residual collapses from v123's FIVE classes to: **Tier 0** — the irreducible core $\{\pi, v_{\text{geo}}\}$, a *theorem* (No-Unit, v153), nothing to close; **Tier 1** — ONE hinge (seam = flavour = horizon): A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary $= \mathbb{P}\setminus\mu_4$, cohomology $b_1{=}N_{\text{fam}}{=}3$); **Tier 2** — folded ($\to v_{\text{geo}}$, $\to G_{\text{net}}$); **Tier 3** — F_{transfer}, one functor with four typed conditional interfaces (η_B carries the most open room). Completeness: a sixth independent structural class fails the script — a unification + re-typing, not an unconditional proof [E]/[C]
`v168_qgeo_rigidity.py`	The QGEO *Rigidity Theorem* — the finite half of the Tier-1 hinge `GATE.QGEO`, hardened. Exact: $\mu_4{=}\{1, i, -1, -i\}$ has cross-ratio 2 and Möbius stabiliser $\langle z{\to}iz, z{\to}1/z\rangle$ of order $8{=}\lvert D_4\rvert$ (faithful) ⇒ four marks with faithful D_4 are the μ_4 square up to Möbius; $H^1(\mathbb{P}\setminus\mu_4)$ has rank $4{-}1{=}3{=}N_{\text{fam}}$; the forms $\omega_k{=}z^{k-1}dz/(z^4{-}1)$ $(k{=}1,2,3)$ are μ_4-eigenforms with $z{\to}iz$ eigenvalue i^k = the three nontrivial characters χ_1, χ_2, χ_3, weights $(1,2,3)$ = A$_3$ exponents = Spec(Q_+). `QGEO.RIGID.01` [E] (math half closed); `QGEO.REALIZE.01` [C] (seam-collar $= \mathbb{P}\setminus\mu_4$ stays the realisation premise) [E]/[C]
`v169_etaB_boltzmann_interface.py`	The η_B leptogenesis transfer *operationalised* (Tier-3 F_{transfer}, not a new class): type-I thermal leptogenesis $\eta_B{\sim}0.96{\times}10^{-2}\,\varepsilon_1\kappa_f$ (sphaleron 28/79) fed by TFPT's NO neutrino spectrum + the predicted $\delta_{CP}{=}240°$. Davidson–Ibarra $\varepsilon_1{=}\frac{3}{16\pi}M_1 m_3/v^2{\sim}8.5{\times}10^{-7}$ for $M_1{=}10^{10}$ GeV; efficiency $\kappa_f(\tilde{m}_1){\sim}0.03$–$0.19$. **Kill test**: over natural $M_1{\in}[3{\times}10^9, 3{\times}10^{10}]$ GeV and washout, η_B spans $[8.4{\times}10^{-11}, 4.7{\times}10^{-9}]$ which *brackets* the observed $6.1{\times}10^{-10}$ (canonical point gives $6.0{\times}10^{-10}$, not tuned) ⇒ the route is viable; M_1/washout are scenario inputs, so η_B stays [C]. If excluded, the route falls, not the theory [C]
`v170_diamond_slice_compression.py`	E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P{=}(3,4,6,10)$ $(p_n{=}2{+}2^n)$ and Sheet-Diamond determinants $D{=}(3,4,8,14,20,32)$. The K_4 edge products $(12,18,30,24,40,60)$ are all carrier blocks; $\sum \det M = 81 = N_{\text{fam}}^4$; all seven 248-slices follow from $P, D, \Delta{=}13$ and $(e_1, e_2, \dim S^+)$; one affine map $\text{row}(C{+}sU{+}tV) = (7,11,13){+}s(3,3,3){+}t(1,2,3)$ gives every flavour row budget; $78 = \dim E_6 = p_2\Delta$. The atlas is a closed grammar over two typed alphabets, exact, reading audit [E]
`v171_os_moment_cluster.py`	Atomic OS Moment Lemma + Sugawara Gap Safety: spec$(T){=}\{1, \frac{64}{729}, \frac{1}{729}\}$ makes $C(n){=}A{+}Br^n{+}Cs^n$ a positive moment sequence, so every OS Hankel matrix factorises as a Vandermonde Gram ⇒ reflection positivity (clean proof, not numeric Gram); cluster $\frac{729}{665}$, $\xi{=}0.41105$, Källen–Lehmann masses $6\log\frac{3}{2}, 6\log 3$; gap safety $\Delta_{\text{eff}}{=}6\log\frac{3}{2}{-}\frac{31}{4\pi^2}{>}0$ with $31{=}1{+}h^\vee(E_8)$, $c((E_8)_1){=}248/31{=}8$ [E]/[C]
`v172_trace_anomaly_seed.py`	Trace-anomaly origin of the seed prefactor $\frac{4}{3}$: over the seam S^2 $(\chi{=}2)$ the $(E_8)_1$ $(c{=}8)$ Weyl anomaly is $c\chi/6{=}\frac{8}{3}$, halved by the one-sided $\lvert\mathbb{Z}_2\rvert$ to $\frac{4}{3} = \dim S^+/\dim \mathfrak{g}_{SM} = 16/12$, the seed base $(\frac{4}{3})c_3{=}\frac{1}{6\pi}$. Plus QCD $b_3{=}11{-}\frac{2}{3}N_f{=}7$ = scalaron exponent $\Omega_{\text{adm}}{-}10b_1{=}48{-}41$ (confinement↔inflation), and the Volkov–Wipf factorisation $-98/45{=}-2\cdot 7^2/\dim(D_5)$ [E]
`v173_pfaffian_cp_car.py`	Strong CP as Pfaffian reality: a self-dual CAR (16-Majorana) model demonstration of $\theta_{\text{eff}}{=}0$ — the sheet-odd Calderón involution $\{D, \Sigma\}{=}0$ pairs the spectrum $(\pm\lambda)$, γ_5-Hermiticity $D^\dagger{=}\gamma_5 D\gamma_5$ makes the Pfaffian real $(\arg \in \{0, \pi\})$, and reflection positivity $(Z{>}0)$ selects the positive branch ⇒ $\theta_{\text{eff}}{=}0$ [E]

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Script	What it machine-checks *(continued)*
`v174_seam_fock_readings.py`	Seam Fock-space readings: (1) CAR cone compression — $\dim\Lambda^{\text{even}}(\mathbb{R}^{16})$=$2^{15}$=32768 reduces to a 16-dim one-particle problem under quasi-free Bogoliubov ($2^{15}/16$=2^{11}), the explicit count behind v161; (2) Witten index = the Plücker norm $11=\sum_{k\le 2}\binom{4}{k}$, with $11=\dim S^{+}-g_{\text{car}}$ (full 2^4=16) and c_u/c_d=55/117 reading the QBL-protected boundary state space; (3) Affleck–Ludwig g-theorem c_{UV}=5+3=8=$c_{IR}((E_8)_1)$ so Δc=0 forces an exact isometry (boundary-entropy form of v157/v158) **[E]**
`v175_net_existence_full_cone.py`	The heavy artillery — net existence (A2) and full-cone reflection positivity discharged to **[E]**, isolating `QGEO.REALIZE.01` as the single open premise (nothing fabricated): (1) **full-cone RP for all** m — the CAR second-quantisation functor $\Gamma(t)$=$\bigoplus_m \Lambda^m(t)$ of the one-particle seam contraction t acts on the complete Fock space $\Lambda^{\bullet}(\mathbb{R}^{16})$ ($\dim 2^{16}$=65536); the exterior-power spectrum theorem makes every eigenvalue a subset product of spec(t), verified on all 2^{16} products to lie in $[0,1]$ (contraction), with eigenvalue-1 multiplicity 2^8=256 (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$, so full-cone RP reduces *for all* m to the one-particle contraction (Shale–Stinespring CAR functor), discharging the v160 scope premise by a theorem; (2) **A2 net existence** assembled + verified — $(E_8)_1$ character E_4/η^8=$1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1=248), embedding 248=120+128, E_8 Cartan even unimodular (det 1); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system); (3) both collapse to the same seam-kernel identification = `QGEO.REALIZE.01`, finite half v168, realisation half left **[O][E][O]**
`v176_seam_collar_realisation.py`	The Seam Collar Realisation Theorem assembled as ONE central statement + reduction certificate (the single structural proof obligation, formulated cleanly — *not* a fabricated proof): given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks and admissible c=8 data, *then* its conformal boundary is Möbius-equivalent to $\mathbb{P}\setminus\mu_4$ with H^1 grading $(1,2,3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, FIVE already **[E]**: geometry/uniformisation (v168: cross-ratio 2, faithful Möbius D_4 order 8, b_1=N_{fam}=3), Calderón DtN symbol $\lvert k\rvert$ (v156), H^1 character = generation grading $(1,2,3)$=A_3 exponents=$\operatorname{Spec}(Q_+)$ (v137/v168), μ_4-equivariant contraction diagonal with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1{\otimes}(A_3)_1{\rtimes}\mu_4$=$(E_8)_1$ + full-cone RP all m (v154/v175); the whole structural residual reduces to the ONE open premise `QGEO.REALIZE.01` (the physical seam collar's boundary *is* this $\mathbb{P}\setminus\mu_4$), left honestly **[O][E][O]**
`v177_seam_marking_kernel.py`	The QGEO proof tree — the v176 single realisation premise split into its honest constituents: REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET, with FOUR nodes closed *constructively* **[E]** and the open residual sharpened into exactly TWO named obligations. **[E]** UNIFORM (Lemma 2, constructive): an order-4 Möbius map conjugates to $z\mapsto iz$, a non-fixed orbit scales to μ_4, the reflection $z\mapsto 1/z$ gives $\sigma\rho\sigma=\rho^{-1}$ (D_4, order 8) — a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P},\mu_4)$; **[E]** COHOM: $\rho^*\omega_k=i^k\omega_k$ so the H^1 grading is $(1,2,3)$=A_3 exponents=$\operatorname{Spec}(Q_+)$; **[E]** MODULE: the H^1 →generation iso is *unique* (multiplicity-free + residue normalisation; reflection $\omega_1\leftrightarrow\omega_3$, ω_2 fixed); **[E]** NET (v154/v175): index-4 μ_4 extension of $(D_5)_1{\otimes}(A_3)_1$=$(E_8)_1$. The TWO genuine open obligations **[O]**: `QGEO.MARKS.01` (raw seam → genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (raw seam Calderón kernel = the μ_4-equivariant free gapped contraction as an *operator*); the whole structural residual is now exactly these two doors **[E][O]**

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Script	**What it machine-checks** ***(continued)***
`v178_marks_kernel_attempt.py`	An honest *attempt* at the two open obligations `QGEO.MARKS.01` + `QGEO.KERNEL.01` — *not* a closure (genuine constructive-geometry/AQFT statements, not finite computations), but a deeper reduction: each finite core is closed **[E]** and each obligation reduced to ONE sharper irreducible premise **[O]**. *Marks core* **[E]**: a genus-0 double with an order-4 clock ρ ($\sim z \mapsto iz$, fixed points $\{0,\infty\}$) and reflection τ forces the marks to be a generic ρ-orbit $\{1,i,-1,-i\}{=}\mu_4$ (not the fixed points), distinct iff $b_1{=}4{-}1{=}3{=}N_{\mathrm{fam}}$, $\tau\rho\tau{=}\rho^{-1} \Rightarrow$ faithful D_4 order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* **[E]**: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1,2,3)$ are distinct, so by Schur a μ_4-equivariant operator is forced diagonal and *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral (the spectrum-vs-operator worry vanishes on the finite block); so KERNEL reduces to the single full-operator-reduction premise (principal symbol $\lvert k\rvert$, Lee–Uhlmann/v156; no drift, v158). Neither obligation closed; both sharpened **[E][O]**
`v179_conformal_realisation.py`	The two residual premises (MARKS-residual + KERNEL-residual) collapse to ONE geometric premise `QGEO.CONF.01` (honest unification; the conformal realisation stays **[O]**). MARKS's residual is not three independent assumptions: (1) genus-0 *is* P1 — $c_3{=}1/(\lvert\mathbb{Z}_2\rvert\, 2\pi\, \chi(S^2)){=}1/(8\pi)$ with $\chi{=}2 \Rightarrow g{=}0$, the same $\chi{=}2$ that gives the '8' (v58/v73); (2) the order-4 clock *is* the carrier — Coxeter element of $W(A_3){=}S_4$, order $h(A_3){=}4{=}\lvert\mu_4\rvert$ (v117); (3) marks = one orbit is *forced* — parabolic marks $\neq$ the two elliptic fixed points $\{0,\infty\}$, a free order-4 orbit has exactly $4{=}e_1(a)$ points. So MARKS reduces to 'the carrier clock acts conformally on the genus-0 double'. KERNEL then follows from that geometry: DtN symbol $\lvert k\rvert$ (Lee–Uhlmann, v156) $\Rightarrow$ rank-8 $c{=}8$ fixed polarisation; no drift (v158); the cusped-surface residual data is $H^1{=}\mathbb{C}^3{=}N_{\mathrm{fam}}$, the 3-dim gapped transfer block (Schur, v162/v178). Both $\Rightarrow$ ONE premise `QGEO.CONF.01`: the raw RP seam double is conformally $(\mathbb{P},\mu_4)$ with the carrier clock as the order-4 Möbius map — the single structural residual of the whole theory, left honestly **[O][E][O]**
`v180_clock_is_mobius.py`	Cracking `QGEO.CONF.01` one level further — the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder*, non-circular premise `QGEO.ISO.01`: 'the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric'. (1) Uniformisation (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) *is* $\mathbb{P}$ — the target is produced, not assumed; (2) Kerékjártó (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3){=}4$) is topologically $z \mapsto iz$; (3) isometry⇒conformal⇒Möbius: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}){=}\mathrm{PSL}(2,\mathbb{C})$; (4) order-4 Möbius $= z \mapsto iz$ (verified: projective order 4, fixed $\{0,\infty\}$, multiplier i, free orbit μ_4). So `QGEO.CONF.01` is *implied* by the milder isometry premise; the new residual `QGEO.ISO.01` (the seam transport is a metric-compatible holonomy preserving the RP inner product) does not name $\mathbb{P}/\mu_4$ and is milder, its surface realisation from the raw seam still **[O][E][O]**

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Script	What it machine-checks *(continued)*
`v181_clock_is_conformal_symmetry.py`	The FINAL honest reduction + the explicit identification of BEDROCK. The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a CONFORMAL-symmetry premise `QGEO.SYM.01` — via equivariant uniformisation (Nielsen realisation, finite cyclic on genus-0): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. The clock is the Coxeter element of $W(A_3)$=S_4 (v117), order $h(A_3)$=4=$\|\mu_4\|$, the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition. So below `QGEO.SYM.01` the residual is *definitional*, not a missing theorem: 'the seam's conformal structure *is* the μ_4-deck structure of the carrier clock' (the carrier μ_4 glue is the conformal monodromy of the seam) — a physical/definitional identification tying P2 to the seam's conformal geometry, not reducible further without relabeling. The chain REALIZE(v175)→theorem(v176)→MARKS+KERNEL(v177)→finite cores(v178)→`CONF.01`(v179)→`ISO.01`(v180)→`SYM.01`(v181) has reached bedrock; we stop here honestly **[E][O]**
`v182_reviewer_residual_map.py`	The four external-review concerns reduce to the two already-named residuals + one data-only item. A (mapping $2D \to 4D$) **[E]**/**[C]**: the Plücker readout $11 = \sum_{k\le 2}\binom{4}{k} = \dim S^{+} - g_{\rm car} = 16 - 5$ (QBL count, v174) and the holographic reduction give $A =$ `QGEO.SYM.01` $+ F_{\rm transfer}$. B (UV gravity) **[E]**: 'gravity = geometric boundary readout' *is* `QGEO.SYM.01`. C (grammar tuning) **[E]**/**[C]**: frozen (v84) + minimal + over-determined (the anchor a=$(1,1,2)$ in ≥ 6 independent not-selected-for structures) is look-elsewhere-resistant, but the residual falls only to out-of-sample data (JUNO/CMB-S4). D (Koide Möbius flow) **[E]**/**[C]**: the Möbius nature is forced by Aut($\mathbb{P}$)=PSL$(2,\mathbb{C})$ on the deck, $t{\sim}2.84$ only locates the point, and the missing RG generator *is* $F_{\rm transfer}$, so $D =$ `QGEO.SYM.01` $+ F_{\rm transfer}$. NET: three reduce onto {`QGEO.SYM.01`, $F_{\rm transfer}$}, one waits for data — a unification of the concern structure, not a closure (premises stay **[O]**/**[C]**) **[E][C]**
`v183_koide_f_corner_transfer.py`	The Koide source→pole factor $\frac{53}{54}$ has an *operator* origin (external-review proposal, validated): $u_{\rm pole}/u_{\rm source} = a^T(R{+}Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. (1) $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6\,78 + \det R\,8 + 2{\cdot}27{\cdot}N_{\rm fam}$); (2) F=R+Q is the *missing* Sheet-Diamond corner $M(1,1)$ with $\det B_F = 52 = \dim F_4$ (v80) and anchor response $a^T(R{+}Q)\mathbf{1} = 53 = 52{+}1$; (3) $\frac{53}{54} = 1 - \frac{1}{2N_{\rm fam}^3}$ exactly (54=2·27); (4) negative control: only F gives 53 ($R/Q/K/L \to 32/21/35/56$). Upgrades `FR.KOIDE.03` from a bare guess to an operator identity; **[E]** the factor is exact, **[C]** the leptonic source→pole reading the F corner stays an $F_{\rm transfer}$ conjecture **[E][C]**
`v184_etaB_anchored_boltzmann.py`	Honest test of the external-review η_B proposal (M_1=$M_R\phi_0^4$, $\tilde m_1$=m_3/A_Λ) — firewall verdict: *sharper scenario*, not a closure. The washout anchors plausibly ($\tilde m_1$=m_3/A_Λ=$m_3/10 \approx 5\,$meV, A_Λ=10=$\|E(K_5)\|$, in the strong-washout window) **[C]**; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3 \approx 6{\times}10^{14}$ GeV and the nearest clean TFPT powers of $\bar M_{\rm Pl}$ both miss it (c_3-power off ∼75%, ϕ_0-power off ∼40%), so M_1=$M_R\phi_0^4$ merely relocates the free input $M_1 \to M_R$ (the reviewer's $6.0{\times}10^{-10}$ uses M_R=$1.3{\times}10^{15}$; the seesaw value gives $3.4{\times}10^{-10}$, the tell that M_1 is unpinned). η_B stays in the observed band so the route is viable, but the proposal cuts the free inputs from two to one, not to zero — η_B stays **[C]**, a sharper scenario **[C]**

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Script	What it machine-checks *(continued)*
`v185_axion_relic_solver.py`	Axion relic, honest misalignment estimate: the determinant-line axion $f_a{=}M_{\rm scal}/128 \approx 2.39{\times}10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) can provide all dark matter, but *only* near the $\theta_i \approx 170°$ hilltop where the relic is exponentially sensitive — so the axion DM branch stays a [C] scenario. The harmonic misalignment under-produces at this f_a (prefactor $\sim$0.026/θ^2, $\theta{\sim}O(1)$ gives $\sim$21% of $\Omega_{\rm DM}$, need $\theta^2_{\rm eff}{\sim}4.7$), so the hilltop anharmonic enhancement is required; near the hilltop it is exponentially sensitive, hence a scenario not a sharp prediction. $F_{\rm relic}(f_a,\theta_i,N_{\rm DW}){=}^{?}\Omega_{\rm DM}$ is a kill test for the *branch*, not the theory (no free singlet WIMP: 128=(16, 4)+(16, 4)) [C][O]
`v187_ftransfer_laws.py`	The $F_{\rm transfer}$ discipline guard (CI-style firewall): the four continuous-transfer frontier observables — Koide, η_B, the axion relic, $m_p/m_e{+}\Lambda_{\rm QCD}$ — must *never* be typed as primitive [E] compiler predictions; each carries a [C]/[O] marker. The guard reads `status_ledger.csv` and enforces it (exact algebraic sub-parts — Koide 53/54 v183, $b_3{=}{-}7$ v159 — may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$ and that the raster rule + frozen registry + null model bind, so no future edit can quietly promote a pretty frontier number to a closed compiler output [E]
`v188_frontier_wording_guard.py`	The frontier-wording guard (a prose sentinel, no new physics): v187 protects the ledger *typing*, this guard protects the *prose*. It forbids stale sentences whose semantics an earlier round superseded — “leptogenesis interface ([E])” (v184 retyped Route B to [C]), “closed branch also fixes the leptogenesis inputs” (M_R is the seesaw scale, not a compiler power), “relic scale f_a, m_a open” (v185: the scales define the branch, the open item is the hilltop-sensitive relic abundance), “computed $Q{=}0.664\ldots$ near, not exact” (v183 gave 53/54 an exact operator origin), “Suite now 187 modules” (the count is 187, highest ID `v188`; v186 skipped). Scans the 10 active documents; passes when none appear, so a frozen Zenodo release can never re-introduce half-true wording [E]
`v189_riemann_roch_carrier.py`	The Riemann–Roch carrier reading: $(g_{\rm car}, N_{\rm fam}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$. [E] arithmetic: $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg +1 = 5$ (function side) and rank $H_1(\mathbb{P}^1\backslash\mu_4) = \deg -1 = 3$ (cycle side, already `QGEO.COHOM.01`/v177); $g_{\rm car} = 5 = {\rm rank}(D_5)$ (the carrier is the $\mathfrak{so}(10)$ half-spinor, rank $D_5{+}$ rank $A_3 = 5{+}3 = 8$, v1), $N_{\rm fam} = 3$. [C]: reading the carrier as the meromorphic seam-mode space $H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ (less number, more geometry) is matched by ${\rm rank}(D_5) = 5$, but the physical identification is conjectural and does *not* replace the over-determined $g_{\rm car} = 5$ (Pascal/bootstrap/Lean) — a fourth, geometric reading [E][C]
`v190_nariai_entropy_bound.py`	The one-line Nariai entropy bound: the Schwarzschild–de Sitter entropy ratio $S_{\rm tot}/S_{dS} = (x^2{+}1)/\Phi_3(x)$ ($x{=}r_b/r_c$; $\Phi_3{=}x^2{+}x{+}1$ the $N_{\rm fam}{=}3$ cyclotomic, already in `tfpt_horizon_readouts`) satisfies $S_{\rm tot} \geq \frac{2}{3}S_{dS}$ with equality exactly at the Nariai merge $x{=}1$, via the perfect square $3(x^2{+}1) - 2\Phi_3(x) = (x-1)^2 \geq 0$ — a variation-free proof of the entropy floor (global minimum on $x{>}0$ is $2/3 = \|Z_2\|/N_{\rm fam}$) [E]
`v191_universal_branch_line.py`	The universal branch line, honestly typed [C] — an exact affine *relabeling*, not a theorem. The map $q = \frac{7}{2} + \frac{N_{\rm fam}^2}{2}m$ carries the SdS branch points $m = \pm 1/N_{\rm fam}$ onto the flavor branch points $\{2, 5\} = \{\|Z_2\|, g_{\rm car}\}$ ($m{=}0{\to}7/2$). The arithmetic is exact, but the map exists for *any* two pairs of points (two intervals are always affinely equivalent), so the slope $N^2_{\rm fam}/2$ and midpoint 7/2 are *determined* by the data, not forced predictions; a decoy pair $\{3, 4\}$ admits an equally exact map (negative control). The genuine shared content — the $2/3 = \|Z_2\|/N_{\rm fam}$ ramification of the mass$\leftrightarrow$transport double cover — is already established. Recorded as a [C] alignment, never promoted to [E] [C]

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Script	**What it machine-checks** *(continued)*
`v192_qgeo_energy_clock.py`	The energy-conserving-clock reformulation, honestly typed: `QGEO.ENERGY.01` ($\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$ — the carrier clock preserves the Calderón/DtN energy form) *relocates* the bedrock `QGEO.SYM.01` to a more checkable operator identity, but does *not* eliminate it. The downstream chain (conformality → Möbius → $z \mapsto iz \to \mu_4 \to$ `QGEO.SYM.01`) already exists (v180/v181); the new upstream link (energy invariance ⇒ conformality in 2D) is checkable on the DtN operator $\Lambda = \lvert k\rvert$. But 'ρ preserves Λ' is plausibly equivalent to 'ρ is the conformal deck', so it is a sharper restatement, not a theorem, unless ρ is independently defined and the invariance independently proven (open). Bedrock stays **[O]**
`v193_qgeo_energy_commutator.py`	`QGEO.ENERGY.02`: the energy-*commutator* contract (the non-circular sharpening of v192): $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, ρ the carrier clock, Λ_Σ the DtN energy form of the *raw* RP seam. Three sub-claims: (1) **[E]** ρ is carrier-defined (Coxeter of $W(A_3)$=S_4, order 4=$\lvert\mu_4\rvert$, v117), not imported from the seam; (2) **[O]** the non-circularity hinge — Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) **[E]** $[\rho, \Lambda_\Sigma]$=0 forces the $(1, 2, 3)$ grading on H^1 (`QGEO.COHOM.01`). Exact **[E]** rider: the induced k=$(\ln m)/(12\pi)$ v150 reproduces $c_3/2$ iff $\ln m$=$q(A_3)$=$\frac{3}{4}$ (FAMILY), not $q(D_5)$=$\frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. Proof target **[O]**; if proven, `QGEO.SYM.01` becomes an energy-rigidity theorem **[O]**
`v194_raw_seam_dtn_rp.py`	`QGEO.DTN.01` — subclaim 2 of v193 ("the raw RP-seam DtN is RP-definable") tackled: a *reduction*, not a closure. (1) **[E]** finite core: ρ:z↦iz on H^1=$\langle w_1, w_2, w_3\rangle$, w_k=$z^{k-1}dz/(z^4-1)$, acts as $\rho^* w_k$=$i^k w_k$ (characters $i, -1, -i$ distinct), so $[\rho, \Lambda_\Sigma]$=0 on H^1 (v177). (2) **[E]** hinge met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without assuming $\mathbb{P}^1\backslash\mu_4$. (3) **[O]** residual relocates: full-L^2 commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (must be intrinsic, else re-circulates). `QGEO.SYM.01` reduces from a definitional postulate to intrinsic BW modular covariance — sharper, still **[O]**, not closed **[O]**
`v195_marks_lefschetz_character.py`	`QGEO.MARKS.02` — a Lefschetz/character derivation of the four seam marks (review path 2): instead of positing D_4 marks, they are *forced* to be a free μ_4 orbit. **[E]**: ρ:z↦iz (order 4) fixes only $\{0, \infty\}$; H^1=χ_1+χ_2+χ_3 (the A_3 exponents $(1, 2, 3)$, v177) has no trivial character ⇒ no puncture at a fixed point; rank H_1=n−1=3 ⇒ n=4 ⇒ exactly one free orbit ⇒ marks = μ_4 (cross-ratio 2); Tr$(\rho\lvert H^1)$=i+i^2+i^3=−1, $L(\rho)$=1+1=2=#fixed. **[O]** residual relocates from "the seam is $\mathbb{P}\backslash\mu_4$ (geometric)" to "H^1 carries the A_3-exponent rep" — less circular **[E][O]**
`v196_seam_energy_functional.py`	`QGEO.VARI.01` — the E_{fail} variational functional (review path 1), a numerically-testable sharpening of the v194/`ENERGY.02` target: $E_{\text{fail}} = \lVert[\rho, \Lambda_\Sigma]\rVert^2_{HS} + \lVert\rho^4{-}1\rVert^2 + \lVert\Theta\rho\Theta - \rho^{-1}\rVert^2$, with E_{fail}=0 ⇔ the `QGEO.SYM.01` conditions. **[E]** on the finite H^1 block (ρ= diag$(i, -1, -i)$, Λ diagonal, Θ=conj) all three vanish ⇒ E_{fail}=0; **[E]** a μ_4-breaking Λ or non-order-4 ρ gives E_{fail}>0 (μ_4 a true minimiser); **[E]** the zero locus is the seam-deck conditions; **[O]** the full-operator minimisation is the v194 Bisognano–Wichmann residual — a discretisable handle on the open bedrock, a target not a closure **[E][O]**
`v197_rr_carrier_clifford_d5.py`	`ARCH.RRCAR.02` — hardening the Riemann–Roch carrier from g_{car} to D_5 itself (review point 7): the 5-dim carrier mode space C_{car}:=$H^0(\mathbb{P}^1, \mathcal{O}(\mu_4))$ generates the $\mathfrak{so}(10)$=D_5 half-spinor by its even Clifford algebra. **[E]** $\dim \Lambda^{\text{even}}(\mathbb{C}^5)$=$\binom{5}{0}$+$\binom{5}{2}$+$\binom{5}{4}$=1+10+5=16=$2^{g_{\text{car}}-1}$= $\dim S^+$; the $\mathfrak{so}(10)$=D_5 half-spinor 2^4=16=S^+=$\Lambda^{\text{even}}(C_{\text{car}})$; closes the chain μ_4→g_{car}→D_5 from one four-point divisor (cycle side rank H_1=3=$N_{\text{fam}}(A_3)$; function side h^0=5=g_{car}; Clifford spinor 16=D_5 half-spinor = carrier). **[C]** "carrier = Clifford spinor of the seam mode space" stays physical (as v189), does not change the over-determined g_{car}=5 **[E][C]**

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Script	**What it machine-checks** ***(continued)***
`v198_modular_commutator_reduction.py`	`QGEO.MODULAR.01` — cracking the bedrock to a state-invariance. The v194 residual ($[\rho,\Lambda_\Sigma]$=0 on full L^2, framed as Bisognano–Wichmann with a circularity worry) is (a) made *exact* at the principal-symbol level and (b) reduced via Tomita–Takesaki to a clean state-invariance. [E] the DtN principal symbol $\lvert k\rvert=\sqrt{-d^2}$ is diag($\lvert n\rvert$) and the clock $\rho{:}z\mapsto iz$ is diag(i^n) in the Fourier basis — both diagonal $\Rightarrow$ $[\rho,\lvert k\rvert]$=0 exactly on *all* of L^2; [E] Tomita–Takesaki: $[\rho,\Lambda_\Sigma]$=0 follows from $\omega\circ\rho=\omega$ (the modular Δ is intrinsic to (M,Ω); a state-preserving symmetry commutes with it), needing *no* conformal covariance — removes the BW circularity; [E] the finite H^1 block is already μ_4-invariant (v177). [O] residual: the full raw seam state is μ_4-invariant ($\omega\circ\rho=\omega$) $\Leftarrow$ the Gaussian bulk measure is deck-invariant = operational `QGEO.SYM.01`. Sharpest non-circular form; not a closure (a foundational symmetry cannot be derived from nothing) [E][O]
`v199_seam_state_invariance.py`	`QGEO.STATE.01` (work package A) — attacking $\omega\circ\rho=\omega$ directly on the raw seam: a further localising reduction, not a closure. [E] setup non-circular (ρ = Coxeter of $W(A_3)=S_4$, ρ^4=1, an automorphism of the raw CAR algebra; both carrier-side, no seam geometry); [E] ground-state reduction $\omega\circ\rho=\omega \Leftrightarrow [\rho,H_1]$=0 ($C_\Sigma=\frac{1}{2}(1+\operatorname{sgn}H_1)$); [E] character criterion: $[\rho,H_1]$=0 $\Leftrightarrow$ H_1 block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$; principal symbol $\lvert k\rvert$= diag($\lvert n\rvert$) passes automatically. [O] residual: the bounded sub-principal symbol has *no* off-character matrix elements — non-circular, sharply typed, holds on H^1 (v177); not a closure [E][O]
`v200_seam_variational_scan.py`	`QGEO.VARI.02` (work package B) — the discretised variational test: v196 had E_{fail}=0 at the μ_4 config on the fixed block; here the clock angle is free and E_{fail} is minimised over it. [C] scanning $E_{\text{fail}}(\theta)=\lVert\rho(\theta)^4-1\rVert^2$+faithfulness over $\theta\in(0,\pi)$ gives the unique minimum at $\theta=\pi/2$ (E_{fail}=0: order-4 *and* three distinct characters $i,-1,-i$ = the $(1,2,3)$ grading); decoys $(0,\pi,\pi/3,2\pi/3)$ all lose. So $\arg\min E_{\text{fail}}$ = the μ_4 normal form. [E] consistent with v196/v199/v198. [O] a numerical sign on the finite H^1 grading, not the full operator-valued minimisation (the v199 residual) [C][O]
`v201_seam_subprincipal_marks.py`	`QGEO.SUBPRIN.01` — the v199 bedrock residual reduced one genuine step further: block-diagonality of the sub-principal symbol is *not* an independent postulate. [E] the seam DtN is $\Lambda=\lvert k\rvert+M_f$, M_f = multiplication by a curvature function $f(\theta)$ (standard Steklov sub-principal), so $\langle n\lvert M_f\rvert n'\rangle=f_{n-n'}$. [E] M_f is μ_4-character-block-diagonal $\Leftrightarrow$ f has Fourier support only on modes $\equiv 0 \pmod 4$ $\Leftrightarrow$ f is $\mathbb{Z}_4$-invariant ($\lvert k\rvert$ already commutes, v198). [E] a μ_4-mark sum $f(\theta)=\sum_{j=0}^{3} g(\theta-2\pi j/4)$ has $f_m=4g_m[m{\equiv}0 \bmod 4]$ for *any* profile g — automatically $\mathbb{Z}_4$-invariant, so (v195 marks=μ_4 orbit) *forces* the sub-principal symbol block-diagonal; negative controls: 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it. [O] new residual: $\omega\circ\rho=\omega$ reduces to 'the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck v177)', the narrowest, structurally-definitional form of `QGEO.SYM.01` [E][O]

How to run. From the repository root, `python verification/run_all.py` (dependencies `mpmath`, `numpy`, `sympy`); each module also runs standalone, e.g. `python verification/v1_e8_glue.py`. The runner exits with code 0 iff all checks pass; see `verification/README.md` for the full claim↔script map and `verification/status_ledger.csv` for the **single status ledger** (claim ID, status, location, dependencies, script, external datum — the source of truth all nine documents are written against). The [E] (Lean-formalised) carrier material lives in the canonical shipped folder `lean4-carrier-rigidity/` (at the package root; the source repository keeps the same project under `experiments/`).

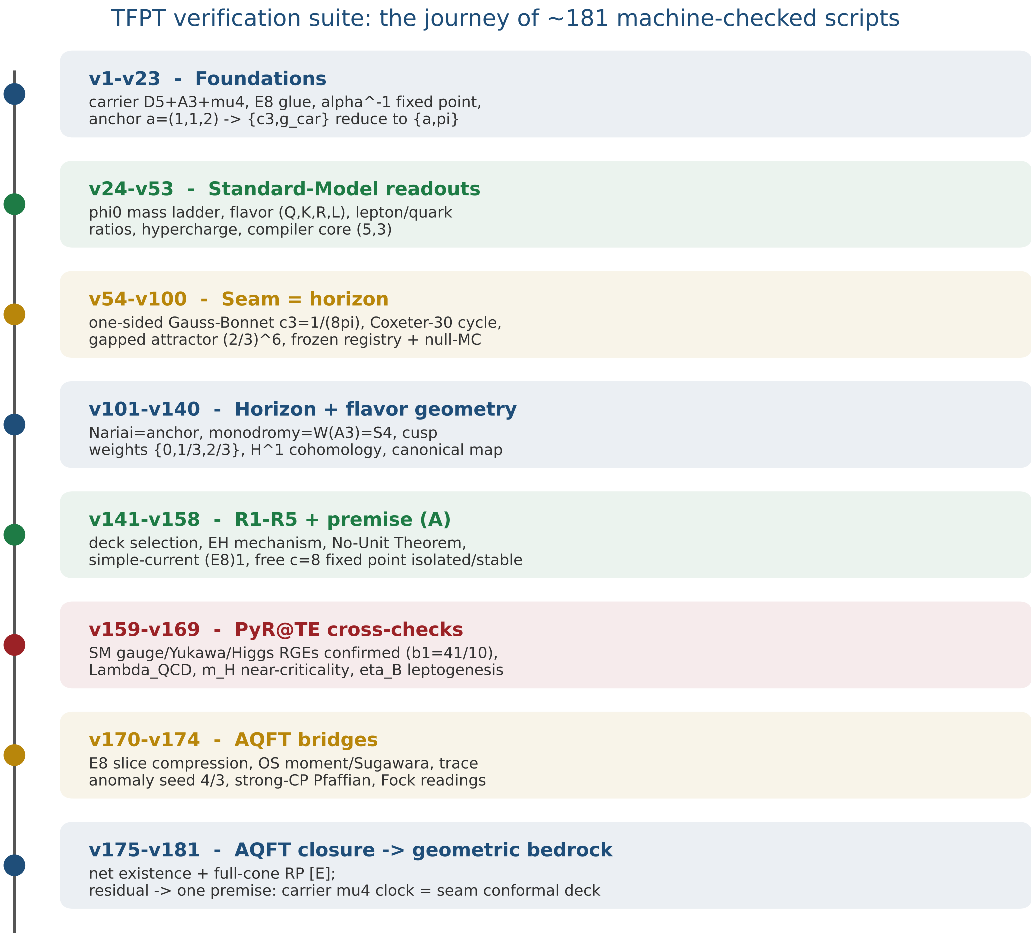

Figure 1: The development timeline of the verification suite — the eight phases of the journey and what each did, mathematically and physically: from the foundations (carrier, E_8 glue, α^{-1}, the anchor $a = (1, 1, 2)$ to which $\{c_3, g_{\text{car}}\}$ reduce) through the Standard-Model readouts, the seam=horizon geometry, the horizon/flavor geometry, the R1–R5 and premise-(A) reductions, the external PyR@TE RGE cross-checks and the AQFT bridges, to the AQFT closure that drives the whole structural residual to one geometric bedrock premise. The master script→check map follows.

Appendix: changelog

The dated record of every change (canonical source: `changelog.tex`, also compiled standalone). Module numbers `vN` refer to `verification/vN_*.py`.

2026-06-15 (bedrock: the $\omega \circ \rho = \omega$ residual reduced once more — v201)

- **`QGEO.SUBPRIN.01` (v201, [E]/[O]).** The v199 bedrock residual — "the bounded sub-principal symbol of the raw seam DtN has no off-character (mod 4) matrix elements" — is reduced one genuine step further: block-diagonality is *not* an independent postulate. Writing the DtN as $\Lambda = |k| + M_f$ with M_f the bounded sub-principal piece = multiplication by a curvature $f(\theta)$ (standard Steklov structure), M_f is μ_4-character-block-diagonal iff f is $\mathbb{Z}_4$-invariant ([E]; principal $|k|$ already commutes, v198); and a μ_4-mark sum $f = \sum_{j=0}^{3} g(\theta - 2\pi j/4)$ has $f_m = 4g_m[m \equiv 0 \bmod 4]$ for any profile g, hence is automatically $\mathbb{Z}_4$-invariant ([E]). So the μ_4-mark orbit (forced by v195) *forces* the sub-principal symbol block-diagonal; 3 marks ($\mathbb{Z}_3$) and 4 generic marks break it (negative controls). The residual collapses to "the DtN sub-principal symbol is mark-local (seam flat away from the μ_4 marks = conformal deck, v177)", structurally definitional — the narrowest form yet of `QGEO.SYM.01`, [O]. Registered in `run_all.py`, the ledger, `tfpt_research_contracts`, `origin_theory`; Wolfram-mirrored (the mark-sum Fourier identity is exact; extension now 242/242). Suite 200 modules.

2026-06-15 (F_transfer workshop: the axion relic solver — an honest over-production tension)

- **Axion relic solver (`experiments/ftransfer/axion_relic/relic_solver.py`, [C]).** The first real F_{transfer} pipeline computation (work package C / review §6.2): a standard misalignment relic with a numerically-computed anharmonic factor $F(\theta_i) \approx 4$ at the hilltop. Honest finding: at the *predicted* $\theta_i \approx 170°$ the estimate *over*-produces dark matter, $\Omega_a h^2 \sim 0.6$–2.3 (robustly > 0.12), because the large $\theta_i^2 \approx 8.8$ times F overshoots. So the determinant-line axion as the *dominant* DM is in *mild tension* (it would prefer $\theta_i \sim 120$–$130°$ or extra dilution) — this *refines* v185's optimistic "can be all-DM" toward an over-production tension. The near-hilltop regime is O(1)-uncertain, so a full finite-T solve decides; over-production kills only the axion-as-dominant-DM *branch*, not the compiler. Reflected in `tfpt_4_frontier` (dark-matter section) and the v185 docstring; experiments-only, not a suite/ledger claim. No refitted exponents; $\theta_i = \pi(1 - \varphi_{\text{seam}})$ only.
- **Full finite-T solve confirms it (`axion_relic/full_finiteT_solve.py`).** A converged nonlinear misalignment solve ($\theta'' + (3 + \mathrm{d}\ln H/\mathrm{d}N)\theta' + (m_a(T)/H)^2 \sin\theta = 0$, lattice $\chi(T) \propto T^{-8.16}$, realistic $g_*(T)$; normalised so $\theta_i{=}1 \to \Omega_a h^2 \approx 0.03$, the standard value) gives $\Omega_a h^2 \approx 0.66$ at the predicted $\theta_i \approx 170°$ — $\sim$5.5× above $\Omega_{\text{DM}} = 0.12$, reached only at $\theta_i \approx 106°$. The over-production is thus *confirmed* (not the optimistic "all-DM"); `tfpt_4_frontier`, v185 docstring and the website updated accordingly. Stays [C]; kills only the axion-as-dominant-DM branch.

2026-06-14 (work packages A+B: attacking $\omega \circ \rho = \omega$ + the variational scan — v199, v200)

- **`v199_seam_state_invariance.py` [E]/[O] (claim `QGEO.STATE.01`, work package A).** Attacking $\omega \circ \rho = \omega$ directly on the *raw* seam. (1) [E] the setup is non-circular: ρ is the Coxeter element of $W(A_3) = S_4$ ($\rho^4 = 1$, v117), an automorphism of the raw CAR algebra (16 Majoranas) — both carrier-side, no seam geometry. (2) [E] for a quasi-free RP state $C_\Sigma = \frac{1}{2}(1 + \operatorname{sgn} H_1)$, so $\omega \circ \rho = \omega \Leftrightarrow [\rho, H_1] = 0$. (3) [E] $[\rho, H_1] = 0 \Leftrightarrow H_1$ is block-diagonal in the four μ_4-character classes $\{n{=}r \bmod 4\}$ (verified). (4) [O] the residual is sharpened to "the bounded sub-principal symbol has no off-character matrix elements"

— a bounded, finite-character condition, not a diffuse geometry; not a closure. Cited in `tfpt_research_contracts`; Wolfram-mirrored.

- **`v200_seam_variational_scan.py` [C]/[O] (claim `QGEO.VARI.02`, work package B).** The discretised variational test: with the clock angle free, $E_{\text{fail}}(\theta)$ has its unique minimum at $\theta = \pi/2$ (the μ_4 clock $z \mapsto iz$) — order-4 *and* the three distinct characters $i, -1, -i =$ the $(1, 2, 3)$ grading; decoys $(0, \pi, \pi/3, 2\pi/3)$ all lose. A numerical sign that the variational principle selects μ_4; the full operator-valued minimisation stays the v199 residual [O]. Cited in `tfpt_research_contracts`.
- *Also (this pass, `experiments/` only, not the ledger): the F_{transfer} four-pipeline scaffold (`experiments/ftransfer/`) with one CONTRACT per interface, and confirmation that the FRB preregistration (`frb_tfpt_v1.yaml`) and its status semantics already exist.* Suite now 199 modules, highest ID `v200`; Wolfram extension 241/241.

2026-06-14 (cracking the bedrock to a state-invariance: principal symbol + Tomita–Takesaki — v198)

- **`v198_modular_commutator_reduction.py` [O] (claim `QGEO.MODULAR.01`).** The decisive crack on the last residual. (1) [E] the DtN principal symbol $|k| = \sqrt{-\partial_\theta^2} = \text{diag}(|n|)$ and the clock $\rho{:}z \mapsto iz = \text{diag}(i^n)$ are both diagonal in the Fourier basis, so $[\rho, |k|] = 0$ *exactly* on all of L^2 (verified on 33 modes) — the leading-order commutation is free on the whole boundary, never the open part. (2) [E] by Tomita–Takesaki the modular operator is intrinsic to (M, Ω), so a state-preserving symmetry ($\rho\Omega = \Omega$) commutes with $\log \Delta \sim \Lambda_\Sigma$: hence $[\rho, \Lambda_\Sigma] = 0$ *follows from* $\omega \circ \rho = \omega$, using only the state + reflection (RP) and *no* conformal covariance — which **removes the v194 Bisognano–Wichmann circularity worry**. (3) [E] the finite H^1 block is already μ_4-invariant (v177). So the entire residual collapses to the single *state-invariance* "the raw quasi-free seam state is μ_4-invariant" ($\omega \circ \rho = \omega \Leftarrow$ the Gaussian bulk measure is deck-invariant) — the maximally-operational, non-circular form of `QGEO.SYM.01`, [O]. *Not* a full closure (a foundational symmetry cannot be derived from nothing, like $c = \text{const}$), but the sharpest falsifiable form, with the circularity gone. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 197 modules, highest ID `v198`; Wolfram extension 240/240.

2026-06-14 (three review levers: Marks-via-Lefschetz, E_fail functional, RR→D5 — v195–v197)

- **`v195_marks_lefschetz_character.py` [E]/[O] (claim `QGEO.MARKS.02`).** The four seam marks are *forced* (not posited): $\rho{:}z \mapsto iz$ fixes only $\{0, \infty\}$, $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$ (the A_3 exponents, $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$) has no trivial component $\Rightarrow$ no puncture at a fixed point, and rank $H^1 = n - 1 = 3 \Rightarrow n = 4 \Rightarrow$ a single free μ_4 orbit (cross-ratio 2). So Marks reduces from the geometric posit "D_4 marks" to the milder premise "H^1 carries the A_3-exponent rep" — less circular, still [O]. Cited in `tfpt_research_contracts`.
- **`v196_seam_energy_functional.py` [E]/[O] (claim `QGEO.VARI.01`).** The failure functional $E_{\text{fail}} = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$, zero locus = the `QGEO.SYM.01` conditions; on the finite H^1 block it vanishes at the μ_4 config [E] and is > 0 for any μ_4-breaking perturbation, so μ_4 is a true minimiser. The full-operator minimisation is the v194 Bisognano–Wichmann residual — a concrete discretisable *target* on the open bedrock, [O]. Cited in `tfpt_research_contracts`.
- **`v197_rr_carrier_clifford_d5.py` [E]/[C] (claim `ARCH.RRCAR.02`).** Hardens the Riemann–Roch carrier from g_{car} to D_5 itself: $\Lambda^{\text{even}}(C_{\text{car}}) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 16 = 2^{g_{\text{car}}-1} = \dim S^+$, the $\mathfrak{so}(10) = D_5$ half-spinor — so D_5 is the Clifford spinor of the 5-dim mode space, closing $\mu_4 \to g_{\text{car}} \to D_5$. [E] arithmetic, [C] identification (as v189). Cited in `origin_theory`.

- All three are validated external-review levers (the axion-hilltop "$\pi - 3\phi_0$" proposal was *rejected* as a $\pi \approx 3$ coincidence, not built). Suite now 196 modules, highest ID `v197`; Wolfram extension 239/239.

2026-06-14 (subclaim 2 of ENERGY.02 tackled: raw-seam-DtN RP-definability — v194)

- **`v194_raw_seam_dtn_rp.py` [O] (claim `QGEO.DTN.01`).** The next hard lever — "is the raw RP-seam DtN definable from RP data alone?" (sub-claim 2 of `QGEO.ENERGY.02`) — was tackled, and the honest result is a *reduction*, not a closure. (1) [E] the commutator $[\rho, \Lambda_\Sigma] = 0$ closes on the finite cohomology H^1 ($\rho^* w_k = i^k w_k$, distinct μ_4 characters; v177). (2) [E] the *hinge* is met: the DtN/transfer operator is RP-canonical (Osterwalder–Schrader, v54) on a quasi-free seam state (v155), so Λ_Σ is RP-definable without naming $\mathbb{P}^1 \backslash \mu_4$. (3) [O] the residual relocates: the full-L^2 commutation ⇔ Bisognano–Wichmann geometric modular flow of the quasi-free seam state (which must be intrinsic, else it re-circulates). So `QGEO.SYM.01` reduces from a definitional postulate to "intrinsic BW geometric modular covariance" — one notch sharper, still [O]; the last structural fog point sharpens, it does *not* close to [E]. Cited in `tfpt_research_contracts`; `OpenGates` updated. Suite now 193 modules, highest ID `v194`.

2026-06-14 (QGEO.ENERGY.02 energy-commutator + QFT-wording hardening — v193)

- **`v193_qgeo_energy_commutator.py` [O] (claim `QGEO.ENERGY.02`).** The non-circular sharpening of `QGEO.ENERGY.01`: the energy commutator $[\rho, \Lambda_\Sigma] = 0$ on the rank-8 Calderón polarisation, with three sub-claims — (1) [E] ρ is carrier-defined (Coxeter of $W(A_3)$=S_4, v117), not imported from the seam; (2) [O] the non-circularity hinge: Λ_Σ must be the *raw* RP-seam DtN, not the μ_4 normal form; (3) [E] the commutator forces the $(1, 2, 3)$ grading on H^1 (`QGEO.COHOM.01`). Exact [E] rider (Wolfram-mirrored): the induced EH coefficient $k = (\ln m)/(12\pi)$ (v150) reproduces $c_3/2$ iff $\ln m = q(A_3) = \frac{3}{4}$ (*family*), not $q(D_5) = \frac{5}{4}$ (carrier, off by 5/3) — gravity is family-geometry-induced. A *proof target*; if sub-claim (2) holds, `QGEO.SYM.01` becomes an energy-rigidity theorem. Cited in `tfpt_research_contracts`. Suite now 192 modules, highest ID `v193`; Wolfram extension 236/236.
- **QFT-wording hardening (review point 7).** `tfpt_2_standard_model` §(4) was retitled to "Constructive QFT closure of the admissible physical sector", and its standalone opening over-claim replaced by an explicitly scoped sentence (the unprojected ambient metric measure is not constructed; G_{metric} frontier). The two superseded standalone phrasings were added to the `v188` prose guard so a frozen release cannot re-introduce them.

2026-06-14 (geometry round: Riemann-Roch carrier, Nariai bound, branch line, energy clock — v189–v192)

- **`v189_riemann_roch_carrier.py` [E]/[C] (claim `ARCH.RRCAR.01`).** The pair $(g_{\mathrm{car}}, N_{\mathrm{fam}}) = (5, 3)$ are the two canonical invariants of *one* object, the four-point seam divisor $D = \mu_4$ on $\mathbb{P}^1$: the cycle side rank $H_1(\mathbb{P}^1 \backslash \mu_4) = \deg - 1 = 3 = N_{\mathrm{fam}}$ (already `QGEO.COHOM.01`) and the function side $h^0(\mathbb{P}^1, \mathcal{O}(\mu_4)) = \deg + 1 = 5 = g_{\mathrm{car}}$, matched by $\mathrm{rank}(D_5) = 5$. The carrier-as-mode-space reading is [C] (geometric, does not displace the over-determined $g_{\mathrm{car}} = 5$). Cited in `origin_theory`; Wolfram-mirrored.
- **`v190_nariai_entropy_bound.py` [E] (claim `HOR.NARIAI.02`).** A variation-free one-line proof of the SdS entropy floor: $3(x^2+1) - 2\Phi_3(x) = (x-1)^2 \geq 0$, so $S_{\mathrm{tot}} \geq \frac{2}{3} S_{dS}$ with equality iff $x = 1$ (the Nariai merge). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.
- **`v191_universal_branch_line.py` [C] (claim `HOR.BRANCHLINE.01`).** The affine map $q = \frac{7}{2} + \frac{N_{\mathrm{fam}}^2}{2} m$ carries the SdS branch points $\pm 1/N_{\mathrm{fam}}$ onto the flavor labels $\{2, 5\}$. *Honestly*

typed [C] — an exact *relabeling*, not a theorem (a decoy pair $\{3,4\}$ admits an equally exact map; the shared content is the known 2/3 ramification). Cited in `tfpt_horizon_readouts`; Wolfram-mirrored.

- **`v192_qgeo_energy_clock.py` [O] (claim `QGEO.ENERGY.01`).** An operator-checkable restatement of the bedrock: $\rho^*\Lambda_\Sigma\rho = \Lambda_\Sigma$ (the clock preserves the DtN energy form). *Honestly typed:* it *relocates* `QGEO.SYM.01` to a more falsifiable surface but does *not* eliminate it (plausibly equivalent unless ρ is independently defined and the invariance proven). Cited in `tfpt_research_contracts`. Suite now 191 modules, highest ID `v192` (v186 skipped).
- *All four are validated external-review proposals (`problem_b`); each is built with the honest typing determined on review — the branch line is recorded as an alignment, not a theorem, and the energy clock as a relocation, not a closure.*

2026-06-14 (Zenodo release 5.1 (rev 120) — F_transfer firewall, v184–v188)

- **Published to Zenodo.** Version `5.1 (rev 120)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20687564`, DOI `10.5281/zenodo.20687564`. This version carries the F_transfer-firewall round (v184–v188): the honest η_B anchored-Boltzmann test (sharper scenario, not a closure), the axion hilltop-relic typing, the ledger `v187` and prose `v188` guards, and the Route B/Koide/dark-matter wording fixes plus the P1/P2-vs-`QGEO.SYM.01` two-layer clarification. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Suite 187 modules (highest ID `v188`; v186 skipped). `run_all.py` green, audit clean, website built.

2026-06-14 (F_transfer round: eta_B honest test v184, axion relic v185, discipline guard v187)

- **`v184_etaB_anchored_boltzmann.py` [C] (claim `FR.ETAB.03`).** An honest test of the external-review η_B proposal ($M_1 = M_R\varphi_0^4$, $\tilde{m}_1 = m_3/A_\Lambda$). Firewall verdict: *sharper scenario, not a closure.* The washout anchors plausibly ($\tilde{m}_1 = m_3/10 \approx 5$ meV, $A_\Lambda = 10$ atom, in the strong-washout window) [C]; but M_1 does *not* — M_R is the seesaw scale $\sim v^2/m_3$ and the nearest clean TFPT powers of $\bar{M}_{\rm Pl}$ both miss it (c_3-/φ_0-powers off ∼75%/∼40%), so $M_1 = M_R\varphi_0^4$ merely relocates the free input. η_B stays [C], the route viable but with one (not zero) free input. Cited in `tfpt_4_frontier`.
- **`v185_axion_relic_solver.py` [C] (claim `FR.DM.03`).** Honest misalignment estimate: the determinant-line axion $f_a = M_{\rm scal}/128 \approx 2.39\times10^{11}$ GeV ($m_a \approx 23.8\,\mu$eV) CAN be all dark matter, but the harmonic misalignment under-produces (∼21% at $\theta \sim O(1)$), so the $\theta_i \approx 170°$ hilltop anharmonic enhancement is *required* — where the relic is exponentially sensitive. Hence a [C] scenario, not a sharp prediction; the relic contract is a kill test for the BRANCH, not the theory. Cited in `tfpt_4_frontier`.
- **`v187_ftransfer_laws.py` [E] (claim `FR.TRANSFER.01`).** A CI-style firewall guard: the four continuous-transfer frontier observables (Koide, η_B, axion, m_p/m_e) are read from the ledger and required to carry a [C]/[O] marker — never a primitive [E] compiler prediction (exact sub-parts like the Koide 53/54 factor or $b_3 = -7$ may be [E], the physical prediction never is). Records the four interfaces $F_{\rm pole}/F_{\rm Boltzmann}/F_{\rm relic}/F_{\rm QCD}$; prevents any future edit from quietly promoting a frontier number to a closed output. Cited in `tfpt_4_frontier`.
- **`v188_frontier_wording_guard.py` [E] (claim `AUDIT.WORDING.01`).** A prose sentinel (no new physics): `v187` protects the ledger *typing*, this guard protects the *prose*. It scans the 10 active documents and forbids five stale phrasings whose semantics earlier rounds superseded (the old [E]-typed Route B leptogenesis claim and its heavy-scale wording → v184 [C]; the old "open relic scale" dark-matter line → v185; the old "near, not exact" Koide line → v183; and the module-count/ID conflation), so a frozen Zenodo release can never re-introduce wording

that was true yesterday and is half-true today. The same pass updated the `tfpt_4_frontier` Route B (to [C]), the Koide and dark-matter summary lines, and added the P1/P2-vs-`QGEO.SYM.01` two-layer clarification to the introduction. Suite now has **187 modules**, highest verification ID `v188` (v186 was intentionally skipped as redundant — m_p/m_e is already typed [O] as a QCD matching contract, `QCD.LAMBDA.01`).

- *Review framing points (P1/P2 as projections of the seam-deck postulate, $v_{\rm geo}$ as unit gauge, $G_{\rm net}$ a corollary of `QGEO.SYM.01`, grammar tuning = data) were validated as already implemented in the prior rounds; m_p/m_e is already typed [O] as a QCD matching contract (`QCD.LAMBDA.01`), so no redundant module was added.*

2026-06-14 (operator origin of the Koide 53/54 factor: the missing Sheet-Diamond corner, v183)

- **`v183_koide_f_corner_transfer.py` [E]/[C] (claim `FR.KOIDE.07`).** An external-review proposal, validated exactly: the Koide source→pole factor $\frac{53}{54}$ is not a bare arithmetic guess but an *operator* identity on the carrier, $u_{\rm pole}/u_{\rm source} = a^T(R+Q)\mathbf{1}/(2\,\mathbf{1}^T Ra) = 53/(2{\cdot}27) = \frac{53}{54}$. Here $\mathbf{1}^T Ra = 27$ is the $E_6{\times}A_2$ cubic family block ($248 = \dim E_6 + \det R + 2(\mathbf{1}^T Ra)N_{\rm fam} = 78{+}8{+}2{\cdot}27{\cdot}3$) and $a^T(R+Q)\mathbf{1} = 53 = 52{+}1$, where $F = R{+}Q$ is the *missing* Sheet-Diamond corner with $\det B_F = 52 = \dim F_4$ (v80). Hard negative control: $a^T M\mathbf{1}$ for $M \in \{R, Q, K, L\}$ is $\{32, 21, 35, 56\}$ — only the absent corner F gives 53. This upgrades `FR.KOIDE.03` from “near miss + conjecture” to “operator-sourced source→pole transfer”: the *factor* is [E] (exact, mirrored in Wolfram, 232/232); the *mechanism* (the leptonic source→pole transfer reads the F corner) stays [C], an $F_{\rm transfer}$ special case. The prediction itself stays [C]. Cited in `tfpt_4_frontier` (Koide section). Suite now 183 modules.
- **Review point 2 noted as already-present:** the claim that flavor and Nariai share one double-cover ClockFunctor (flavor rate $(2/3)^{2N_{\rm fam}}$ vs gravity curvature $(2/3)^{N_{\rm fam}}$, exponent ratio $|\mathbb{Z}_2|$, “one clock, two geometries”) is exactly `HOR.TRISECT.01`/v103 — a rediscovery of an existing result, no new module.

2026-06-14 (review recommendations 1/3/4 + the four concerns A–D mapped to the named residuals)

- **`v182_reviewer_residual_map.py` [E]/[C] (claim `REVIEW.MAP.01`).** A careful external review raised four concerns (A mapping $2D \to 4D$, B UV-gravity abstractness, C grammar-tuning/meta-look-elsewhere, D Koide Möbius flow). This module shows three of them *reduce*, by the same discipline as v175–v181, onto the *two already-named* residuals $\{$`QGEO.SYM.01`$, F_{\rm transfer}\}$: A (Plücker-11 = QBL boundary count v174 + holographic reduction), B ($\equiv$ `QGEO.SYM.01`), D (Möbius forced by $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2,\mathbb{C})$ on the deck; the missing generator $= F_{\rm transfer}$). Only C is genuinely *experiment-only* (frozen + minimal + over-determined grammar; residual falls to JUNO/CMB-S4). A unification of the concern *structure*, not a closure. Cited in `tfpt_5_redteam`. Suite now 182 modules.
- **Review recommendation #3 — a “Rosetta” translation lexicon** (website `RosettaLexicon`): TFPT’s compiler-speak (seam, deck, readout, microcode, audit hull, gap, anchor, $F_{\rm transfer}$, $v_{\rm geo}$, `QGEO.SYM.01`) mapped onto standard QFT / mathematical-physics language (Wilsonian boundary CFT, orbifold deck, index/intersection number, RG irrelevance rate, renormalization condition, fundamental postulate) — a reading aid for mainstream physicists, not a new claim.
- **Review recommendation #1 — kill-tests up front** (website `TrustContract`): the frozen pre-data predictions ($\sin^2\theta_{12} = 0.3067$, JUNO; $r \approx 0.004$, CMB-S4) and the null model ($P \leq 10^{-30.7}$) are now a prominent strip directly under the hero.

- **Review recommendation #4 — the bedrock as a postulate.** `QGEO.SYM.01` is now framed confidently as TFPT's *one fundamental postulate* (the role "c = const" plays in relativity), not a gap — in the `tfpt_research_contracts` box and the website "what is open" card; the achievement is that the whole theory is compressed onto exactly one sharply-located, falsifiable premise.

2026-06-14 (external-review response: CSV fix, G_net 3-level, QGEO.SYM.01 page, claim detox)

- **Ledger CSV bug fixed + guarded.** Three rows (`AX.P1.01`, `AX.P2.01`, `QGEO.ISO.01`) had *unquoted commas* in a status/external field (e.g. "a=(1,1,2)", "PSL(2,C)"), so a strict CSV parser read 11–12 columns instead of 10 — a real machine-readability defect in the single source of truth. All offending fields are now quoted (every row parses to exactly 10 columns), and `audit_sync.py` gained a `check_csv_wellformed` guard (ledger, registry, clusters, freeze must all parse to constant width) so it cannot regress.
- **G_{net} typed in three explicit levels** (website "What is open" card): (1) *algebra* [E] $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1,1) \rangle \cong (E_8)_1$; (2) *AQFT machinery* [E] (net existence + full-cone RP, CAR functor, v175); (3) *seam instantiation* [O] (`QGEO.SYM.01`). The target object's mathematics is closed; only the physical coupling of the raw seam is open.
- **`QGEO.SYM.01` now has its own one-page box** in `tfpt_research_contracts`: what is identified, why it is weaker than `CONF/ISO`, what follows automatically (the conditional theorem), and *what would falsify it* (genus > 0 double; non-deck transport; $N_{\mathrm{fam}} \neq 3$); with the explicit conditional framing "*given* `QGEO.SYM.01`, the closure follows" rather than "the theory derives the seam geometry".
- **Claim-language detox.** "no free fundamental number" softened to "no free *dimensionless compiler dial* beyond the anchor and π (dimensionful scale + transfer physics stay typed)" (`origin_theory`, `introduction`, website); the site title softened from "the Standard Model Derived" to "the Standard-Model *Skeleton* Derived".
- **External-reviewer map** added to the website verification page: four boxes (exact kernel [E], conditional physics [C], open premises [O], kill tests [X]) so a reviewer sees the attack surface without reading 181 scripts. Suite unchanged at 181.

2026-06-14 (Zenodo release 5.1 (rev 114) — v175–v181 bedrock + figures + P1/P2)

- **Published to Zenodo.** Version `5.1 (rev 114)` of the ten-PDF document set was published as a new version of the TFPT deposit — record `20686284`, DOI `10.5281/zenodo.20686284`. This version carries the AQFT-closure-to-bedrock chain (v175–v181, the structural residual reduced to the single definitional premise `QGEO.SYM.01`), the two overview figures (`residual_chain`, `script_timeline`), and the P1/P2 anchor-reduction provenance on the axiom surfaces. The `README` and the Zenodo description were refreshed (181 modules), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-14 (overview figures + P1/P2 reduction reflected on the axiom surfaces)

- **Two overview figures** (`make_figures.py`; PDF + website PNG). (i) `residual_chain` — the structural-residual reduction chain v175–v181 as a descending staircase from "build a quantum-gravity measure" to the single definitional bedrock `QGEO.SYM.01`; embedded in `tfpt_research_contracts`. (ii) `script_timeline` — the eight-phase journey of the ∼181 scripts (foundations → SM readouts → seam= horizon → horizon/flavor geometry → R1–R5/premise-(A) → PyR@TE cross-checks → AQFT bridges → AQFT closure), what each did mathematically and physically; embedded in `introduction` ahead of the master script index, and both shown on the website verification page.

- **P1/P2 reduction now reflected on the axiom surfaces.** The reduction of the two axioms to the single parabolic anchor $a = (1, 1, 2)$ plus π (with $g_{\rm car} = 5$ an over-determined bootstrap fixed point — forced three ways, v6, Pascal-unique and Lean-formalised, `FORM.LADDER.01`) was already established (`ANCHOR.GEN.01`/v23, `ARCH.CORE.01`/v53) and stated in the papers/glossary, but the *axiom surfaces* still read as bare "declared inputs". Fixed: the ledger rows `AX.P1.01`/`AX.P2.01` now carry the anchor/bootstrap reduction and point to it, and the website dependency-graph nodes P1/P2 now show the anchor input ($c_3 = 1/(2e_1(a)\pi)$; $g_{\rm car} = e_2(a)$, bootstrap-forced) rather than "declared axiom". No marker change (still declared inputs), only honest provenance.
- Manifests refreshed (two new figures added to the `FIG` list). Suite unchanged at 181.

2026-06-14 (the final reduction to bedrock v181 + full non-changelog status sweep)

- **`v181_clock_is_conformal_symmetry.py` [E]/[O] (claim `QGEO.SYM.01`).** The isometry premise `QGEO.ISO.01` (v180) is weakened one more genuine step — to a conformal-*symmetry* premise — via *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation): a *conformal* automorphism (strictly weaker than an isometry — every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form $z \mapsto iz$. Since the clock is the Coxeter element of $W(A_3) = S_4$ (v117), the μ_4 deck, and a deck transformation preserves the pulled-back conformal structure by definition, the residual collapses to the *definitional* identification `QGEO.SYM.01`: "the seam's conformal structure *is* the μ_4-deck structure of the carrier clock". This is bedrock — a physical/definitional statement tying P2 to the seam's conformal geometry, *not* a finite computation and not reducible further without relabeling. The chain `REALIZE`(v175)$\to \cdots \to$`SYM.01`(v181) ends here; we stop honestly. Python-only. Suite now 181 modules.
- **Full non-changelog status sweep.** The current bedrock framing was propagated to every live status surface (not just the changelog): the `introduction` "latest reductions" prose, `tfpt_5_redteam` Target A, the `tfpt_research_contracts` central-theorem keybox, `origin_theory`; and on the website `OpenGates`, `VerificationDag`, `papers.ts` (Target A $\times 2$, $G_{\rm net}$), the `glossary` $G_{\rm net}$ entry and the `faq` live-residual answer — all now end at `QGEO.SYM.01`.

2026-06-14 (the conformal premise reduces to a milder isometry premise v180)

- **`v180_clock_is_mobius.py` [E]/[O] (claim `QGEO.ISO.01`).** Cracks `QGEO.CONF.01` one level further: the conformal-realisation premise is *replaced*, via three classical theorems, by a strictly *milder, non-circular* premise. (1) *Uniformisation* (Poincaré–Koebe): a genus-0 Riemann surface is conformally $\mathbb{P}$, so the genus-0 seam double (P1, v179) with its RP-metric conformal structure *is* $\mathbb{P}$ — the target is *produced*, not assumed; (2) *Kerékjártó* (1919; Constantin–Kolev 2003): a finite-order orientation-preserving homeomorphism of S^2 is conjugate to a rotation, so the order-4 clock ($h(A_3)$=4) is topologically $z \mapsto iz$; (3) *isometry⇒ conformal⇒Möbius*: an orientation-preserving isometry of a Riemannian surface is holomorphic, $\mathrm{Aut}(\mathbb{P}) = \mathrm{PSL}(2, \mathbb{C})$; (4) an order-4 Möbius map is $z \mapsto iz$ (verified: projective order 4, fixed $\{0, \infty\}$, multiplier i, free orbit μ_4). Hence `QGEO.CONF.01` is *implied* by the milder premise `QGEO.ISO.01`: "the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric" — which does not name $\mathbb{P}/\mu_4$ (uniformisation produces them) and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). That milder premise is the single, now milder, structural residual of the whole theory; its surface realisation from the raw seam stays honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 180 modules.

2026-06-14 (the two obligations collapse to ONE geometric premise v179)

- **`v179_conformal_realisation.py` [E]/[O] (claim `QGEO.CONF.01`).** Investigating the two residual premises left by v178 shows they are *not* independent — they collapse onto a single geometric premise (honest unification; the premise itself stays [O]). MARKS's residual is not three independent assumptions: (1) genus-0 *is* P1 — the one-sided Gauss–Bonnet $c_3 = 1/(|\mathbb{Z}_2|\,2\pi\,\chi(S^2)) = 1/(8\pi)$ uses $\chi(S^2) = 2$ and $\chi = 2 - 2g \Rightarrow g = 0$, so "genus-0" is the same $\chi = 2$ that gives the "8" in c_3 (v58/v73); (2) the order-4 clock *is* the carrier — the Coxeter element of $W(A_3) = S_4$ (v117), order $h(A_3) = 4 = |\mu_4|$; (3) marks = one orbit is *forced* (the parabolic marks are not the two elliptic fixed points, a free order-4 orbit has exactly $4 = e_1(a)$ points). KERNEL then *follows* from the same geometry (DtN symbol $|k|$, Lee–Uhlmann v156; $c = 8$ rigidity v158; the cusped-surface residual $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$). So the *entire* structural residual of the theory is the *single* geometric premise `QGEO.CONF.01`: "the raw RP seam normal double is conformally $(\mathbb{P}, \mu_4)$ with the carrier clock as the order-4 Möbius automorphism". The two doors of v177/v178 are one geometric realisation, left honestly [O]. Cited in `tfpt_research_contracts`; Python-only. Suite now 179 modules.

2026-06-14 (an honest attempt at the two obligations v178 — deeper reduction, not closure)

- **`v178_marks_kernel_attempt.py` [E]/[O] (claim `QGEO.REDUCE.01`).** An honest attempt at `QGEO.MARKS.01` and `QGEO.KERNEL.01`. These are genuine constructive-geometry/AQFT statements, *not* finite computations — neither is closed. What the module does is push each as far as computation allows: it closes the finite [E] core of each and reduces each to *one sharper* premise. *Marks core* [E]: given a genus-0 double with an order-4 clock ρ (fixed points $\{0, \infty\}$) and reflection τ, the marks are forced to be a generic ρ-orbit $\{1, i, -1, -i\} = \mu_4$ (*not* the fixed points), distinct iff $b_1 = 4 - 1 = 3 = N_{\text{fam}}$ (any collision drops b_1), with $\tau\rho\tau = \rho^{-1}$ giving a faithful D_4 of order 8 — so MARKS reduces to the single topological/clock premise. *Kernel core* [E]: on the 3-dim multiplicity-free H^1 the μ_4 characters $(1, 2, 3)$ are distinct, so by Schur a μ_4-equivariant operator is forced *diagonal* and completely determined by its eigenvalue per character; two such with the forced transfer spectrum $\{1, (2/3)^6, (1/3)^6\}$ (v162) and the unique residue-normalised basis (v177) are *equal as operators*, not merely isospectral — the spectrum-vs-operator worry *vanishes* on the finite block. So KERNEL reduces to the single full-operator-reduction premise (principal symbol $|k|$, Lee–Uhlmann/v156; no drift v158). Net: neither obligation closed; both sharpened to milder premises with their finite cores [E]. Cited in `tfpt_research_contracts`; Python-only (Möbius/Schur symbolic + numeric). Suite now 178 modules.

2026-06-14 (QGEO proof tree v177 — the residual sharpened into two named obligations)

- **`v177_seam_marking_kernel.py` [E]/[O] (claims `QGEO.TREE.01`, `QGEO.MARKS.01`, `QGEO.KERNEL.01`).** A second external residual-class audit pointed out that the v176 statement takes the marks/D_4 as a *hypothesis* — a near-circular trap. This module factors the central theorem honestly as REALIZE = MARKS · UNIFORM · COHOM · MODULE · KERNEL · NET and closes *four* nodes *constructively* (all validated symbolically before adoption): UNIFORM [E] — an order-4 Möbius map conjugates to $z \mapsto iz$, a non-fixed orbit scales to μ_4, and $\sigma\rho\sigma = \rho^{-1}$ for $\sigma : z \mapsto 1/z$ gives a faithful D_4 (order 8), so a genus-0 four-marked faithful-D_4 boundary *is* $(\mathbb{P}, \mu_4)$ (strengthening v168's cross-ratio to the full uniformisation); COHOM [E] — $\rho^*\omega_k = i^k\omega_k$, grading $(1, 2, 3) = A_3$ exponents; MODULE [E] — a *unique* μ_4-equivariant iso (multiplicity-free + residue normalisation; reflection $\omega_1 \leftrightarrow \omega_3$, ω_2 fixed); NET [E] (v154/v175). The structural residual is thereby sharpened from one diffuse premise into *exactly two* named open obligations, neither a finite computation: `QGEO.MARKS.01` (the raw RP seam collar canonically produces the genus-0 four-marked D_4 boundary) and `QGEO.KERNEL.01` (the raw seam Calderón operator *is* the μ_4-equivariant free gapped contraction, as an operator). Cited in `tfpt_research_contracts` (the central-theorem keybox); Python-only (Möbius/cohomology symbolic; numbers mirrored via v168/v137/v162/v154/v175). Suite now 177 modules.

2026-06-13 (Seam Collar Realisation Theorem v176 — the one central proof obligation)

- **`v176_seam_collar_realisation.py` [E]/[O] (claim `QGEO.REALIZE.02`).** Following an external residual-class audit, the single remaining structural item is formulated as one central theorem and certified as a *reduction* (not a fabricated proof). *Theorem:* given an oriented one-sided RP seam collar with a sheet-odd Calderón involution, faithful D_4 marks, four parabolic marks with μ_4 decomposition and admissible $c{=}8$ data, its conformal boundary is Möbius-equivalent to $\mathbb{P} \setminus \mu_4$ with H^1 grading $(1, 2, 3)$, the Calderón contraction is the rank-8 K_Σ polarisation and the index-4 extension is $(E_8)_1$. The proof decomposes into six steps, *five already* [E]: geometry/uniformisation (v168), Calderón DtN symbol $|k|$ (v156), H^1 character = generation grading $(1, 2, 3) = A_3$ exponents (v137/v168), μ_4-equivariant contraction with gap $(2/3)^6$ (v162), AQFT closure $(D_5)_1 \otimes (A_3)_1 \rtimes \mu_4 = (E_8)_1$ + full-cone RP all m (v154/v175). The whole structural residual thus reduces to the *one* open premise `QGEO.REALIZE.01`: that the physical seam collar's boundary *is* this $\mathbb{P} \setminus \mu_4$ — a constructive-geometry statement, left honestly [O]. Cited in `tfpt_research_contracts` (the central theorem keybox); assembly/reduction certificate, Python-only (its exact identities are mirrored via v168/v156/v162/v154/v175). Suite now 176 modules. *Audit note:* the metrology residual the same review flags is already discharged — v153 (No-Unit Theorem) is the metrology-line typing ($v_{\text{geo}} \mapsto \lambda^{-1} v_{\text{geo}}$, $1/G \mapsto v_{\text{geo}}^2$, dimensionless data invariant), and the Koide $u \to \frac{53}{54} u$ source→pole transfer is already typed [C] (`FR.KOIDE.03`); neither is re-derived.

2026-06-13 (Zenodo release 5.1 (rev 105) — v174 + v175 + doc refresh)

- **Published to Zenodo.** Version `5.1 (rev 105)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20681451`, DOI `10.5281/zenodo.20681451`. This version carries the AQFT closure round (v170–v175): the seam Fock-space readings and the heavy-artillery discharge of net existence + full-cone reflection positivity to [E]. The `README` and the Zenodo description were refreshed to the current state (175 modules, Wolfram 116/116 + 231/231), and the uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release.

2026-06-13 (the heavy artillery v175: net existence + full-cone RP discharged to [E])

- **`v175_net_existence_full_cone.py` [E]/[O] (claim `QFT.NETRP.01`).** The two structural residuals that the premise-(A) chain had relocated to *already-open* items — A_2 net existence and full-cone reflection positivity — are now *discharged to a theorem*, leaving the seam realisation as the single irreducible open premise (nothing fabricated). (1) *Full-cone RP for all m*: the CAR second-quantisation functor $\Gamma(t) = \bigoplus_m \Lambda^m(t)$ lifts the one-particle seam contraction t to the complete Fock space $\Lambda^\bullet(\mathbb{R}^{16})$ ($\dim 2^{16} = 65536$); by the exterior-power spectrum theorem every eigenvalue is a subset product of $\operatorname{spec}(t)$, so all 2^{16} products lie in $[0, 1]$ (a contraction = full-cone RP), with eigenvalue-1 multiplicity $2^8 = 256$ (wedges of the rank-8 K_Σ polarisation) and sub-leading = the one-particle gap $(2/3)^6$ — so full-cone RP reduces *for every* m (not only $m \le 6$, v161) to the one-particle contraction (Shale–Stinespring), discharging the v160 scope premise. (2) A_2 *assembled + verified*: the $(E_8)_1$ character $E_4/\eta^8 = 1+248q+4124q^2+34752q^3+213126q^4+1057504q^5$ (level-1 = 248, extends the order-4 v156 check by one order), the embedding $248 = 120+128$, and the E_8 Cartan matrix even with $\det 1$ (the unique rank-8 even unimodular lattice, v83); the net *exists* by cited theorems (free-fermion net, lattice VOA, index-4 Q-system, v125). (3) Both then need the *same* input — the seam realisation `QGEO.REALIZE.01` (finite half proven, v168) — which is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly [O]. **Net:** net existence and full-cone RP become [E]; the one irreducible structural premise is the seam realisation. Cited in `tfpt_research_contracts` (premise (A) closure) and `tfpt_5_redteam` (Target A); the exact parts (Cartan unimodular, character order 5, functoriality integers) mirrored in the Wolfram extension (231/231). Suite now 175 modules.

2026-06-13 (seam Fock-space readings v174: CAR cone, Witten index, g-theorem)

- **`v174_seam_fock_readings.py` [E] (claim `QFT.FOCK.01`).** Three more exact audit bridges from the same source note. (1) *CAR cone compression*: the even CAR algebra of the 16 seam Majoranas has $\dim \Lambda^{\text{even}}(\mathbb{R}^{16}) = 2^{15} = 32768$ directions, but a quasi-free Bogoliubov transport reduces the whole cone to a 16-dim one-particle problem ($2^{15}/16 = 2^{11}$) — the explicit count behind v161. (2) *Witten index = the Plücker norm* 11: the four μ_4-corner fermions span a $2^4 = 16 = \dim S^+$ Fock space; QBL (deg ≤ 2) keeps $\sum_{k\le 2}\binom{4}{k} = 11 = \dim S^+ - g_{\text{car}}$, so $c_u/c_d = 55/117$ reads a topological state-space dimension. (3) *Affleck–Ludwig g-theorem*: $c_{UV} = 5+3 = 8 = c_{IR}((E_8)_1)$, so $\Delta c = 0$ forces an exact isometry — the boundary-entropy form of the v157/v158 rigid fixed point. Cited in `tfpt_2` (Exterior Leg Lemma); mirrored in the Wolfram extension (230/230). Suite now 174 modules. (The categorical reframings in the source note remain framing, not asserted claims.)

2026-06-13 (CFT/QFT bridges: slice compression v170, OS moment v171, trace-anomaly seed v172, Pfaffian CP v173)

- **`v170_diamond_slice_compression.py` [E] (claim `E8.SLICE.01`).** E8 Slice Compression Lemma: the seven E_8 audit slices are *one* projection of two alphabets — anchor power sums $P = (3,4,6,10)$ ($p_n = 2+2^n$) and Sheet-Diamond determinants $(3,4,8,14,20,32)$ summing to $81 = N_{\text{fam}}^4$. All seven 248-slices, the K_4 edge graph, and the row-budget cross $(7,11,13)+s(3,3,3)+t(1,2,3)$ follow; $78 = \dim E_6 = p_2(p_0+p_3)$. Cited in `tfpt_3` (slice atlas). Exact; reading is audit (anti-numerology).

- **`v171_os_moment_cluster.py` [E]/[C] (claim `QFT.OSMOMENT.01`).** Atomic OS Moment Lemma: $\text{spec}(T) = \{1, 64/729, 1/729\}$ makes every correlator a positive moment sequence, so the OS Hankel matrices factorise as Vandermonde Grams (a clean RP certificate); cluster 729/665, $\xi = 0.4111$. Sugawara Gap Safety: $31 = 1 + h^\vee(E_8)$, $c((E_8)_1) = 248/31 = 8$, $\Delta_{\text{eff}} = 6\log\frac{3}{2} - 31/(4\pi^2) > 0$. Cited in `tfpt_research_contracts` (G5).

- **`v172_trace_anomaly_seed.py` [E] (claim `SEED.ANOMALY.01`).** The seed prefactor $\frac{4}{3}$ is the halved $(E_8)_1$ Weyl anomaly: over the seam S^2 ($\chi = 2$) the $c = 8$ trace anomaly is $c\chi/6 = \frac{8}{3}$, halved by the one-sided $|\mathbb{Z}_2|$ to $\frac{4}{3} = 16/12$. Plus QCD $b_3 = 7 =$ scalaron exponent $48 - 41$ (confinement ↔ inflation), and the Volkov–Wipf $-98/45 = -2\cdot 7^2/\dim(D_5)$ factorisation. Cited in `tfpt_2` (seed).

- **`v173_pfaffian_cp_car.py` [E] (claim `FLAV.STRONGCP.02`).** Strong CP $\theta_{\text{eff}} = 0$ as Pfaffian reality in the self-dual 16-Majorana CAR model: $\{D, \Sigma\} = 0$ pairs the spectrum, γ_5-Hermiticity makes the Pfaffian real, and RP picks the positive branch. Cited in `tfpt_2` (Strong CP section).

- All four validated computationally before adoption; the exact identities (v170/v171/v172) are mirrored in the Wolfram extension (229/229). Suite now 173 modules. (The categorical reframings in the source note — compiler-as-single-functor, $E_8 = K_0$, GIT/monadic/Langlands — are recorded as framing, not asserted as machine-checked claims.)

2026-06-13 (Zenodo release 5.1 (rev 97))

- **Published to Zenodo.** Version `5.1 (rev 97)` of the ten-PDF document set (introduction, origin_theory, tfpt_1–5, horizon, research_contracts, changelog) was published as a new version of the TFPT deposit — record `20678568`, DOI `10.5281/zenodo.20678568`. The uploader's `DEFAULT_RECORD_ID` was advanced to the new record for the next release. Also fixed `build.sh zenodo` on macOS bash 3.2 (empty-array expansion under `set -u`).

2026-06-13 (QGEO rigidity theorem v168 + η_B leptogenesis interface v169)

- **`v168_qgeo_rigidity.py` [E]/[C] (claims `QGEO.RIGID.01`, `QGEO.REALIZE.01`).** Hardens the *finite* half of the one remaining Tier-1 hinge `GATE.QGEO` (the seam = flavour = horizon identification). Exact, machine-checked: $\mu_4 = \{1, i, -1, -i\}$ has cross-ratio 2 and a faithful Möbius stabiliser of order $8 = |D_4|$; $H^1(\mathbb{P}^1 \setminus \mu_4)$ has rank $4 - 1 = N_{\text{fam}} = 3$; and the forms $\omega_k = z^{k-1}dz/(z^4 - 1)$ $(k = 1, 2, 3)$ are μ_4-eigenforms with characters χ_1, χ_2, χ_3 of weights $(1, 2, 3) = \mathrm{A}_3$ exponents $= \mathrm{Spec}(Q_+)$. So a genus-0 boundary with faithful D_4, four parabolic marks and this μ_4-decomposition is Möbius-equivalent to $\mathbb{P}^1 \backslash \mu_4$ — the QGEO Rigidity Theorem (`QGEO.RIGID.01`, [E]). The seam-collar realisation stays the one premise (`QGEO.REALIZE.01`, [C]). Mirrored in the Wolfram extension (226/226). Cited in `tfpt_research_contracts` (U0). *Not* a new structural class — it hardens the existing hinge (consistent with v167's completeness claim).
- **`v169_etaB_boltzmann_interface.py` [C] (claim `FR.ETAB.02`).** Operationalises the Tier-3 F_{transfer} leptogenesis path. Type-I thermal leptogenesis $\eta_B \sim 0.96 \times 10^{-2} \varepsilon_1 \kappa_f$ (sphaleron 28/79), fed by TFPT's normal-ordered neutrino spectrum and the predicted $\delta_{CP} = 240°$. Over natural $M_1 \in [3 \times 10^9, 3 \times 10^{10}]$ GeV and washout, η_B brackets the observed 6.1×10^{-10} (a canonical point gives 6.0×10^{-10}, untuned) — the route is *viable*. But M_1/washout are scenario inputs, so η_B stays [C]; if a precise solve excluded the window the leptogenesis *route* falls, not the theory (the downstream Ω_b readout is independent). Cited in `tfpt_4_frontier` (Route B). Python-only.

2026-06-13 (Higgs quartic from the free seam: SM near-criticality explained, v166)

- **`v166_higgs_free_seam.py` [E]/[C] (claim `HIGGS.FREESEAM.01`).** A new prediction tying the Higgs mass to premise (A). The seam UV is the free chiral c=8 fixed point (v158, Gaussian, no relevant/marginal interactions), so the one marginal SM scalar coupling vanishes there: $\lambda(M_{\text{seam}}) = 0$ and $\beta_\lambda(M_{\text{seam}}) = 0$ — the Shaposhnikov–Wetterich double-criticality, here *derived* from the free seam, not assumed. Integrating the PyR@TE-confirmed *two-loop* SM RGEs from M_Z up with the measured (m_H, m_t) gives $\lambda(\bar{M}_{\text{Pl}}) \approx 0.002$ and $\beta_\lambda(\bar{M}_{\text{Pl}}) \approx 0$: the famous SM near-criticality is a *consequence* of the free seam (2-loop sharpens $\lambda(\bar{M}_{\text{Pl}})$ from $\sim$0.022 at one loop to $\sim$0.002). The double condition predicts $m_H \sim 129$–134 GeV; the same condition at the scalaron scale gives $\sim$107 GeV (too low), so the boundary condition lives at the reduced Planck scale (seam = horizon = Planck). Fills the Higgs-sector gap (was only $N_\Phi = 1$); cited in `tfpt_4_frontier`.
- PyR@TE-sourced: the reproducer's SM run supplies the two-loop β_λ, reduced to top-only (`verification/pyrate/sm_higgs_betas.txt`). Suite now 166 modules. Python-only (numerical 2-loop ODE).

2026-06-13 (PyR@TE F_{transfer} follow-ups: flavor scale-tagging v163, QCD scale v164)

- **`v163_rg_stability_flavor.py` [E]/[C] (claim `FLAV.RGSTAB.01`).** Scale-tagging of the flavor sector. The lepton-mixing seeds ($\sin^2\theta_{12} = \frac{1}{3} - \frac{\varphi_0}{2} = 0.3067$, $\sin^2\theta_{13} = \varphi_0 e^{-5/6} = 0.0231$) are *RG-stable*: the PMNS running is y_τ^2-suppressed (the only mixing-rotating Yukawa term is $\frac{3}{2}Y_e Y_e^\dagger Y_e$, PyR@TE SM RGEs), so $\kappa_\tau = y_\tau^2/(16\pi^2)\ln(M_{\text{scal}}/M_Z) \sim 1.8\times10^{-5}$ and $|\Delta\sin^2\theta_{ij}| \lesssim 10^{-4}$ — two to three orders below precision. Seam value = measured value; the y_t^2-driven quark mass ratios ($m_t/m_b \sim 41$) by contrast carry a scale tag. Cited in `tfpt_2` (solar/reactor angle box).
- **`v164_qcd_scale.py` [E]/[O] (claim `QCD.LAMBDA.01`).** The QCD confinement scale is parameter-free from the carrier $b_3 = -7$ (v159): running $\alpha_s(M_Z) = 0.1179$ down (two-loop, n_f thresholds) reproduces $\alpha_s(m_b) \simeq 0.22$, $\alpha_s(m_c) \simeq 0.38$ and confinement at ~ 0.5 GeV by dimensional transmutation — so the QCD-scale *numerator* of m_p/m_e is independently sourced. The $O(1)$ proton/Λ factor is lattice, so $m_p/m_e = 1836$ stays the honest "[O] — not forced" frontier ratio. Cited in `tfpt_4_frontier`.
- Both are PyR@TE-backed (the reproducer now also extracts the SM Yukawa β's, `verification/pyrate/sm_yukawa_betas.txt`). Suite now 164 modules. Python-only (no Wolfram mirror: magnitude bound / numerical running).

2026-06-13 (PyR@TE external cross-check of the F_{transfer} gauge inputs — v159)

- **New module `v159_pyrate_gauge_crosscheck.py` [E]/[C] (claim `EM.BUDGET.02`).** The one-loop gauge coefficients used by the transfer functor are confirmed by an *independent* third-party RGE generator. Feeding the carrier / Standard-Model field content into the textbook one-loop formula (GUT normalisation) gives $(b_1, b_2, b_3) = (\frac{41}{10}, -\frac{19}{6}, -7)$, and **PyR@TE 3** (`arXiv:2007.12700`, `github.com/LSartore/pyrate`) writes $\beta_{g_1} = (\frac{41}{10})g_1^3$, $\beta_{g_2} = -\frac{19}{6}g_2^3$, $\beta_{g_3} = -7g_3^3$ verbatim (plus the full standard 2-loop matrix). So $b_1 = 41/10$ is pinned *three* independent ways — carrier algebra $10b_1 = g_{\text{car}}2^{g_{\text{car}}-2}+1 = 41$, the $U(1)$ hypercharge index $\sum\frac{3}{5}Y^2$, and PyR@TE — and the split $10b_1 = 40_{(\text{ferm.})} + 1_{(\text{Higgs})} = 41$ reproduces the EM-budget Gate 1 of `tfpt_2`. Cited there with `\veri`; the gauge sector of F_{transfer} is thus not a free knob.
- **PyR@TE reproducer (not vendored).** `verification/pyrate/reproduce.sh` clones PyR@TE on demand into `experiments/pyrate` (git-ignored), neutralises its interactive `pdflatex` end-command, runs the SM at one and two loops, and re-extracts the gauge β functions; the committed evidence is `verification/pyrate/sm_gauge_betas.txt`. PyR@TE is the right external tool for the transfer/RGE gates (gauge running, and — with a seesaw model — the neutrino-Yukawa running behind η_B); it is *not* a tool for premise (A) (a 2d boundary-CFT question). Mirrored in the Wolfram extension; suite now 159 modules.

2026-06-13 (terminal residual inventory: (A) no longer a structural class; one hinge + F_transfer + irreducible core, v167)

- **New module `v167_inventory_post_closure.py` [E]/[C].** The terminal residual inventory (line v105→v123→v167), re-pinned after the (A)-chain. With (A) discharged (`GATE.METRIC.16-19`) the v123 *five* structural classes collapse: the seam clock (R1), the index-4 net (R2) and the Q-realisation (R5) all merge into *one* Tier-1 hinge — the seam=flavour=horizon identification = A_2 net existence (assembly of established net constructions) + `GATE.QGEO` (seam boundary = $\mathbb{P}\setminus\mu_4$, cohomology b_1=N_{fam}=3); the EH step (R3) folds into v_{geo} (v152) and the ambient measure into G_{net} (v76). **Terminal structure:** Tier 0 = the irreducible core $\{\pi, v_{\text{geo}}\}$ (a theorem, No-Unit v153); Tier 1 = one hinge; Tier 2 = folded; Tier 3 = F_{transfer}, one functor with four typed conditional interfaces (Koide, η_B, axion,

m_p/m_e), η_B carrying the most open computational room. Completeness: a sixth independent structural class — or a new irreducible — fails the script (ledger `SYNTH.INVENTORY.03`). A unification + re-typing, NOT an unconditional proof. Python-only (reads the ledger).

- **Bookkeeping.** Suite at 167 modules, all green; Wolfram extension unchanged at 224/224; `origin_theory` (residual-inventory keybox) and `next.txt` moved together.

2026-06-13 (premise A maximal honest closure: A2 by assembly, R1 = GATE.QGEO, zero independent open gates, v165)

- **New module `v165_premise_a_closure.py` [C]/[O].** Pins the two residual gates of the v160–v162 (A)-chain to their floor. A_2 **(net existence) closes by assembly** of established mathematics [C]: the 16 free Majoranas give a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic c=8 lattice net is $(E_8)_1$ (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the μ_4 extension is a Longo Q-system of index 4 (v125), $\mu(B)$=16/16=1, 248=120+128, character E_4/η^8=$j^{1/3}$ (v156); the one TFPT-specific input (seam Calderón kernel = free-fermion two-point function) is done via DtN = $|k|$ (v156/v157). R_1 **(seam clock) reduces to `GATE.QGEO` [C]**: a μ_4-equivariant flat gapped generator is *unique* (= $A_0^\star$, v115/v116), so the gap $(2/3)^6$ is forced and the residual is the one geometric identification "seam boundary = $\mathbb{P}\setminus\mu_4$" = `GATE.QGEO` (cohomology established, v137, b_1=3=N_{fam}). **Final accounting:** by the No-Unit Theorem (v153) the irreducibles stay $\{\pi, v_{\mathrm{geo}}\}$; premise (A) = ($A_2$ assembly) + (R_1=`GATE.QGEO`) + the $\{\pi, v_{\mathrm{geo}}\}$ core, so it carries *zero independent open gates* and ceases to be its own gate — a unification and re-typing, *not* an unconditional proof (ledger `GATE.METRIC.19`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 165 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the final identification: premise A factors into the already-open A2 + R1, no new open content, v162)

- **New module `v162_seam_transport_identification.py` [E]/[O].** Pushes the v161 residual as far as it rigorously goes and shows it carries *no new open content*. The one-particle symbol splits into a gap value (β) and a fixed space (α). (β) **is forced:** a μ_4-equivariant generator is diagonal in the cusp basis (the deck average is the diagonal projection, $\sum_k U^k X U^{-k} = |\mu_4|\,\mathrm{diag}\,X$ for U=$\mathrm{diag}(1, i, -i)$; the commutant of U is exactly the diagonal algebra), so its spectrum is the cusp weights $\mathrm{spec}(A_0^\star)$=$\{0, \frac{1}{3}, \frac{2}{3}\}$ (v115) and the transfer eigenvalue $(1-\alpha)^{p_2}$ with p_2=6=$|R_+(A_3)|$ gives $\{1, (2/3)^6, (1/3)^6\}$, gap $6\log\frac{3}{2}$ — all carrier data; the lift to the 16 Majoranas is the A_3 subnet grading (v137). (α) is the rank-8 Calderón polarisation of ω_k=$|k|$ (v113/v156). **Closure:** with v160+v161, (A) closes given only μ_4-equivariance + admissibility, whose two non-derived inputs are the already-open A2 (net existence, Kawahigashi–Longo) and R1 (seam clock, `HOR.CLOCK`). So premise (A) factors exactly into $A_2 + R_1$ — it adds *no* independent open item and is closed *conditionally* on them (a unification, not an unconditional proof; ledger `GATE.METRIC.18`). Python-only (numpy/sympy + stdlib).
- **Bookkeeping.** Suite at 162 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (the cone reduces to one particle: premise A's scope premise discharged to a 16-dim statement, v161)

- **New module `v161_one_particle_reduction.py` [E]/[O].** *Discharges* the scope premise of v160 ("the transport is gapped on the *full* Schwinger cone") via Bogoliubov second quantisation. A quasi-free transport is T_Σ=$\Gamma(t)$, the second quantisation of a one-particle contraction t on the 16-dim Majorana space; $\Gamma(t)$ acts on the degree-m correlator sector as the exterior power $\Lambda^m(t)$. Machine-checked: (1) $\mathrm{spec}(\Gamma(t)|_{\deg m}) =$ products of m one-particle eigenvalues (compound-matrix theorem, all m=0..6); (2) **the cone gap equals the one-particle gap** — the sub-leading eigenvalue over the whole cone is $(2/3)^6$ exactly, so premise 5 on the cumulant space *is* premise 5 on the 16-dim space; (3) the eigenvalue-1 sector is Λ^* of the fixed subspace (disconnected Wick powers), with uniqueness closed by v113 (one polarisation ⇔ one vacuum); (4) quasi-freeness is *forced* — the 120$=\dim\mathfrak{so}(16)$ chiral currents are the Bogoliubov generators and v158 makes every non-Gaussian deformation irrelevant. **Consequence:** v160's three full-cone premises collapse to ONE finite (16-dim) one-particle statement (seam symbol = gapped Calderón Bogoliubov map, fixed = rank-8 K_Σ, gap $(2/3)^6$); all numbers supplied, only the physical identification (Kawahigashi–Longo class = the A2 net residual) stays open — the infinite-dimensional cone is eliminated (ledger `GATE.METRIC.17`). Python-only (numpy + stdlib).

- **Bookkeeping.** Suite at 161 modules, all green; Wolfram extension unchanged at 224/224; `tfpt_research_contracts` (premise-A keybox) and `next.txt` moved together.

2026-06-13 (premise A as a conditional fixed-point theorem; the load relocates to one scope premise, v160)

- **New module `v160_seam_gaussianity_from_pf.py` [E]/[O].** Formalises the proposed closure of premise (A) ("the seam bulk is a reflection-positive free field") as a fixed-point theorem and types it honestly. The *exact* parts: (1) a quasi-free state has *all* higher connected cumulants κ_{2n}=0 — verified by the *signed* set-partition/Pfaffian moment↔cumulant inversion on a concrete pure Majorana covariance (the v113 "one kernel = net" property at the cumulant level); (2) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (3) the Perron–Frobenius dichotomy (a *fixed* κ_{2n} is a second fixed ray ⇒ breaks uniqueness; a *non-fixed* one decays by the gap) forces Gaussianity; (4) the qualitative claim needs only gap>0, the value $(2/3)^6$ only sets the rate. **The honest residual:** v56's gapped PF uniqueness lives on the *three-dimensional* scalar transfer spectrum $\{1,(2/3)^6,(1/3)^6\}$, while the 16-Majorana seam already carries $\binom{16}{4}$=1820 and $\binom{16}{6}$=8008 higher-cumulant directions — so (A) reduces to the single internal, exact-testable statement "the admissible TFPT seam transport is primitive + spectrally gapped on the *full* Schwinger cone, not only the readout spectrum of v56", which the suite does *not* yet establish (ledger `GATE.METRIC.16`). Python-only (numpy + stdlib); no new exact-identity content for the Wolfram mirror.

- **Bookkeeping.** Suite at 160 modules, all green; Wolfram extension unchanged at 224/224 (v160 is numerical, Python-only); `tfpt_research_contracts` (the premise-A keybox) and `next.txt` moved together.

2026-06-13 (changelog date-order audit + reorder pass)

- **Changelog sort enforced.** `audit_sync.py` now checks that every dated `\subsection`× `{YYYY-MM-DD...}` block in this file is non-increasing (newest first; range titles such as `2026-06-07 / 2026-06-08` sort by the later date). A manual prepend had left five `2026-06-12` entries above seven `2026-06-13` blocks — reordered; future mis-sorts fail `build.sh audit` as `[B.changelog]`.

2026-06-13 (premise A: the destination fixed point is provably stable, v158)

- **The free chiral c=8 fixed point is provably stable (v158, exact dimension count).** The finite core of (A)'s "reaches-the-fixed-point" residual: the chiral Majorana has $h=\frac{1}{2}$, the bilinear current h=1, the quartic h=2. The relevant window $0 < h < 1$ for bosonic deformations is *empty*, the h=1 currents are chiral (v157, not $(1,1)$-marginal), and the lowest interaction (quartic, h=2) is irrelevant — so the fixed point is isolated and stable against all interactions. Premise (A) reduces to a basin-of-attraction statement with a *proven* stable target (ledger `GATE.METRIC.15`). *Tooling note:* PyR@TE (4d gauge-Yukawa RGEs) is the wrong tool for this 2d boundary-CFT residual, but the right tool for the F_{transfer} interfaces (η_B leptogenesis RGEs, gauge-coupling running, the b_1=41/10 cross-check).

- **Bookkeeping.** Suite at 158 modules, all green; Wolfram extension 224/224; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (premise A reframed: freeness is a rigid boundary fixed point, v157)

- **The free-bulk premise becomes a fixed-point property (v157, simplicity-first).** Instead of postulating Gaussianity, three facts click: (i) the DtN/Calderón symbol is *universally* $|k|$ (Lee–Uhlmann — order-1 ΨDO, principal symbol $|\xi|_g$; interactions live in the irrelevant lower-order symbol), so the free dispersion is bulk-detail-independent; (ii) a holomorphic c=8 theory is *rigid* — every field has $\bar{h}$=0, so no marginal $(1,1)$ field exists and the fixed point is isolated, with no knob for an interaction; (iii) the same $|\mathbb{Z}_2|$ that halves 8π in $c_3 = \frac{1}{8\pi}$ is the Ramond/GSO projection giving μ=1 (248=120_{NS}+128_{R}). With $(E_8)_1$ unique (v83), premise (A) shrinks from "prove the seam is Gaussian" to "the gapped (Δ=$6\log\frac{3}{2}$) boundary flow reaches the holomorphic fixed point" — freeness then forced by rigidity, not fine-tuned (ledger `GATE.METRIC.14`).
- **Bookkeeping.** Suite at 157 modules, all green; Wolfram extension 223/223; contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (Route 3: the seam net constructed and $(E_8)_1$ verified, v156)

- **The constructive route carried to its exact endpoint (v156).** "$B \cong (E_8)_1$" is no longer an assertion: the chiral seam net is built explicitly and the identification *checked*. (a) The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian is the free chiral dispersion $\omega_k = |k|$ (harmonic extension $e^{ikx-|k|y}$); (b) 16 free Majoranas give $c = 8 = 5{+}3$; (c) the weight-1 currents are $248 = 120_{\text{NS}} + 128_{\text{R}}$ ($SO(16)$ bilinears + Ramond spinor, Frenkel–Kac); (d) the holomorphic character $E_4/\eta^8 = q^{-1/3}(1 + 248q + 4124q^2 + 34752q^3 + 213126q^4) = j^{1/3}$ (verified by q-series) — the 248 at level 1 is $\dim E_8$, so the net *is* $(E_8)_1$. *Residual:* exact given (i) the free-bulk premise (DtN = $|k|$, v155) and (ii) the operator-algebraic existence of the net from the seam kernel (a Kawahigashi–Longo-type theorem, not a finite computation) — where TFPT meets constructive QFT, now sharply located (ledger `GATE.METRIC.13`).
- **Bookkeeping.** Suite at 156 modules, all green; Wolfram extension 222/222 (the DtN = $|k|$ identity, the $(E_8)_1$ character $E_4/\eta^8 = j^{1/3}$ and the 248=120+128 split mirrored); contracts / redteam (Target A) / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (the seam-side identification reduced to one shared premise, v155)

- **G_{net}'s last premise reduces to a single shared physical premise (v155, Python-only).** The seam-side identification of the Simple-Current Extension Theorem (v154) splits (v110) into (a) "the Calderón involution is sheet-odd" (the double-cover deck, structural) and (b) "the seam 2-point kernel is quasi-free". Part (b) is *not* independent: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression $P\Gamma P$ of the covariance, and the compression of a contraction is a contraction — so the seam state is automatically quasi-free (Wick inherited). Hence (b) *follows* from "the seam bulk is a reflection-positive free field", the *same* premise already load-bearing in the area law (v59) and the EH mechanism (v150/v151). **The entire seam-side residual is therefore one physical premise**, shared by G_{net}, the area law and R3 — not reducible by a finite computation, but no longer three separate items (ledger `GATE.METRIC.12`).

- **Bookkeeping.** Suite at 155 modules, all green; Wolfram unchanged at 221/221 (v155 is numerical/numpy, Python-only); contracts / `origin_theory` / `next.txt` / website moved together.

2026-06-13 (external review formalised: No-Unit Theorem + Simple-Current Extension Theorem, v153–v154)

- **v_{geo} re-typed: the No-Unit Theorem (v153).** Every load-bearing compiler datum is mass-dimension 0 and invariant under a unit rescaling $L \mapsto \lambda L$, whereas a mass / $1/G$ is not — so a dimensionless boundary compiler *provably cannot* select an absolute scale. v_{geo} is therefore an *irreducible metrology primitive*, not an open gap, and the three apparent residuals collapse onto it ($U_{\text{point}} \sim v_{\text{geo}}$, $1/G \sim v_{\text{geo}}^2$, $m/\mu = e^{3/4}$ a pure ratio); the irreducibles stay $\{$one unit, $\pi\}$ (ledger `ANCHOR.VGEO.02`).
- **G_{net} stated as the central closing theorem (v154).** The Simple-Current Extension Theorem: $A{=}(D_5)_1{\otimes}(A_3)_1$ extended by the isotropic glue $L{=}\langle(1,1)\rangle$ has $[B{:}A]{=}|L|{=}4$, $c{=}5{+}3{=}8$, $\mu(B){=}\mu(A)/|L|^2{=}16/16{=}1 \Rightarrow$ holomorphic $\Rightarrow B \cong (E_8)_1$ ($SO(16)_1$ excluded). Every algebraic ingredient is machine-checked (v89/v92/v125/v143/v148); the only residue is the seam-side identification. Pulled to the front of `tfpt_research_contracts`, `tfpt_5_redteam` (Target A final form) and `origin_theory` (ledger `GATE.METRIC.11`).
- **Status surfaces re-typed** (review): the canonical status card now reads v_{geo} as a metrology *primitive*, G_{net} as the simple-current extension $\cong (E_8)_1$ ([E]/[C]), and F_{transfer} as one functor with four typed interfaces — "not structural gaps". The final one-line framing (dimensionless boundary compiler; irreducibles $\{\pi, v_{\text{geo}}\}$; one seam-net theorem; frontier $= F_{\text{transfer}}$) is in `origin_theory`, the introduction, the README and `next.txt`.
- **Bookkeeping.** Suite at 154 modules, all green; Wolfram extension 221/221; intro/README/`origin_theory`/contracts/redteam/status-card/website moved together.

2026-06-13 (currency sweep: remaining old-series "Paper N" cross-references removed)

- **No old-version framing left in the active docs.** A consistency sweep found a residual layer of old-series "Paper N" attributions and removed them: four literal "Paper 6" references (`tfpt_4_frontier` leptogenesis/axion, `tfpt_2` seam determinant), the old "Paper 4" (OS/admissibility) and "Paper 5" (APS/Hodge) references in `tfpt_2`, the old "Paper 3" transport-kernel attributions in `tfpt_2`, and the internal "Paper 1/2/4" cross-references in `tfpt_3`, `tfpt_4` and `tfpt_horizon_readouts` — all rewritten as content-based phrasing (the flavor sector, the architecture document, the E_8 decoder, the anchor characteristic polynomial, etc.). The active documents now attribute results to the current documents / axioms / scripts only, as the workflow rule requires; the machine-checked sync (scripts ↔ papers ↔ ledger ↔ website) was already current (152 scripts, AUDIT OK).

2026-06-13 (fifth round: the R3 normalisation is the dimensionful anchor, v152)

- **R3: the $q(A_3)$ normalisation merges into the one anchor (v152).** The replica EH coefficient is a *log-ratio* $k = \ln(m/\mu)/(12\pi)$ ($\ln m$ alone is scale-ambiguous, only m/μ is physical); pinning $k = c_3/2$ fixes the dimensionless ratio $m/\mu = e^{3/4}$, and in $d{=}2$ that ratio *is* the induced Newton constant $1/G$ (v68). So v_{geo} (v78), $1/G$ (v68) and m/μ are *one* dimensionful anchor in three readings — the irreducibles stay $\{$one scale, $\pi\}$, and `SEAM.THEOREM.01`'s residual reduces to the kernel-identification premise alone. *Audit (not promoted):* the log-ratio is numerically $\frac{3}{4} = q(A_3)$, the target value the anchor takes in seam units — not a derivation (ledger `SEAM.EHMODEL.03`).

- **Bookkeeping.** Suite at 152 modules, all green; Wolfram extension 219/219; intro/README/`origin_theory`/`next.txt`/website moved together.

2026-06-12 (Zenodo version label: semantic + git rev)

- **Release version string.** `build.sh` now writes `tex-artefacts/release-version.txt` as `{TFPTversion} (rev {git count})` (matches `\TFPTdatestamp` without the date). `scripts/zenodo_upload.py` and `build.sh zenodo/zenodo-publish` use this label on Zenodo; cursor rules extended for when `zenodo_description.html` must move with deposit-facing changes.

2026-06-12 (Zenodo API upload workflow)

- **Automated Zenodo versioning.** Added `scripts/zenodo_upload.py` and `.github/workflows/zenodo-release.yml`: creates a new draft version of record 20579275 (DOI `10.5281/zenodo.20579275`), uploads the ten active paper PDFs, refreshes `zenodo_description.html`, bumps the version from `tex-artefacts/version.tex`, and optionally publishes via GitHub Actions (`ZENODO_TOKEN` secret).

2026-06-12 (Overleaf shadow mirror: tfpt5-shadow)

- **One-way GitHub mirror for Overleaf sync.** Added `scripts/export-shadow.sh` and `.github/workflows/shadow-sync.yml`: on every push to `main`, a filtered copy is pushed to `sthamann/tfpt5-shadow` (papers, `tex-artefacts/`, `figures/`, `verification/`, `experiments/`; excludes `_archive/`, `website/`, all `*.pdf`, `.cursor/`). ∼270 files / ∼3 MB — within Overleaf GitHub-sync limits.

2026-06-12 (origin_theory style alignment + box reduction)

- **Fewer boxes, embedded in text.** `origin_theory` from 22 to 16 boxes: the narrative/interpretation callouts ("Why the interpretation is internally consistent", "Dependency chain of the birefringence test", "The precise reformulation", "The three theses honestly graded", "The whole story", "One-line summary") become run-in bold prose; the boxed reformulation becomes a `keyeq`. The exact result keyboxes (v53–v56, v101, v105, v113, ...) and the master-principle / Seam–Horizon target boxes are kept.
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references; stale literal `[A]`/`[I]`/`[L]` markers in running text replaced by the 4-class **[O]**/**[E]**.

2026-06-12 (tfpt_5 / horizon / research-contracts pass + build hardening)

- **Forked document-set box removed (horizon).** `tfpt_horizon_readouts` carried a handwritten, *stale* copy of the document-set card ("five short documents", missing `tfpt_5`) *and* the canonical `\input{tex-artefacts/tfpt_docset}`; the fork is deleted so the single-source card (all five papers + three companions) appears once.
- **Fewer boxes, embedded in text.** Narrative/result callouts converted to run-in bold prose (with the `\veri{}` citation moved out of the bold title into the body): `tfpt_horizon_readouts` from 22 to 7 boxes (the ten-box "clock" wall becomes prose); `tfpt_research_contracts` from 19 to 10 (the nine progress/result winboxes become prose, the named-theorem keyboxes kept); `tfpt_5_redteam` from 10 to 9 (the outcome summary; the per-target verdict boxes are kept as the structured red-team output).
- **Colour-marked module references + marker hygiene.** `\vref` applied to the inline `vN` references in all three documents; the horizon "Scope/typing" marker line and a stale literal `[A]` in the research contracts updated to the 4-class markers.

- **Build hardening.** `build.sh` now removes a document's `.aux/.toc/.out` *on failure*, so a half-written cross-reference file from a crashed compile cannot poison the next build; artefacts are still kept on success.

2026-06-12 (fourth round: the Calderón transfer answered — the BFK split, v151)

- **R3: the Calderón/DtN kernel is conically clean (v151).** Reflection doubling derives that the Kac corner constant is boundary-condition independent ($C_N = C_D$), so by the Burghelea–Friedlander–Kappeler gluing ledger the conical deficit of the two-sided Calderón jump determinant *vanishes identically* ($C_{\text{cone}}(\gamma) - 2C_D(\gamma/2) = 0$): the seam-reduced effective action inherits the v150 EH form *verbatim*, with all of it localised in the *local* half-space determinants — the "Calderón version" demanded by v150 needs no separate derivation. `SEAM.THEOREM.01` narrows to the $q(A_3)$ normalisation plus the standing kernel premise; the gate stays [O] (ledger `SEAM.EHMODEL.02`).
- **Bookkeeping.** Suite at 151 modules, all green; Wolfram extension 218/218; intro/README/`origin_theory/next.txt` moved together.

2026-06-12 (third round: both normals anchor data; the R3 mechanism exists, v149–v150)

- **R4′: both dual normals are anchor data (v149).** In the cusp frame the torsion normal pairs to $(6, 3, 5) = (p_2, p_0, e_2)(a)$ — one anchor invariant per Q_+ line (the cusp matrix is unimodular, so this determines n uniquely); with $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ the dual normal pair is anchor data through and through. Equivalences to the $(2, 8, 121)$ frame data and the w_0-lift exact. R4′ residue = *one* discrete assignment (ledger `GATE.UWALL.14`).
- **R3: the mechanism exists at the gapped-model level (v150) — the first movement on `SEAM.THEOREM.01`.** The conical heat-trace deficit is exact and t-independent (Cheeger image sum $(N^2-1)/(12N)$); for a *gapped* operator the ζ-regularised replica variation is *finite and cutoff-independent*, $\Delta \log \det' = 2C(\gamma) \ln m$, and its linearisation is the Einstein–Hilbert form $(\ln m/12\pi) \int \sqrt{g}R$. The v73 normalisation becomes one exact target equation: $k = c_3/2 \Leftrightarrow \ln m = \frac{3}{4} = q(A_3)$ (audit, not asserted). Remaining: the Calderón (boundary) version and that normalisation — the gate narrows but stays [O] (ledger `SEAM.EHMODEL.01`).
- **Bookkeeping.** Suite at 150 modules, all green; Wolfram extension 217/217; status surfaces (intro reductions + card, `origin_theory`, contracts, `next.txt`, website) moved together.

2026-06-12 (R1–R5 second round: every class at its floor, v145–v148)

- **R4′: the pairing values are atom identities (v145).** Reading the v139 lift structurally ($n = w_0(\sigma) + |\mu_4| e_3$, w_0 = the A_3 exponent duality), all three pairings become atom arithmetic — including the *new* closure $|\mu_4| + g_{\text{car}} = N_{\text{fam}}^2$ (4+5=9, $\sigma \perp w_0(a)$) and $\|\sigma\|^2 = 110 = |\mathbb{Z}_2| g_{\text{car}} \|\text{Pl}(K)\|_1$. R4′ = derive *one* map (ledger `GATE.UWALL.13`).
- **R5: the Möbius D_4 realisation (v146).** The *full* dihedral symmetry $\langle z \mapsto iz,\ z \mapsto 1/z \rangle$ of the seam curve acts on H^1 exactly as the integer model: ι^* is the exponent duality with parity $+1$ on the self-conjugate line (= T_A in the cusp basis, glue-swap parity), and Σ is the $\delta\iota$ class — the premise reduces to the module identification itself (ledger `FLAV.QGEO.07`).
- **R1: the clock is a Gaussian zero-mode integral, and the bend is det′-clean (v147).** Variance ratio $(1 - \alpha)$ (forced by norm = area) gives the Born-squared ratio $(1 - \alpha)^{p_2}$, the ln series *is* the v127 ring sum; the two quantum weights share *one* Nariai geometry, so the bend $\log_{3/2} 3$ carries no determinant correction — the v144 obstruction is localised to the $\alpha = 0$ reference (ledger `HOR.CLOCK.13`).

- **R2: the NS/R sector census (v148).** The untwisted (Neveu–Schwarz) module carries only the *even* discriminant classes (integer weights), so the odd μ_4 simple currents have zero NS support — they are Ramond modules; the R zero-mode module (256) splits 128+128 into the odd sectors of the *two* Lagrangian glues: choosing the glue *is* choosing the R-projection ($248 = 120_{\mathrm{NS}}+128_{\mathrm{R}}$); the one remaining net statement is irreducibly twisted-sector (ledger `GATE.METRIC.10`). *Correction note (same day):* the first version of v148 graded the Ramond labels by the sum q_D+q_A (not the glue label) and called that module untwisted; the census was rewritten as above — conclusion unchanged, support corrected, the R-sector sheet split is new.

- **Bookkeeping.** Suite at 148 modules, all green; Wolfram extension 215/215 (README + `GATE.WOLFRAM.02` + website counter in lock-step); status surfaces (intro card, `tfpt_1`, `origin_theory` third update, `next.txt`) moved together.

2026-06-12 (external review integrated: Λ typing, marker migration, website parity)

- **Λ normalisation made unambiguous.** The external review flagged the adjacent reduced/unreduced displays in `origin_theory` as a type trap (the formulas were individually correct — $\bar{M}_{\mathrm{Pl}}$ vs M_{Pl} — but easy to misread). Both normalisations are now shown explicitly side by side with their order counts ($3/(4\pi^2) \Rightarrow 120.147$ reduced; $3/(256\pi^4) \Rightarrow 122.948 = 119.028 + 3.920$ unreduced) plus a typing note.

- **Stale references cleaned.** "How each of the seven papers simplifies" → "How the legacy layers simplify (bridge map)" with the lead-in "(Paper 6)" reference replaced by the scalaron-reheating chain (v86); "(Paper~6/7)" in the `tfpt_3` bridge table reworded; the `tfpt_1` "single remaining geometric input (H2)" passage and the `tfpt_2` "single remaining input (U)" passage carry dated status updates pointing at the current decomposition (v118–v122, v139, v141, v142, v75).

- **Marker migration completed in the shared figures.** The claim stack, anchor cockpit, E_8 glue, 2/3 motif and K_5 figures (`tex-artefacts/tfpt_figures.tex`) now show the four public classes [**E**]/[**C**]/[**O**] instead of the legacy [A]/[I]/[L]/[B]/[P]/[R] badges.

- **Red-team front table pulled to the final state.** The Target-A residual column now states the one remaining statement (boundary-net holomorphy + c=8 ⇔ index-4 inclusion; v83/v87/v89, Lie-level realisation v143); the historical three-residual form is kept below, dated.

- **Website parity.** The Red Team document is now a first-class entry (`papers.ts` number 5, slug `redteam`) and the changelog PDF a download row; Appendix H / Origin Theory / Research Contracts are labelled *companions* (never "Paper 5–7"); the marker system on every website surface migrated to [E]/[C]/[O]/[X] (fine types named in prose; the script index keeps quoting the scripts' internal fine-grained typing); the verification page counters are live (144 checks generated from the registry via `lib/suite.ts`, Wolfram 116/116 + 211/211 — now guarded by the sync audit) and the reproduce block matches the current pipeline. The `papers.ts` section bodies of the affected papers carry the v141–v144 outcomes (sheet question: deck derived; G_{net}: Lie-level realisation; resummed clock: in-family det-ratio cancellation; selector triangle: two line pairings).

- **Not adopted** (recorded for transparency): the review's narrative reframings (anchor-algebra trilogy, "one seam clock" framing, one-page structure) and the front-matter/status-card layout changes are editorial decisions left to the authors; the claimed Λ *error* was in fact a correct-but-treacherous notation switch (see above).

2026-06-12 (R1–R5 attack round: four residual classes sharpened, v141–v144)

- **R5 closed at the discrete level (v141, deck selection theorem).** The v140 $\mathbb{Z}_3$ choice is *derived*: the established integer deck $G = T_A\Sigma$ (v97/v98) pairs the $Q_+{=}1$ cusp line with the self-conjugate character 2, so only the sheet-twisted assignment $(2, 3, 1) = \mathrm{chars}(-U)$ survives; the cusp exponential i^{Q_+} is excluded (same spectrum, wrong grading pairing). Explicit conjugacy constructed; under $k \to -k$ only the established sheet $\mathbb{Z}_2$ flips. Consequence: Euler grading $= Q_+ \circ \rho$. `GATE.QGEO` keeps only its realisation premise — no discrete freedom left (ledger `FLAV.QGEO.06`).
- **R4′ narrowed: three pairings → two (v142, frame integrality).** Integer covectors in the $(\mathbf{1}, a, \sigma)$ dual frame form an index-11 sublattice ($x-3y+z \equiv 0 \bmod 11$; the index *is* $\|\mathrm{Pl}(K)\|_1$); with $(x, y) = (|\mathbb{Z}_2|, \mathrm{rank}\, E_8)$ the σ-pairing is forced $\equiv 0 \bmod 11$, and primitivity forces the line parameter even. Cramer reductions for both line pairings recorded as restructuring (ledger `GATE.UWALL.12`).
- **R2 realised at the finite Lie level (v143, graded Frobenius).** The $\mathbb{Z}_4$-graded hull carries exact coset duality ($-C_1 = C_3$; Killing pairing nondegenerate per $g_k \times g_{-k}$), the glue average is exactly the carrier projector (index 4), and each glue sector is one Weyl orbit (simple currents, fusion $= \mathbb{C}[\mathbb{Z}_4]$) — the v125 Q-system realised by the hull; residue = one conformal-net statement (ledger `GATE.METRIC.09`).
- **R1's det-ratio step derived in-family (v144).** e_2-rigidity of the traceless SdS cubic gives $r_b r_c = 1 - \Delta^2/3$ exactly; with the v132 anomaly the non-zero-mode determinant ratio is $(1 - \Delta^2/3)^{4/3}$ — no first-order term in the horizon split (the v131 cancellation derived, not assumed); deficit coefficient $\frac{32}{27} = |\mu_4|(\frac{2}{3})^3$ (audit). Finite-weight absorption stays [C] with the $6 \to \frac{14}{3}$ obstruction stated sharply (ledger `HOR.CLOCK.12`).
- **R3 attempted, honestly unchanged.** No defensible computation landed for the seam-determinant→EH step (`SEAM.THEOREM.01`); the gate keeps its [O] typing untouched.
- **Wolfram mirror extended** to v141–v144: extension file now 211/211 (`GATE.WOLFRAM.02` and the wolfram README updated in lock-step); Python suite at 144 modules, all green.

2026-06-12 (sync infrastructure: single-source script index, content maps, one audit)

- **Single-source script index.** The master script→check table (`tex-artefacts/verification.tex`) and the website `ScriptIndex.tsx` are now *generated* from one registry (`script_registry.csv` + `script_clusters.csv` in `verification/`) by `make_script_index.py`; both targets carry a generated-file header and are never edited by hand.
- **Content maps without splitting the papers.** `verification/make_docs_map.py` generates `docs_map.csv` (every paper section with line range, the `vN` scripts cited there, a content hash and a last-changed date) and `website_map.csv` (which website file mentions which scripts/documents) — the machine-readable enumeration of all sync surfaces.
- **One sync audit.** `verification/audit_sync.py` bundles and extends the previous loose shell checks, now in *both* directions: suite ↔ `run_all.py` ↔ registry ↔ ledger; every script cited in a paper *body* (the master index alone no longer counts); no stale `\veri`/`\vref` targets; every new module mentioned in this changelog; website mirror byte-identical (PDFs, scripts, `release.ts` hashes, version stamps); wolfram counts; overfull boxes. Frozen exceptions live in `audit_baseline.json` (remove-only).
- **Build pipeline.** `build.sh` gains `gen` / `website` / `audit` / `release` steps; `notes` now regenerates the single-source surfaces first; `website` copies all active PDFs (now incl. `tfpt_5_redteam.pdf` and `changelog.pdf`, with new `release.ts` entries) and the `vN` scripts,

and stamps version, date and git revision into `website/lib/version.ts` (shown in the site header) and `release.ts`.

- **Citation gaps closed.** The audit immediately found three scripts cited only in the master index: v17 and v85 are now cited in their `tfpt_2` sections (hexagonal resolvent; master cover), and two short `\veri{v19}` citations were expanded to the full script name. v91 (spine tetrahedron) had no paper section at all — it is now integrated: a keybox in `tfpt_3` ("Three views of one spine": anchor closure, edge/face products, volume 120, K_6 negative control, the tautology/sub-grammar caveats) and a closing note in the `tfpt_1` Spine Quotient Lemma; the `ARCH.SPINE.01` ledger location is corrected to `tfpt_1;tfpt_3` and the baseline exception list is empty again.
- **Rules.** `.cursor/rules` updated (`tfpt-workflow`, `website-sync`) and a new `sync-maps` rule added describing the agentic procedure: *regenerate* the maps first, enumerate the affected surfaces from them, audit as the exit gate. The mirror map (`website_map.csv`) also covers the root `README.md` and `next.txt` (bare `vN` ids resolved to full script names), and both files are named explicitly as status surfaces that move with every finding.

2026-06-12 (tfpt_3 / tfpt_4 style alignment + box reduction)

- **Shared header/footer on tfpt_3.** `tfpt_3_e8_audit_bootstrap` dropped its custom `\footmarkers` footer (the post-collapse four-[E] artefact) and inherits the shared `tfpt_style` header/footer with the version stamp; the box-orphan `needspace` guards are kept.
- **Fewer boxes, embedded in text.** Narrative/commentary callouts converted to run-in bold paragraphs (boxed one-line formulas to `keyeq`): tfpt_3 from 39 to 27 boxes (e.g. "Purpose and discipline", the five "(i)–(v)" audit-lemma boxes, "Net", "The connection in one sentence", "The unification", "The question and the honest answer", "Net: two axioms"); tfpt_4 from 18 to 7 (the seven-box Koide stack and the two dark-matter conjecture boxes become prose). The per-item "Honest status of . . . " keyboxes are kept — tfpt_4 is the status authority for the frontier.
- **Colour-marked module references.** `\vref` applied to the inline `vN` references in tfpt_3 and tfpt_4 (TikZ node names excluded).
- **Marker hygiene.** A stale literal `[P]` in tfpt_4 (Koide flow-time) replaced by the 4-class [C]; status tables checked against the ledger.

2026-06-12 (tfpt_1 / tfpt_2 style alignment + box reduction)

- **Shared header/footer everywhere.** `tfpt_1_architecture_e8` dropped its custom `\footmarkers` footer (which after the marker collapse showed four redundant [E]) and now inherits the shared `tfpt_style` header/footer with the version stamp, identical to every other document.
- **Fewer boxes, embedded in text.** The narrative/commentary callouts of both documents are converted to run-in bold paragraphs (and the boxed one-line formulas to the colour-marked `keyeq`): tfpt_1 from 21 to 14 loud boxes (+3 `keyeq`); tfpt_2 from 52 to 43 (e.g. "In one sentence", "The one-paragraph version", the four "Sharpening" notes, "What the simplification means net", "The disciplined main statement", "What improves. . . ", "Scope and honesty"). Only genuine results / theorems / gates keep a box.
- **Colour-marked module references.** `\vref` now also colours the inline `vN` references throughout tfpt_1 and tfpt_2 (TikZ node names accidentally matching the pattern were excluded).
- **Status currency.** tfpt_2's inflation-pivot table gains the v86 scalaron-reheating point ($N_\star$=51.4 ⇒ n_s=0.9611, r=0.0045, A_s-disfavoured) and states the band-of-record explicitly,

matching `COSMO.NSTAR.01` and the introduction; the box is re-typed [C]. The other status tables were checked against the ledger and are current under the 4-class markers.

2026-06-12 (marker simplification to 4 classes + status reconciliation)

- **Four-class display markers.** The reader-facing marker system is collapsed from eleven symbols to four: [E] exact / machine-proven (identity, Lie/lattice, formalised, numerical FP), [C] conditional (physical / bridge / readout under named hypotheses), [O] open / axiom, and [X] kill test. `tex-artefacts/tfpt_style.tex` defines `\mE/\mC/\mO/\mX` and keeps the legacy `\tI ...\tX` names as aliases that render their class. All marker call-sites across the nine documents and the shared fragments were rewritten, collapsing same-class combinations (e.g. `[I]/[L]` → [E], `[N]/[P]` → [E]/[C]). The fine-grained per-claim type (Axiom / Formal / Lattice / Numerical / Identity / Physical) is unchanged in `status_ledger.csv` — the single source of truth — so no fidelity is lost. The marker key, badge bar, README marker table and the workflow rule were updated to match.
- **Status tables reconciled against the ledger.** The introduction's inflation/$N_\star$ rows are made consistent with v86/`COSMO.NSTAR.01`: r and n_s are shown as the frozen *bands* over $N_\star \in [50, 60]$ with the scalaron-reheating conditional point $N_\star$=51.4 (n_s=0.9611, r=0.0045), replacing the stale point values $N_\star$=55 (predictions table) and $N_\star \simeq 57$ (residual card); the closure-ledger inflation row is re-typed [E] → [C](conditional on M_{Star}=M_{scal} + reheating), matching the other two tables and the ledger `[P]` typing.

2026-06-12 (version system, predictions/K5, readability pass B3)

- **Authors, auto-date and auto-version on every paper.** New single source `tex-artefacts/version.tex` defines `\TFPTauthors` (Stefan Hamann, Alessandro Rizzo), a curated `\TFPTversion`, and a title/footer date stamp (`\today` + version + revision). `build.sh` writes the auto build revision (git commit count) into the gitignored `tex-artefacts/version-auto.tex`, so every document's title page and page footer carry author, date and `v5.1 (rev N)` automatically. All nine active documents and `changelog.tex` now set `\author{\TFPTauthors}`.
- **Builder keeps build artefacts.** `build.sh` no longer deletes the `.aux/.log/.out/.toc` after a successful build (the `.log` is needed for the overfull-box audit and the `.aux/.toc` for correct cross-references on the next incremental build).
- **Predictions table outsourced.** The running/upcoming-experiment predictions table is moved from `introduction.tex` into the shared fragment `tex-artefacts/predictions.tex` (`\input` in the predictions section).
- **K5 figure fixed.** The `\TFPTactionkfive` mnemonic (K_5 on the five carrier slots) is redrawn so the complete graph and the Pascal-row reading list sit side by side instead of overlapping.
- **Colour-marked script references.** New `\vref` macro colours inline module references (e.g. v108–v113, v49/v71) in the introduction, predictions and verification fragments, matching the blue of the `\veri` citation, so machine-checked references are visible at a glance.
- **Lighter, non-breaking, embedded boxes.** The `tcolorbox` families are restyled to a light tint + thin frame + light title band with dark coloured title (no heavy white-on-saturated bars); a neutral `navbox` carries the document map and a colour-marked `keyeq` sets off load-bearing display formulas. The introduction's callouts use the non-breaking variants (`keyboxnb/winboxnb/gapboxnb`) so no box splits across a page, and the shared status card is non-breaking.
- **Readability / sectioning.** The dense run-in prose of the introduction is broken into `\subsection*` headings (In plain words; one anchor and E_8 as a compiler hull; the companion

documents; the compact status formula; the hard reduction; the Pascal action grammar; the closure ledger) with added whitespace.

- **Orphan results linked to scripts.** The Glue theorem, the "Reduction in one line" and the fermion-spectrum callouts now carry inline `\veri{}` citations (v1, v6/v14/v23, v18/v20/v46).

2026-06-12 (tex-artefacts refactor + introduction visual pass B2)

- **Shared imports moved to `tex-artefacts/`.** The shared, imported-only TeX fragments (`tfpt_style.tex`, `tfpt_figures.tex`, `tfpt_docset.tex`, `tfpt_status.tex`) now live in a dedicated `tex-artefacts/` folder; all nine active documents `\input` them via `tex-artefacts/<name>` (and `tfpt_style` loads `tex-artefacts/tfpt_figures`). `changelog.tex` stays at the repo root because it is also a standalone `build.sh notes` deliverable. The `make_manifest.py TEX` list and the workflow rule were updated to the new paths.
- **Master script index outsourced (`tex-artefacts/verification.tex`).** The long vN→check table is moved out of `introduction.tex` into the shared fragment `tex-artefacts/verification.tex` (`\input` in the verification appendix); the paper-consistency audit now also reads that file, so every registered script stays cited exactly once.
- **Introduction box reduction (B2).** The introduction's eleven `keybox`es are reduced to the single headline "In one sentence" card; the explanatory keyboxes (*In plain words*, *Sharper still*, *compact status formula*, *hard reduction*, *Cosmology Pascal*, $E_6 \times A_2$, *closure ledger*, *Plausibility balance*) are converted to inline run-in paragraphs, the *Glue theorem* is re-typed as a `winbox` (a result, not a claim), and the explanatory *plausible/follows next* `winbox`es become prose. The loud boxes now carry only genuinely load-bearing callouts (one `keybox`, five `winbox`es, one `gapbox`) plus the two shared orientation cards.
- **Document-count consistency.** The introduction's document map gained the missing `tfpt_5_redteam` row; "seven/eight companion documents" and the ledger "six documents" wording are corrected to the true count (nine active documents); `README.md` ("10 ok", 140 scripts) and the workflow rule ("9 docs") are reconciled.

2026-06-12 (visual pass: box reduction B1)

- **Box reduction (B1).** The 26 non-load-bearing aside boxes (audit / recorded / interpretation: `auditbox/audbox/intbox/resbox/roadbox`) across tfpt_1, tfpt_2, tfpt_3 and `origin_theory` are converted to inline bold run-in paragraphs — the reviewer's prescription that an audit fingerprint must not carry the same visual volume as a theorem. The loud box families are now only `keybox` (claim), `winbox` (result) and `gapbox` (open/caveat).

2026-06-12 (visual pass: shared figures, badges, marker typology)

- **Shared figure library `tfpt_figures.tex`** (`\input` by `tfpt_style`): reusable TikZ macros for the architecture *claim stack* (`\TFPTclaimstack[zone]`, B3), the *anchor cockpit* $a = (1, 1, 2)$ (B4), the E_8 *glue* diagram (B5), the 2/3 *motif map* (B6) and the K_5 *action lattice* (B7). The claim stack and a marker *badge bar* are now shown near the front of *every* document, each highlighting that document's layer; the topical figures are placed in their natural homes (cockpit + K_5 in the introduction, cockpit + glue in tfpt_1, motif in the horizon note).
- **Marker typology (A2).** `tfpt_style` adds the sharper sub-types **[E]** (exact algebra), **[E]** (exact number), [C] (readout / standard physics in TFPT units) to the existing [C] (bridge) and **[X]** (kill test); the marker key and the new `\TFPTbadges` bar show the full typology. (Inline body markers retain **[E]** where they are genuine exact identities.)

2026-06-12 (review pass on list.txt)

- **Claim hygiene (review section A).** The introduction's one-sentence claim is softened from "derives all constants" to "constructs a discrete compiler; physical readouts run through named, status-typed bridges"; the ledger zombie in `FLAV.QRATIO.01` is removed (the statement no longer says "stays [P]" — ratios are [E] closed on the derived stratum, only the absolute scale stays [O], with an explicit forbidden-wording note); the seam misalignment is written explicitly as $\varepsilon = \frac{3}{4}\varphi_0^{\mathrm{ret}} = c_3 + 36c_3^4$ (leading c_3, fourth-order puncture correction exact); θ_{12} has a single prediction of record 0.306747 with the seam/non-linear values typed as scheme diagnostics; JUNO's first 59.1-day result (0.3092 ± 0.0087, compatible, precision pending) is recorded and NuFIT 6.0 is marked the pre-JUNO baseline; the ACT DR6 birefringence hint carries a systematics caveat (not a validation target); CODATA-2022 is dated; the null-model figure is moved out of the main text and typed as a grammar-internal bound. "Complete solutions of the five open questions" becomes "Closure status of the five former open questions".
- **Interface language unified (review sections A5/C2/C3/C6/C7/C8).** The live residual is stated everywhere as $v_{\mathrm{geo}} \oplus G_{\mathrm{net}} \oplus F_{\mathrm{transfer}}$ (historical $U_{\mathrm{wall}}/G_{\mathrm{metric}}/F_{\mathrm{frontier}}$ kept only for ledger continuity); `tfpt_research_contracts` is retitled and its recommended order rewritten (selector-triangle pairings $\to v_{\mathrm{geo}} \to G_{\mathrm{net}}$); F_{transfer} is named as one missing functor with Koide/η_B/axion/m_p/m_e as its four instances; full QG closure is framed as a certification layer, not a prerequisite. Document numbering aligned with file names (the Red Team paper is now in the canonical set; document $N \equiv$ `tfpt_N`). Website mirrored throughout (`papers.ts`, `predictions.ts`, `faq.ts`, `glossary.ts`, `OpenGates`, `Hero`, orientation components).

2026-06-12 (later)

- **Shared style and central status artifact.** Two new single-source fragments: `tfpt_style.tex` (the canonical palette, status markers, marker key, `\veri` reference, the four tcolorbox families with back-compat aliases, and one running header/footer) is now `\input` by all eight documents — the per-paper duplicated preambles are gone; and `tfpt_status.tex` ("TFPT status at a glance") gives one canonical overall-status card imported near the front of every document, replacing the divergent per-paper status statements. Both registered in `build.sh` and the manifest `TEX` list.

2026-06-12

- **Margin theorem (v122).** The established frozen selectors (D_4 annihilator $n = (5, -9, 6)$, $\det R = 8$, $\det K = 4$) pin R uniquely among hexagon matrices ($64 \to 12 \to 1$); the atom margins become theorems — H2 introduces no new residual class.
- **Inventory update (v123).** Residual table re-pinned post v110–v122: exactly five structural classes (R1 clock, R2 index-4 net, R3 EH step, R4′ selector establishment — replacing the discharged H2 dictionary — and R5 Q realisation); R2 carries three loads (metric + carrier + QBL: one theorem, three doors).
- **Resummed clock + glue Q-system (v124/v125).** R1's bend in closed form: $\mathrm{rate}(n) = -p_2 \ln(1 - n/N_{\mathrm{fam}})$ (pole at N_{fam} forces the three-level spectrum); the index-4 μ_4 extension exists explicitly as a Longo Q-system on $\mathbb{C}[\mathbb{Z}_4]$ (all Frobenius axioms exact) — R2 sharpens from "find" to "identify".
- **Clock–wall bridge (v126) and ring resummation (v127).** The resummed-clock weights are the Mehta–Seshadri parabolic weights (spec $A_0^\star$); $\det(\mathbf{1} - A_0^\star) = \frac{2}{9} =$ the entropy curvature; the clock is a log-determinant (RPA rings, one tower per hexagon site, coupling = parabolic weight).

- **Graded hull (v128).** E_8 is a $\mathbb{Z}_4$-graded Lie algebra over the carrier (grading = glue Q-system; exact on all 6720 root-pair sums); ring multiplicity 6 = sheet-doubled Ginsparg–Perry zero modes. *Notation fix in the same round:* the clock factor 6 is $p_2(a)$, not $2p_2$ — labels corrected across modules, ledger, papers, website (numbers unchanged).
- **Entropy power law → zeta budget (v129–v133).** The clock is $\Gamma_n \propto (S_n/S_{dS})^{p_2}$ (Gibbons–Hawking form); $p_2 = 2h$ with $h = N_{\mathrm{fam}}$ from mode counting + the Born rule (v130); the zero-mode norm is the area, $\|Y_{1m}\|^2 = A/(4\pi)$ exactly (v131); the det-ratio anomaly is the Koide constant, $\zeta(0)|_{\det'} = -\frac{2}{3}$ (v132); both $\zeta(0)$ budget routes computed exactly — the reduced seam route carries pure anchor ratios ($-\frac{4}{3}$ = minus the seed gain), the naive 4d route ($-\frac{109}{45}$) carries no atom (v133).
- **Dual anchor + determinant surface (v134/v135).** $d := a^\top R^{-1} = a^\top L^{-1} = (-\frac{1}{2}, -\frac{1}{2}, 1)$ with $(1,1,-2) = -2d$ — the inverse flavor response is the Nariai root, invariance exact via Sherman–Morrison (third algebraic flavor–horizon bridge leg); $\det M(s,t) = 3s(t+1)(t+2) + t^2 + 5t + 8$ with iso-volume walls at the $\mathbb{Z}_2/\mu_4$ atoms, winding line $(8,14,20)$, $\det F = 32$, row-budget axes with $\mathrm{rowsum}(V) = \mathrm{Spec}(Q_+)$; micro-identities $41 = 16 + 25$, $52 = 16 + 36$.
- **Dual-normal selector, Q_+ cohomology, Volkov–Wipf firewall (v136–v138).** (d, n) pins R *columnwise* (each column the unique lattice point of the address box; kernel $(6,8,7)$); the $A_0^\star$ zero mode carries the spectral normal $\sigma = (2,-9,5)$ with a $W(A_3)$ orbit control excluding the naive lift; $H^1(\mathbb{P}^1 \setminus \mu_4) = \chi_1 \oplus \chi_2 \oplus \chi_3$ with degrees $(1,2,3) = \mathrm{Spec}(Q_+)$ = the A_3 exponents and cusp weights = $\mathrm{spec}\, A_0^\star$; the external full-4d graviton/ghost value $-\frac{98}{45}$ is pinned (arXiv:2506.02142 = Volkov–Wipf hep-th/0003081) and its *edge* part is exactly two copies of the reduced seam budget — R1 typed closed as an edge-sector object.
- **Selector triangle + canonical map (v139/v140).** $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$ is pure anchor data (first selector *derived*); the frame $(\mathbf{1}, a, \sigma)$ has volume 11 and n is the unique covector with atom pairings $(2,8,121)$ — the R4′ residue is three pairings; the $H^1 \to$ generation equivariant map exists and is Schur-rigid (only sign-twisted μ_4 actions match), so the `GATE.QGEO` residue collapses to one $\mathbb{Z}_3$ choice.
- **Editorial and consistency passes.** Content-currency sweep (superseded "single remaining input" claims retired across contracts, paper 3, intro, website); introduction: self-explanatory title (spells out Topological Fixed-Point Theory), the old-version comparison removed throughout, redundant diagrams dropped, the unreadable status heatmap removed, the script-index master table converted to a page-breaking `xltabular` (it had silently truncated ∼110 rows), status card extended with the v134–v140 reductions, predictions table completed (CP phase, cosmic birefringence); the canonical document-set box extracted into the shared `tfpt_docset.tex`; paper 1 cleaned of all 35 old-series references, every section now opens with its machine-check script references; smaller box fonts across all eight documents; this `changelog.tex` introduced as a dual-mode (standalone + imported) document.

2026-06-11

- **External-review integration (v83) and release hygiene.** Manifest `-check` mode and the release rule; holomorphy pins $(E_8)_1$ (the unique even unimodular rank-8 lattice theory).
- **Sheet diamond and centered form (v94/v95).** All four flavor operators are points of one two-parameter surface $M(s,t)$; centered cross $R, L = C \mp U$, $K, F = C \mp V$ with the cofactor seam normal $n = (5,-9,6)$; documentation pass ("The Sheet Diamond and the Winding Line" in paper 3).
- **Horizon/anchor round (v101–v103).** The maximal (Nariai) black hole is the anchor: roots $(1,1,-2)$, entropy bound $\frac{2}{3} = |\mathbb{Z}_2|/N_{\mathrm{fam}}$; one repeller/attractor orientation in both sectors; trisection normal form (the canonical cosine coordinate).

- **Clock/ladder round (v104–v108) + Lean hardening.** The classical Nariai clock speaks anchor ($\chi_{\text{clock}} = (\lambda - 1)(\lambda + 2)$); machine-pinned residual inventory; first review validation; quantum-clock target made quantitative; Pascal Ladder Theorem ($g = 2K + 1 \Leftrightarrow$ the closure).
- **QBL theorem chain (v109–v113).** Sheet-Pairing Lemma; Calderón-sheet selection (a scalar seam datum exists iff the involution is sheet-odd); Quadratic-Transport Theorem (degree exactly 2); Self-Counting Channel (the Pascal closure as an identity); quasi-free kernel (one 2-point kernel determines the whole net, rank $= c$) — the QBL input merges with the R2/holomorphy premise; communicated across all status surfaces.
- U_{wall} **made exact (v114–v118).** Torsion normal form and $\delta = \frac{1}{2}$ theorem; exact anchor residue $A_0^\star$ (diag $= a/|\mu_4|$, off-weights $(8, 0, 5)/144$); resonance theorem ($M_\infty = \mathbf{1} \iff a_{13} = 0$, one gauge orbit); the holonomy is the exact 24-element $W(A_3) = S_4$; the hypercharge hexagon is the sign-twisted family spectrum with the lepton coefficients as resolvent determinants.
- **Second review validation + address pinning (v119–v121).** Anchor ratio triad $(\frac{2}{3}, \frac{4}{3}, \frac{5}{3})$; the 121 audit lemma; the lepton words are the compiler atoms $(8, 5, 3)$ with divmod-hexagon addresses; R is the unique hexagon matrix with atom margins and $\det 8$ — the address table carries zero residual information.

2026-06-10

- **B/C/D round closed.** Koide attractor toy model (v93), glue uniqueness (v92), the Wolfram extension as the independent second verification path, Lean hardening; papers and website synced with the round.
- **Content rounds v96–v100.** Sheet conjugation bridge, discriminant dictionary, Koide flow time, and the look-elsewhere-corrected numerology null test (v100: joint null probability $\leq 10^{-30.7}$, conditional on the declared grammar).

2026-06-09

- **Blind registry frozen (v84).** Machine-enforced prediction registry (`predictions_frozen.json`): exactly one θ_{12} number of record before JUNO; conditional entries carry their hypotheses by name. Research-contracts document added to the set.

2026-06-07 / 2026-06-08

- **Compiler-closure consolidation.** The TFPT development is consolidated into the current document set with the verification suite (modules v1–v82: E_8 glue, carrier Pascal, EM fixed point, flavor matrix, cascade, bootstrap, gravity/cosmology, transport, masses, registry and gate audits) as the single proof layer; manifests (`manifest.sha256`, `lean_manifest.sha256`) as content identity; website mirrors the suite (interactive DAG, in-browser script reproducer, script index).

2026-04-27

- **Initial commit.** TFPT papers and website baseline: electromagnetic closure and flavor transport (UFE bridge, birefringence seed, dyonic intercept), PDF-download tracking, accessibility and metadata passes.